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INSTITUTE OF ENGINEERING

THAPATHALI CAMPUS

DEPARTMENT OF CIVIL ENGINEERING

A Report on

"DETAILED PROJECT REPORT ON ENGINEERING SURVEY DESIGN, COST ESTIMATION OF THE ROAD SECTION FROM SURYABINAYAK GANESH TEMPLE TO GUNDU PATHIVARA TEMPLE"

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A final year project in the partial fulfillment of the requirement for Degree of Bachelor's in Civil Engineering

Supervisor:

Dr. Ram Chandra Tiwari

Submitted to:

Department of Civil Engineering

Thapathali Campus

Submitted By:

Sandip Lamsal [THA076BCE100]
Sanket Mudhbari[THA076BCE101]
Santosh Kumar Rokaya[THA076BCE104]
Spandan Poudel[THA076BCE122]
Sudip Thapa[THA076BCE127]
Swagat Adhikari[THA076BCE138

Abstract

This project is concerned with the detailed design and cost estimation of a road section from Ganesh temple to Gundu pathivara temple that lies in Suryabinayak Municipality, Bhaktapur district.

It plays a vital role in providing transport services to people, materials, and goods as well as promoting tourism through the gateway to various religious heritage and natural landscapes of Gundu, Suryabinayak. The total length of the road is 2.55 km. The designed bituminous pavement thickness comprises of 200 mm Granular Subbase, 250 mm granular base and 50 mm Asphalt concrete. The total filling volume is 12129.24 cum and total cutting volume is 23502.55 cum. For the stability of road, Gabion Retaining Wall and Stone Masonry Breast wall are used with total volumes of 770 cu.m and 480.9264 cu.m respectively. The road has maximum gradient of percent 9.97% and minimum gradient of 0.7%. The work has been carried out through map study and topographic survey. Design has been carried out using SW Road software. Pavement design has been conducted considering ordinary soil sample. The total cost of the project is Rs 171713381.81 including VAT and Contingencies and the cost per km is Rs 57485587.94.

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Project Members:

Sandip Lamsal [THA076BCE100]

Sanket Mudhbari [THA076BCE101]

Santosh Kumar Rokaya [THA076BCE104]

Spandan Poudel [THA076BCE122]

Sudip Thapa [THA076BCE127]

Swagat Adhikari [THA076BCE138]

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Acronym and Abbreviations

ADT	Average Daily Traffic
BC	Beginning of Curve
BVCS	Beginning of Vertical Curve
BVCE	Beginning of Vertical Curve Elevation
CA	Catchment Area
CBR	California Bearing Ratio
DOR	Department of Road
DOLIDAR	Department of Local Infrastructure Development and Agriculture Road
EC	End of Curve
EG	Existing Ground
ESAL	Equivalent Single Axle Load
EVCS	End of Vertical Curve Station
EVCE	End of Vertical Curve Elevation
GPS	Global Positioning System
ISD	Intermediate Sight Distance
LC	Length of Curve
LVC	Length of Vertical Curve
MC	Middle of Curve
NRS	Nepal Road Standard
NRRS	Nepal Rural Road Network
OMC	Optimum Moisture Content
PC	Point of Curvature
PC	Premix Carpet

PCU	Passenger Car Unit
PI	Point of Intersection
PT	Point of Tangency
PVI	Point of Vertical Intersection
RL	Reduced Level
Rm	Running meter
SRN	Strategic Road Network
SSD	Slopping Sight Distance
VDC	Village Development Committee
VDF	Vehicle Damage Factor
VFD	Vehicle Per Days

Salient Features

Table 1: Salient features of road

Features	**Description**
Name of the Road	Suryabinayak Ganesh Mandir-Gundu Pathivara
Road Classification	District Road Core Network
Scope	Detailed Engineering Survey, Design and Cost Estimation
Location	
Province	Bagmati
District	Bhaktapur
Municipality/VDC	Suryabinayak
Length	
Starting point	Suryabinayak Ganesh Mandir
End point	Gundu Pathivara Mandir
Geographical Features	
Terrain	Mid Hill
Altitude Range	
Climate	Mild, warm and temperate
Geology	Basically, characterized by phyllite and quartzite and common soil types include colluvial and residual soils
Meteorology	Unevenly Distributed Precipitation Controlled by Monsoon
Design Standard	
Standard	NRRS-2071
Existing surface	Earthen surface
Proposed Pavement	Flexible Pavement
Geometrics	
Design Speed	20k m/hr
Right of way	10m on either side
Formation width	5.25m
Carriage Way Width	3.75m
Shoulder Width	0.75m on either side

Maximum Gradient	9.97%
Minimum Gradient	0.7%
Lane	Single
Drainage Structures	
Side drain	1785.835
Causeway	Null
Slab Culvert	Null
Pipe Culvert	5 Nos.
Structures	
Stone Masonry Work	1528.41 cum
Earth Work	
Embankment/Filling	12129.24 cum
Excavation/Cutting	23502.55 cum
Pavement	
Sub base	200mm granular material
Base course	250mm granular material
Surface course	50mm Asphalt Concrete
Costing	
Earthwork	Rs 3679851.67
Sub-Grade	Rs 5316988
Sub base and Base	Rs 44784364.98
Bituminous surface and base course	Rs 26471030.12
Concrete for structures	Rs 6634397.28
River training and protection works	Rs 7230013.96
Masonry for structures	Rs 30850119.98
Grand Total (Incl. 13% VAT and 4% contingency)	Rs 171713381.81
Cost per KM	Rs 57485587.94

1. Introduction

1.1 General

Transportation engineering is a branch of civil engineering that involves the application of scientific and engineering principles to plan, design, and optimize transportation systems, ensuring the safe and efficient movement of people and goods. Transportation system includes roadways, railways, waterways and various intermodal operations.

Roads have been instrumental in the development of human civilization, fostering migration, trade, and cultural exchange. From early paths facilitating human movement to the strategic Roman roads enabling empire expansion, road networks have played a crucial role in societal organization and connectivity. In the modern era, roads are essential for economic activities, emergency response, and the integration of diverse regions, symbolizing a foundational element in the ongoing development of human civilization. For a developing country like Nepal, roadways play crucial role in the upliftment of country's economy.

In a country like Nepal, characterized by challenging terrain and diverse landscapes, the significance of roads and roadways is paramount along with a possession of various hurdles throughout the construction process. To serve as an access the road network has to pass through various topographical complications. Basically, hill roads may be defined as the one which passes through a terrain with a cross slope of 25% or more. There may be a section along hill roads with the cross slopes less than 25% especially when the road follows a river route. Even then these sections are also considered as hill roads. Hence, the overall consideration of terrain is very crucial for the design and establishment of hill roads.

The need of this road approach between the two points are as follows:

- Helps to improve the geometric design since the existing road does not have proper geometric design by considering the future traffic condition.
- Prevents the inundation of road surface by the help of proper drainage design considering the future extreme discharge condition.
- Helps in the safe, efficient, and fast movement of people and goods.
- Acts as a main serving route for the tourist flow to Ghyampe Danda.

- Improves the structural strength of the existing road by the design of Retaining wall and Breast wall and maintain slope stability.

1.2 Site Location Information

The Ganesh Temple to Gundu Pathivara road section lies in the Bhaktapur district under the jurisdiction of Suryabinayak Municipality. Suryabinayak Municipality (Nepali: सूर्यविनायक नगरपालिका) was created in December 2014 through the merger of the Village development committees of Kautunje (the municipality's centre), Sipadol, Nankhel and Chitpol. The municipality's name stems from the local Suryavinayak Temple.

Table 1-1: General Information of the site

GENERAL INFORMATION	
District	Bhaktapur
Province	Bagmati Province
Population (2011)	78490
Latitude	85° 25′ 26″ E
Longitude	27° 39′ 24″ N



Figure 1-1: Satellite view of the proposed site

1.3 Problem Statement

After observation of the existing road condition and having conversation with locals ,it was found that the road was constructed without proper supervision and proper design. Due to the lack of proper engineering design and the lack of proper geometric design the road, detrimental effects on overall road safety, efficiency, and sustainability was observed. The presence of steep gradients also impacted the movement of people and goods increasing the risks of accident and severity. It was observed that no proper drainage arrangements were present over the road. Inadequate drainage can affect the performance of road in various ways viz. : reduction of pavement and subgrade strength, inundation of road surface, loss of vehicle control, embankment erosion, loss of slope stability, etc. Inadequate design of such steep drainage line may cause serious damage tothe roads and culvert which they cross on their way downhill. Therefore, this road needs to be designed by applying knowledge of geometric design and considering future extreme discharge condition, future traffic volume, etc. Proper design is needed for the development of safe and efficient roads. Proper drainage facilities are to be provided for the better performance of the roadand to maintain the strength of pavement.

1.4 Objectives

1.4.1. General Objectives

To perform detail engineering survey, design and cost estimation of the road section from Suryabinayak Ganesh Temple to Gundu Pathivara Temple.

1.4.2. Specific Objectives

i) To be familiar with geometric design, pavement design and drainage design following various design norms.

ii)To prepare drawings including alignment plan, design profile, design cross section and typical drawings.

iii)To prepare detailed design report including detailed quantity and cost estimate.

1.5 Scopes.

Phases	Details
Geometric Design	L-Section, Cross Section
Detail Engineering Design	Design of pipe culvert and side drain Design of Horizontal and Vertical curve Design of pavements
Estimating and Costing	Cost Estimate Quantity Estimate Rate Analysis Abstract of Cost

2. Literature Review.

Literature review is a critical analysis and review of published research articles, books or any other relevant sources according to the nature of the project. Literature review helps in developing context and background with accordance to the project. It also helps to generate new ideas thereby supporting the methodology and design procedure of the project. Following existing literatures were thoroughly reviewed for the DPR project of road:

i)Nepal Road Standard (NRS),2070

ii) Nepal Urban Road Standard (NURS),2076

iii) Bagmati Province (Ministry of Physical Infrastructure Development) Transport Infrastructure Directorate, Draft Guidelines for Works including Preparation of Detailed Project Report (DPR) of Roads,2021.

iv) Nepal Rural Road Standard (NRRS),2071

2.1 Selection of Optimum Road Alignment

2.1.1. Requirement of Road Alignment

The ideal alignment must have the following requirements:

Safe

The road should be safe for every road user. Appropriate sight distance, super elevation, widening and infrastructures should be provided.

Easy

The alignment should be chosen near the location where the construction materials required for the road construction are available with easy access. Also, the alignment should be simple in design with minimum curves and no steep gradients and curves.

Short

Our alignment should be as short as possible as it saves our time, fuel and transportation costs but economical point of view should also be considered.

Economical

For economical design of road, our alignment should not have steep slopes or heavy cut and fill. Alignment should be such that it should have minimum drain structures. And operation and maintenance should also be considered for economic point of view.

Comfort

The alignment should be fixed in such a way that it provides comfort to the road users.

2.1.2. Special Consideration in Hill Road Design

1. Steep slopes are avoided, and gradient should be kept below the ruling gradient.
2. Unstable slope, ground with settlement problem, high flood land is avoided as far as possible.
3. Hair pin bends are minimized as much as possible.
4. Unnecessary rise and fall in alignment should be avoided.

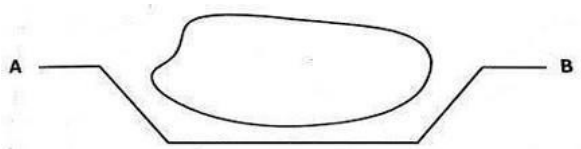
2.1.3. General Guiding Principle for Highway Alignment

The road alignment selection is influenced by the following factors:

• Alignment should be selected in such a way that it connects the starting point and end point with a short distance and with fewer curves.

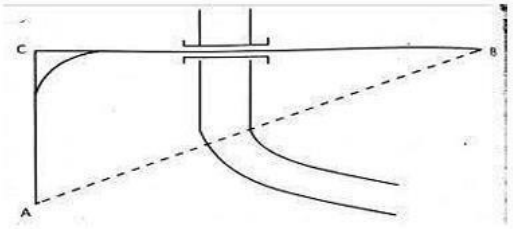
• The location through which alignment needs to be passed should give an easy gradient with easy curves. Sharp curves and steep slopes are avoided.

• Obligatory points are those through which road alignment should or should not pass through. Alignment should pass nearby the village or town. This helps in the development of rural as well as urban areas. The alignment should not pass through national parks, schools, hospitals, or offices, where a peaceful environment is needed.



• If the alignment is passed crossing the water supply pipeline, then the alignment needs to be shifted.

• The alignment crossing the bridge should be perpendicular to the river stream flow. The alignment line should pass through a stable bank. Bridge alignment should not deviate from a great distance with adjoining road alignment. Entry and exit alignment of the bridge structure should be kept straight as far as possible to avoid accidents due to not sufficient sight distance in the hilly region.



• The alignment should be placed such that the operating road condition does not affect the surrounding environment.

• The alignment should be placed near the location where the availability of road construction materials is high.

• Hard soil strata and balanced slope for cut and fill are perfect for the best alignment

• Locations with drainage problems, high water table levels, weak soil strata, and susceptibility to flood and landslide should be avoided.

• If the alignment is passing parallel to the river bank, then the toe erosion condition due to river water should be considered for the design. Also, the HFL for the past 50 years shall be kept in view.

2.2 Engineering Survey Procedure

2.2.1. Reconnaissance Survey

The Reconnaissance Survey is the preliminary exploration or study of an entire area that might be used for a road. Existing topographic maps and aerial photographs are collected and studied thoroughly to find the best possible alignment. Its purpose is to find fast, short and best cost-effective possible alignment. The reconnaissance survey can be performed in following manners.

Study of topographic maps and meteorological maps.

The reconnaissance survey begins with the analysis of all relevant maps. A number of cost-effective orientations are possible after studying the topographical features on the maps

Aerial Reconnaissance survey wherever possible and needed.

Aerial reconnaissance will provide you with a bird's eye perspective of the alignments in the issue as well as the surrounding area. The aerial survey could be used to make the final choice regarding which alignments should be explored in-depth on the surface.

Ground reconnaissance survey.

Ground reconnaissance encompasses a broad assessment of the topography, which includes traveling or riding along probable routes and collecting all essential data for review. In the case of hill parts, it could be a good idea to start the reconnaissance from a necessary point of view up to the summit.

2.2.2. Preliminary Survey

The main objective of the preliminary survey is to collect all the necessary physical information and details of topography, drainage, and soil of various established alternatives, calculating the earthwork, material quantity and ultimately cost of alternatives proposal and to compare all the alternative and select the best feasible route.

Survey Procedure

• Primary traverse is established, and the center line length should be measured with high precision using Total Station with least accuracy of 1 in 1000.

• All geographical and artificial features along the traverse and at a certain distance in either side are surveyed and plotted.

• Leveling work is also carried out to give profile and cross sections.

• Hydrological data are collected to estimate appropriate type, number, and size of hydraulic drainage structure.

• The soil survey is conducted in working out stability of slopes, pavement type and its approximate thickness.

• Material availability survey was done, and location of required materials was surveyed.

• Traffic survey at the starting point of alignment was done to know traffic lane distribution, load calculation and thickness calculation.

• After survey is completed, various alternatives are compared based on cost, length, comfort, feasibility and best possible alignment is selected.

Map preparation

• Plans and longitudinal sections are prepared as per data obtained from survey.

• Scales for the maps should generally be the same as adopted for the final drawings. The following scales are suggested:

a) Built-up areas and stretches in hilly terrain 1:1,000 for horizontal scale and 1:100 for vertical scale.

b) Plain and rolling terrain 1:2500 for horizontal scale and 1:250 for vertical scale.

c) For difficult terrain, we use larger than above recommendation and contour of 2m.

2.2.3. Determination of Final Centre line in the design office

Determination of the final center line is done in the office and it involves following steps:

• Making use of the maps from preliminary survey showing the longitudinal profile,

cross-sections, and contours, a few alternatives are selected and studied, and best route is selected.

• In the case of improvements of an existing road, the existing road levels are also kept in consideration.

• Horizontal curves including transitions with adequate provision for visibility are designed and final center line marked on map

• Vertical curves are defined, and profiles are determined.

2.3 Geometric Design Study

2.3.1. Background

Geometric design refers to the dimension and arrangement of physical elements of a roadway which include road widths, horizontal and vertical alignments, intersections and other features. The primary goal of the geometric design is to ensure the safe, efficient, and comfortable traffic flow, taking into account factors like terrain, soil characteristic, traffic volume, and driver behavior. There are several factors affecting the design of highways and they are explained below:

Design speed: Design speed is the single most important factor that affects geometric design as it directly affects the sight distance, horizontal curve, and the length of vertical curves. Since the speed of vehicles vary with driver, terrain, etc., a design speed is adopted for all the geometric design. According to NRRS 2071, design speed allocated is as follows:

Table 2.1: Design of speed

Road categories	Hill in kmph		Terai in kmph	
	Ruling	Minimum	Ruling	Minimum
District road	25	20	50	40
Village road	15		30	

Topography: Topography significantly influences geometric design by dictating the horizontal and vertical alignment of roads. The geometric design of roads is heavily influenced by the topography of the terrain, and adjustments to standards are often made based on economic considerations to align with the specific characteristics of the landscape.

Traffic Characteristic: Traffic characteristics significantly influence the geometric design of roads. Factors such as traffic volume, vehicle types, speed expectations, and patterns of flow play a pivotal role in determining road dimensions and layout.

Design Hourly Volume and Capacity: The general unit for measuring traffic on highway is Annual Average Daily Traffic Volume(AADT). The traffic flow (or) volume keeps fluctuating with time, from a low value during off-peak hours to the highest value during the peak hour and the road's capacity is its ability to handle traffic without excessive congestion. Geometric design must accommodate the projected AADT, ensuring sufficient lane capacity, appropriate intersection layouts, and suitable road widths to prevent congestion and maintain smooth traffic flow during peak. It will be uneconomical to design the roadway facilities for the peak traffic flow.

Environmental and other factors: Environmental factors like natural terrain, climatic condition, ecological condition, air pollution, noise pollution, aesthetics, and other global conditions should be given due considerations in the geometric design of roads.

2.3.2. Cross Section Elements

Carriageway width: The width of carriage way depends upon: The dimension of vehicles using the road, speed of travel, Traffic volume, width of shoulder. For district road (core network) with low volume of traffic (<100 vpd) single lane operation is adequate as there will be only small probability of vehicles meeting. Carriageway width of district road is 3.75m but can be reduced to 3m where traffic volume is less than 100 motorized vehicles per day. According to NRRS 2071, Carriage way width is shown in table 2.2.

Shoulder Width: Shoulders are provided on both sides of pavement in case of undivided carriageway and on outer edge of carriageway in case of divided carriageway. They provide structural stability and support to the edges of the flexible pavement. They can also act as service lanes and can also be used to prevent accidents by providing enough cross section for

vehicles. Minimum width of shoulder is adopted as 0.75m on either side of road. Shoulder width is shown in table 2.2 for different road categories and terrain.

Roadway Width: The total width of carriageway including separator and shoulder on both sides of the road is called roadway width. Roadway width for different road categories and terrain is shown in table 2.2.

Table 2.2: Carriageway width, Shoulder width and Roadway width

Road Categories	terrain	Carriageway width(m)	Shoulder width(m)	Roadway width(m)
District Road	Hill	5.5 (traffic > 400 Vpd)	0.75	7
		3.75 (traffic > 100 Vpd)	0.75	5.25
		3 (traffic < 100 Vpd)	0.75	4.5
	Terai	5.5 (traffic > 400 Vpd)	1	7.5
		3.75 (traffic > 100 Vpd)	1.5	6.75
		3 (traffic < 100 Vpd)	1.5	6
Village Road	Hill	3	0.5	4
	Terai	3	0.75	4.5

Right of Way: Right of way for a road project is the land area that is acquired or reserved for the construction and maintenance of the road. It includes the space for the road itself, as well as for any future expansion, drainage, utilities, and other facilities that are related to the road. Various ROW is recommended according to the type of roads which are listed as below.

Table 2.3: Right of way

Road categories	Total right of way(m)	Set back distance from ROW to building line in either side	Comments
District road	20	6	10 m ROW on either side from the center line
Village road	15	3	7.5 ROW on either side

Cross Slope of Camber: Cross slope is the transverse slope with respect to the horizon. Cross slope is provided to give a drainage gradient so that water will run off the surface to a drainage system. Various camber values are recommended which are shown below in table 2.4.

Table 2.4: Carriageway cross slope (%)

Camber		District road (core network)		Village Road	
		Hill	Terai	Hill	Terai
Carriageway Cross slope (%)	Earthen (existing)	5	5	5	5
	Gravel	4	4	4	4
	Bituminous seal coat	3	3	-	-

Lateral Clearance: It is the distance between edge of shoulder and roadside object. Hill roads normally have 1.0m but can be reduced to 0.5m in case of steep and difficult areas lateral clearance. In Terai road- normally 1.5 m is kept but can be reduced to 1 m in minimum.

Vertical Clearance: A vertical clearance of 5m should be ensured over the full width of roadway at all underpasses, and similarly at overhang cliffs. It should be measured with reference to the highest point of the carriageway i.e. the crown or super elevated edge of the carriageway. But in case of overhead wires as well as poles, clearance should be at least 7.0m above the surface of the road.

Super Elevation: Super elevation is the transverse slope provided to counteract the effect of centrifugal force and reduce the tendency of vehicle to overturn and to skid laterally outwards by raising the pavement outer edge with respect to inner edge. As per NRS, maximum super elevation to be provided is limited to:

• In plain and rolling terrain: 7%

• In snow bound areas: 7%

• In hilly areas not bound by snow: 10% • Minimum value of superelevation to be provided is equal to rate of camber of pavement

Widening of Curves: When the horizontal curve is not of greater radius, it is a common practice to widen the road slightly more than normal width which is called widening of curves. Recommended increase in width is shown in table below:

Table 2.5: Widening Width

Curve radius(m)		Up to 20	20-60	Above 60
Increase in width	For 3m carriageway	1.5m	0.6m	Nil
	For 3.75m carriageway	0.9m	0.6m	Nil

Sight Distance: The actual distance along a road surface which a driver sitting at specified height in a vehicle can see object either or stationary on the road surface. It can be Stopping Sight Distance, Safe Overtaking Sight Distance, Intermediate Sight Distance or Sight Distance at Intersection.

Stopping Sight Distance (SSD): Adequate stopping sight distance must be provided on 100% of the street and highway system so a driver with the standard eye height can see an object of standard height with sufficient time to stop safely. Factors influencing stopping sight distance are Perception time of driver. Speed of vehicle, Efficiency of brakes, Frictional resistance between road and wheel of vehicle and Longitudinal gradient of road. By considering total reaction time of driver as 2.5 seconds, SSD is calculated as:

$SSD = 0.278 \times v \times t + (v2 / 254 f)$ ..... Equation 2.1

Where,

v=speed of vehicle in kmph.

f=coefficient of longitudinal friction which varies according to speed.

Table below represents the stopping sight distance:

Table 2.6: Safe stopping sight distance

Speed(kmph)	Coefficient of longitudinal Friction	Safe SSD
15	0.4	15
20	0.4	20
25	0.4	25
30	0.4	30
40	0.38	45
50	0.37	60

Intermediate Sight Distance: ISD is related to design speed. However, the design speed can vary with different vehicle types.

Mathematically,

$$ISD = 2 \times SSD \quad \text{... Equation 2.2}$$

Overtaking Sight Distance (OSD): The minimum distance on highway required by a vehicle to overtake slow moving vehicle ahead i.e. travelling in the same direction safely against approaching traffic in the opposite direction. OSD calculations consider factors such as the speeds of the overtaking and overtaken vehicles, as well as the speed of oncoming traffic.

2.3.3. Horizontal Alignment

It is series of horizontal curves, tangents and transition curves used for roadway geometry. There are several elements of a horizontal curve. A Simple Horizontal curve is shown in figure below.

Layout of a Simple Horizontal Curve

R = Radius of Circular Curve
BC = Beginning of Curve
(or PC = Point of Curvature)
EC = End of Curve
(or PT = Point of Tangency)
PI = Point of Intersection
T = Tangent Length
(T = PI – BC = EC - PI)
L = Length of Curvature
(L = EC – BC)
M = Middle Ordinate
E = External Distance
C = Chord Length
Δ = Deflection Angle

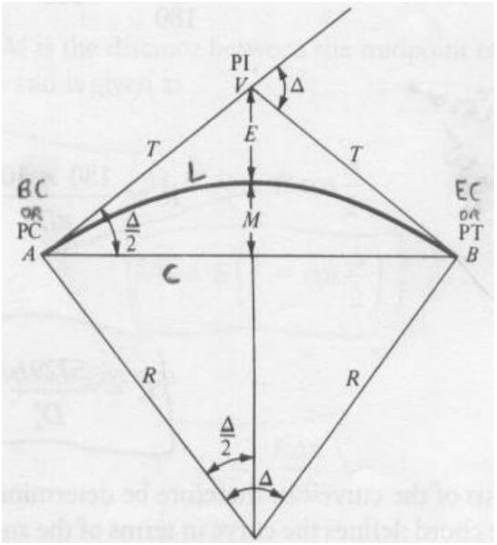


Fig: Simple Horizontal Curve

PI (Point Of Intersection): The point of intersection is the point where the back and forward tangents intersect. Sometimes, the point of intersection is designated as V (vertex).

Intersecting Angle: The intersecting angle is the deflection angle at the PI. Its value is either computed from the preliminary traverse angles or measured in the field.

R (Radius): The radius of the circle of which the curve is an arc, or segment. The radius is always perpendicular to back and forward tangents.

PC (Point of Curvature): The point of curvature is the point on the back tangent where the circular curve begins. It is sometimes designated as BC (beginning of curve) or TC (tangent to curve).

PT (Point of Tangency): The point of tangency is the point on the forward tangent where the curve ends. It is sometimes designated as EC (end of curve) or CT (curve to tangent).

POC (Point of Curve): The length of the curve is the distance from the PC to the PT, measured along the curve.

T (Tangent Distance): The tangent distance is the distance along the tangents from the PI to the PC or the PT. These distances are equal on a simple curve.

LC (Long Chord): The long chord is the straight-line distance from the PC to the PT.

E (External Distance): The external distance (also called the external secant) is the distance from the PI to the midpoint of the curve. The external distance bisects the interior angle at the PI.

M (Middle Ordinate): The middle ordinate is the distance from the midpoint of the curve to the midpoint of the long chord. The extension of the middle ordinate bisects the central angle.

D (Degree of Curve): The degree of curve defines the sharpness or flatness of the curve. Radius of Circular Curve: Mathematically, minimum radius of circular curve is given by,

$R_{min} = v2 / 127(e + f)$ …… Equation 2.3

Transition Curve Length (Ls): Transition curve is a curve provided outside of circular curve such that radius at outer end is infinity and at the circular curve end is equal to circular curve radius. It provides comfort to passengers. Transition curve length can be determined by using three criteria which are as follows:

Rate of change of centrifugal acceleration

Ls = V3 / 47 CR … Equation 2.4

Here, C is rate of change of centrifugal acceleration,

C = 80 / (75+V) … Equation 2.5

Where, V is design velocity in kmph

R is radius of circular curve in m

Rate of introduction of super elevation Pavement is rotated about Centre line of pavement so,

Ls = (W+We) / 2 --- Equation 2.6

Where, e is super elevation

1/N is rate of introduction of super elevation,

N = 60-150 (for hilly N=60)

W is normal width of carriage way

W_e is extra widening of carriage way

Maximum of them will be taken into the design.

Tangent Length (TI): It is the space or distance from its corresponding IP which is required for curve. It is given by:

TI = (R+S) tan Δ/2 + Ls/2 … Equation 2.7

Where, R is radius of circular curve in m

Δ is deflection at IP in degree

Ls is transition length in m S is lateral shift

S = Ls / 24R … Equation 2.8

Apex Distance:

Apex distance (E) = (R+S) (sec {Δ /2} − 1) +S … Equation 2-9

Where, R is radius of circular curve in m

Δ is deflection at IP in degree

Ls is transition length in m and

S is lateral shift as in Equation 2.8

Length of circular curve (LC)

It is the curve length of circular portion of the curve which is given by the following equation:

Lc = $\frac{\pi R \Delta S}{180}$ … Equation 2.10

Where,

$\Delta s = \Delta - 2s$

$\theta s = (180 * ls) / 2\pi R$ … Equation 2.11

Note: if Lc <0, Take Lc=0

Hairpin Bend:

A hairpin bend, also known as a hairpin turn or switchback, is a type of road curve characterized by a tight and acute change in direction. These bends are named for their resemblance to a hairpin or bobby pin. Hairpin bends are commonly found in mountainous or hilly terrains where the topography requires the road to ascend or descend steep slopes. The sample figure of hairpin bend is as below.

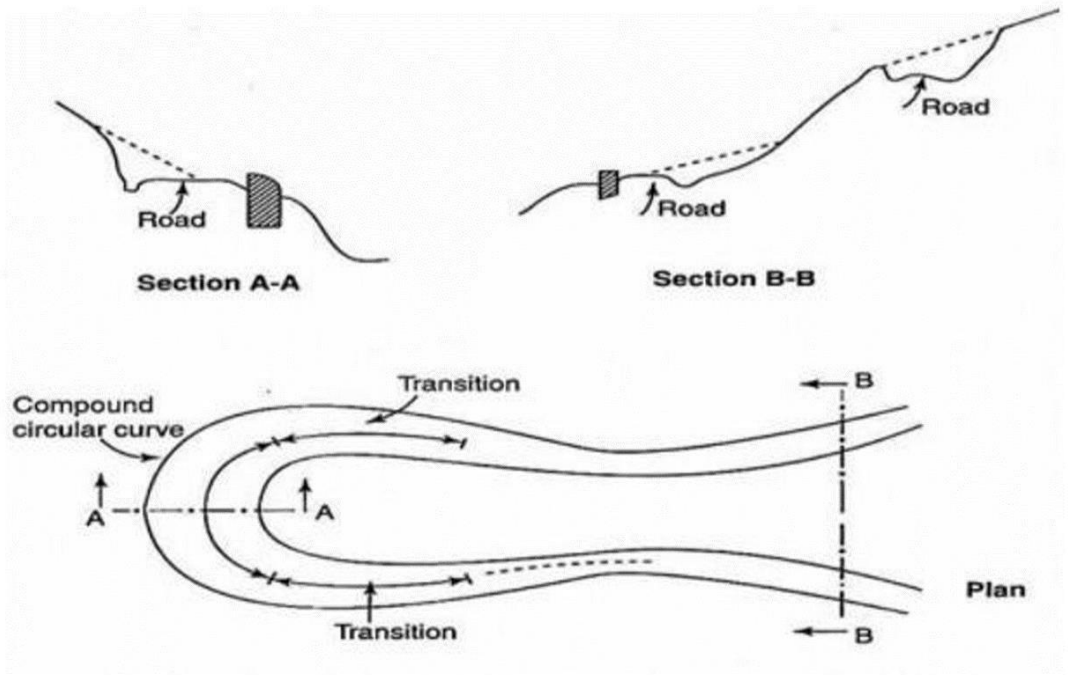


Fig 2.2: Hairpin Bend

Design criteria is mentioned as below:

Table 2.7: Hairpin Bend Design Criteria

S.N	Design Standard	District Road	Village Road
		Hill	Hill
1	Minimum Spacing between hairpin bend curves(m)	100	100
2	Minimum Radius of Curve(m)	12.5	10
3	Minimum roadway width at apex	5.5 for 4.5m roadway width 6.25 for a 5.25 m roadway width	5 for 4 m roadway width
4	Maximum Gradient (%)	4	4
5	Minimum Gradient (%)	0.5(max 1)	0.5 max 1)
6	Maximum Superelevation(%)	10	10
7	Minimum Trasnsition curve length(m)	15	15

2.3.4. Vertical alignment

Vertical alignment, within the realm of road design and transportation engineering, refers to the arrangement and structure of a road in the vertical dimension. The primary goal of vertical alignment is to create a road that provides smooth transitions between different grades (slopes) and minimizes abrupt changes in elevation. The vertical profile should be compatible with the horizontal alignment and properly coordinate with it.

Gradient:

Recommended gradient for different terrain condition are as below:

Table 2.8: Gradient for different terrain.

S.N	Design Standard	District Road (Core)		Village Road	
		Hill	Terai	Hill	Terai
1	Ruling Gradient (%)	7	5	7	5
2	Limiting Gradient (%)	10	6	10	6
3	Exceptional Gradient (%)	12	7	12	7
4	Limiting of maximum gradient length (m) above average gradient of 7%	300	-	300	-
5	Maximum recovery gradient (%) to be applied after gradient in excess of 7% for a minimum recovery length of 150m	4	-	4	-
6	Maximum gradient at bridge approach (%)	6	5	6	5
7	Minimum gradient on hill road (for better drainage) (%)	0.5 (max 1)	-	0.5 (max 1)	-

Vertical curve:

Both summit curves and valley/sag curves should be square parabolas. The length of the vertical curves is governed by sight distance requirements. Minimum length of vertical curves is shown below:

Table 2.9: Minimum length of vertical curve.

Design Speed (kmph)	Maximum grade change (%) not requiring a vertical curve	Minimum length of vertical curve
Up to 35	1.5	15
40	1.2	20
50	1.0	30

Summit Curves: Length of summit curve is determined by choice of sight distance that can be Stopping Sight Distance or Intermediate Sight Distance.

Table 2.10: Length of Summit Curve for SSD

Case	Length of Summit Curve (m)
	For safe stopping sight distance
When the length of the curve exceeds the required sight distance (i.e. L>S)	$L=(NS^2)/(4.4)$
When the length of the curve is less than the required sight distance. (i.e. L<S)	L=2S-(4.4)/N

N = Deviation angle, i.e., the algebraic difference between the two grades

L= Length of parabolic vertical curve (m)

S= stopping sight distance (m)

The above formula has been derived based on the following assumption

Height of driver's eye (H) = 1.2m (above the pavement surface)

Height subject above the pavement surface = 0.15m

In case of intermediate sight distance

Table 2.11: Length of summit curve for ISD

Case	Length of Summit Curve (m)
	For intermediate sight distance
When the length of the curve exceeds the required sight distance (i.e. L>S)	$L=(NS^2)/(9.6)$
When the length of the curve is less than the required sight distance. (i.e. L<S)	L=2S-(9.6)/N

Valley Curve: The length should be such that, for night travel, the headlight beam distance is equal to the stopping sight distance. Ready-made charts for the length of vertical curves—both summit and valley curves are given in IRC: 52-1982. According to the comfort criteria,

$L = 0.38\sqrt{NV^3}$ … Equation 2.12

Also, according to safety criteria,

Table 2.12: Length of Valley Curve for SSD

Case	Length of valley Curve (m)
	For stopping sight distance
When the length of the curve exceeds the required sight distance (i.e. L>S)	$L=(NS^2)/(1.5+0.035S)$
When the length of the curve is less than the required sight distance. (i.e. L<S)	$L=2S-(1.5+0.035S)/N$

2.4. Soil Survey and Studies

2.4.1. Sample collection of soil

While taking soil sample, upper surface debris should be removed from soil and only soil probe, spade or trowel should be used to take sample. Sample location should be homogeneous.

2.4.2. Required Test

Various tests on soil are to be conducted to decide the quality of soil for construction. Sometests are conducted in laboratory and some at site. Here we only discuss the CBR value testwhich is important test on soil.

CBR Test

CBR (California Bearing Ratio) test is a test used to determine the strength of

subgrade soil and base course materials for the design of highways, airport runways, and other typesof pavements. Here are the basic steps of conducting the CBR test:

- Sample preparation: A soil sample is collected from the field and brought to the laboratory for testing. The sample is then prepared by removing any organic matter and rocks larger than 20 mm in diameter.
- Specimen preparation: The prepared soil is compacted into a cylindrical mold using a standard compaction technique. The specimen is then allowed to cure for 4 days in the mold.
- Testing: A load is applied to the top of the specimen using a loading machine. The load is applied at a constant rate of 1.3 mm/minute until the specimen fails or the maximum load is reached.
- Calculation: The CBR value is calculated as the ratio of the load required to penetrate thesoil to a depth of 2.5 mm to the load required to penetrate a standard material at the samedepth.

2.5 Pavement Design Study

2.5.1. Pavement Design Principle

Pavement consists of one or more layers of subgrade, subbase, base, surface or wearing course. Surfacing course is the top course and is provided to provide a smooth, abrasion resistant, waterproof, and strong layer. Base course is the medium through which stress is distributed uniformly in the underlying layer. Subbase provides additional help in distributing the loads. Subgrade course is the compacted natural earth. The main function of the pavement is to support and distribute wheel load of vehicle over a wide area of theunderlying subgrade layers which leads deformation to the permissible range.

It is classified into three types:

a) Flexible Pavement
b) Rigid Pavement
c) Semi Rigid Pavement

Flexible Pavement: They have low flexural strength and are flexible in structural nature.The flexible pavement layers reflect the deformation of the lower layers on to the surfaceof the layer. No section of the pavement structure is subjected to excessive deformation to form local depression or settlement while designing flexible pavement. Stage construction can be done, and construction cost is low in flexible pavement.

Rigid Pavement: They have flexural strength and are either made of reinforced concreteor prestressed concrete**.** They have a slab action and can transmit the stresses generated through a wider area below.

Semi Rigid Pavement: When bonded materials like pozzolanic concrete, lean cement concrete or soil cement are used in base or subbase course layer, the pavement layer hasconsiderably higher flexural strength than the common flexible pavements layers but don't have flexural strength that of Rigid Pavement. So there pavement is called Semi Rigid Pavement.

2.5.2. Traffic Data

Traffic data is obtained from traffic volume studies like AADT, AAWT, ADT, AWT, etc.

2.6. Retaining Walls

They are rigid walls used for supporting soil laterally so that it can be retained at different levels on the two sides. Retaining walls are structures that are built to hold back soil and prevent erosion. They are typically used in areas where there is a significant difference in elevation, such as in hillsides or along coastlines. Retaining walls can be made from a variety of materials, including concrete, stone, brick, and wood.

2.6.1. Types of Retaining walls

A) Gravity Retaining Walls: They use their own weight to maintain stability and toresist lateral Earth Pressure. The main forces acting on gravity retaining walls are the vertical forces from the weight of the wall, the lateral earth pressure acting on the back face and the seismic loads. Gravity wall with different forces is shown in figure below:

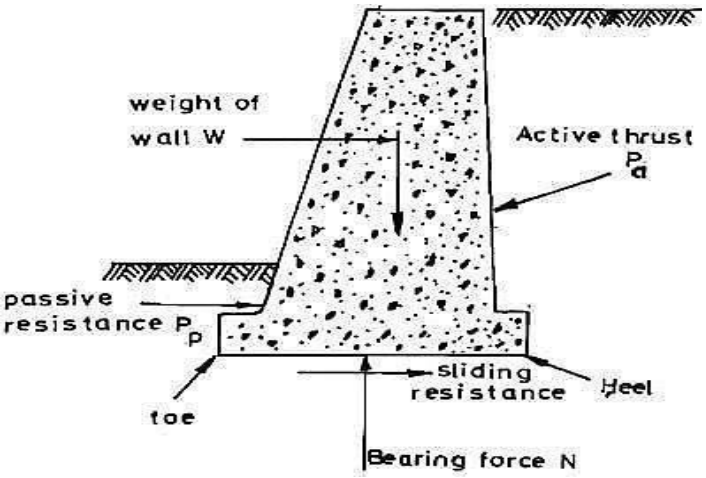


Figure 2- 3: Gravity Retaining Wall

(Source: theconstructor.org)

B) Cantilevered Retaining Wall: A cantilever retaining wall is a type of retaining wall that is designed to hold back soil and other materials by using a cantilevered structure. In other words, the wall is anchored to a base slab or footing, and extends up and outwards from there, without any external support from below. The base slab is typically thicker than the rest of wall and is reinforced with steel. It is more cost effectivethan other types of retaining walls.

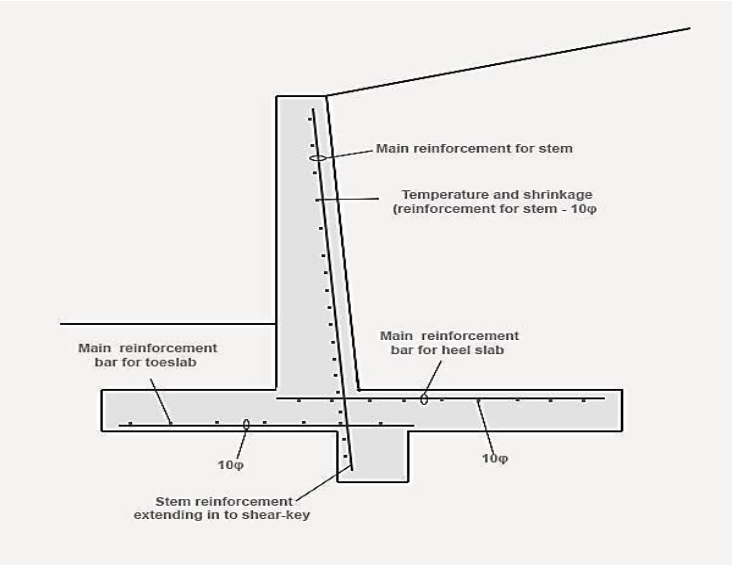


Figure 2- 4: Cantilever Retaining Wall

(Source: rocscience.com)

C) Sheet Piling: They are generally used in soft soils and tighten spaces. Sheet piles are driven down into the ground and are made up of steel, vinyl, aluminum, fiberglass, or wood planks.



Figure 2- 5: Sheet Piling

(Source: rocscience.com)

D) **Bored Pile:** They are retaining structures made by assembling a sequence of boredpiles, followed by excavating away the excess soil. They may include a series of earth anchors, reinforcing beams, soil improvement operations and shotcrete reinforcement layer depending upon nature of project.



Figure 2- 6: Bored Pile

(Source: rocscience.com)

E) **Anchored Retaining Walls:** They are pinned at both the top and bottom using cables or other stays, which are anchored in the rock or soil behind it. Anchors are driveninto the material and then expanded at the end of the cable, either by mechanical means or by injecting pressurized concrete into the hole. It is a technically complex method andis used where very large loads are expected.

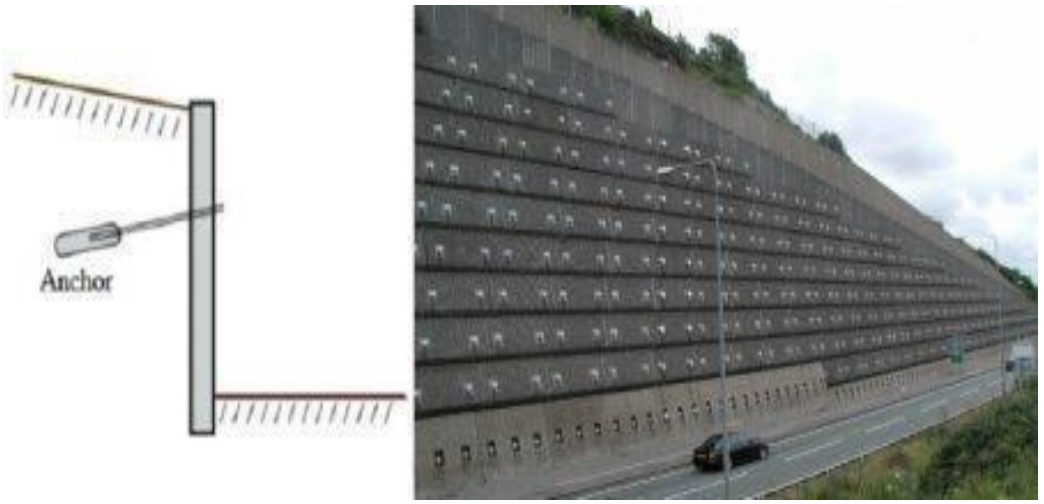


Figure 2- 7: Anchored Retaining Wall

(Source: rocscience.com)

2.7 Traffic Safety Works

2.7.1. Traffic Signs

Warning traffic signs are only used in our designed road. The sign will be positioned on the left-hand side of the road, in advance of the junction. They are:

Sharp bend to right:

This traffic signs contains an arrow showing a right-hand bend (if left symbol is reversed). This sign warns that there is a right hand (left hand) bend ahead which is likely to have a significantly different design speed to the road on which driver is travelling. It can also be used where it is difficult to see bend in advance. Misuse of the sign confuses drivers and makes the sign appear silly and unnecessary.

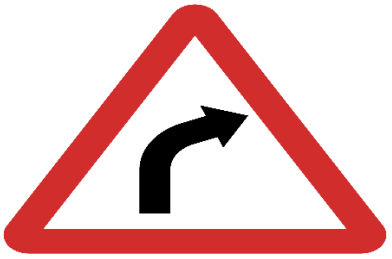


Figure 2-8: Traffic sign showing Sharp bend to right.

Double Bend 1st left:

This traffic sign contains arrow showing a tight left (right) followed by a tight sight (left) bend. This sign warns that there is a double bend ahead first to the left(right) and then to the right (left) which will have a lower design speed then the road on which the driver is travelling. This sign is only used where the distance between the first bend and the second is less than 250 meters.



Figure 2-9: Traffic sign showing double bend left

Steep hill downwards:

This traffic sign contains a symbol a car on a steep downhill gradient. The sign warns that there is a steep downhill gradient ahead.



Figure 2-10: Traffic signs showing steep hill downwards.

Steep hill upwards:

This traffic sign contains a symbol a car on a steep upward gradient. The sign warns that there is a steep upward gradient ahead.



Figure 2-11: Traffic signs showing steep hill upwards.

2.7.2. Road Markings

Edge of Carriageway

- It is denoated by yellow road marking 100/150mm wide (150mm for additional impact where needed) with a 2m mark and 2m gap.
- Define the edge of carriageway and as such are very useful for driverson unlit rural road particularly at right.



Figure 2-12: Edge of carriageway

3. Design Methodology

3.1. Survey

3.1.1. Map Study

Our project focus was on the upgradation of the existing road which had uneven gradients,improper drainage system, graveled road and was not engineered design. So, we were bound on the improvement of same road and had no option to choose other alternative. For the study purpose and to generate general idea of the alternatives, first we studied themap for the possibility of the alignment routes and its feasibility. Map study gives a roughguidance of the routes to be further surveyed in the field.

3.1.2. Reconnaissance Survey

We went on a tour of the place where we were surveying so that all the features that wereshown in the map could be checked and to examine the general character of the area. Wehave noted the deficiencies in the existing alignment that need to be assessed for improvement through visual inspection.

3.1.3. Preliminary Survey

Before conducting detailed engineering survey, preliminary survey was conducted where we have used Google Earth software to see the alignment in more detail and toplan the survey. We have made sure that no obligatory points were missed during thesurvey from where the road must pass.

3.1.4. Detailed Survey

Topography survey was carried out with the help of Total Station for the data required forpreparation of the contour map as well as road corridor points. Topographic survey consists of Horizontal and Vertical Control. For Horizontal control, Road strip survey method was carried out to get horizontal data of alignment. For Vertical control, fly levelling alongside horizontal control was carried out to transfer RL. A temporary benchmark was set at the start of survey to an electric pole. With helpof GPS, the co-ordinates of temporary benchmark were noted and was made as back sightfor an intermediate station with total station from which the transfer of coordinates was done to the first control point or station for the survey. All traverse angles, distances and co-ordinates were doubled checked with reciprocal observations. Traverse and level werecalculated at the site itself for accuracy as well as data validation.

3.2. Geometric Design

This project falls under the Village Road type. So we have designed the geometrical components according to NRRS 2071 for district road.

The sample calculation for the horizontal curve, vertical curve and hair pin bend are givenbelow. And rest of the horizontal curve and vertical curve design are calculated in ANNEX II and ANNEX III.

3.2.1. Sample calculation of elements of Horizontal Alignment (Curve)

Here,

for chainage 0+007.321m

Coefficient of lateral friction (f) =

0.15Efficiency of brake (ŋ) = 100%

Deflection angle (α)=97.9054 (d)

Radius (R) = 92m

Design Velocity (V) = 20 kmph (Village Road)

Length of wheelbase (l) = 6.1m

Reaction time (t) = 2.5 sec

A) Extra widening (We**)=**Wmechanical+ Wpsychological

$$=\frac{nl^2}{2R}+\frac{V}{9.5\sqrt{R}}$$

According to NRRS 2071, only mechanical widening is required for single lane low traffic speed.

So, We $=\frac{nl^2}{2R}$ (where n= no. of lanes)

=0.2022m

The minimum recommended value of widening on single lane for 92m radius and 3.75m wide carriageway is 0.9m.

∴ Adopted widening=0.6m

B) Superelevation (e) = $\frac{(0.75V)^2}{127R}$

$$=\frac{(0.75*20)^2}{127*92}$$

=0.0192<0.1

Check for value of coefficient of lateral friction devloped

or, f = $\frac{(V)^2}{127R} - e$

= 0.0149 < 0.15

C) Minimum Radius = $\frac{(V)^2}{127(e+f)}$

$= \frac{20^2}{(0.1+0.15)*127}$ = 12.59m

D) Transition curve

a) By rate of change of centrifugal acceleration

C = $\frac{80}{75+V} = \frac{80}{75+20}$ = 0.8 (0.5 < C < 0.8)

Where c is the rate of change of centrifugal acceleration given by an empirical formula suggested by IRC.

Length of transition curve (L_s) = $\frac{V^3}{47CR} = \frac{20^3}{47*0.8*92}$ = 2.31m

b) By rate of introduction of superelevation

N= 60(for hilly terrain), e = 0.0192

W= w+w_e = 3.75 + 0.2022= 3.9522

$L_s = \frac{Ne(w+we\ )}{2} = \frac{60*0.0192*3.9522}{2}$ = 2.27m

c) By empirical formula for steep and hilly terrain,

$L_s = \frac{V^2}{R} = \frac{20^2}{92}$ = 4.44m

Since, greatest value of L_s from these three methods is 4.44m.

Hence, the designed length of transition curve, L_s = 4.44m

E) Stopping Sight Distance (SSD) = 0.278*V*t + $\frac{V^2}{254f}$

Here, f= coefficient of longitudinal friction= 0.4 (NRRS 2071, for 20Kmph)

= 0.278*20*2.5 + $\frac{20^2}{254*0.4}$= 17.84m

Safe stopping sight distance = 20m for 20Kmph for Reaction time 2.5 sec

(From NRRS 2071) so adopted SSD = 20m

F) Intermediate Sight Distance (ISD) = 2*SSD =2*20 = 40m

G) Shift (S) = $\frac{Ls}{24R} = \frac{4.44^2}{24*92}$ = 0.00898 < 0.25 (So no need for transition curve)

3.2.2. Sample Calculation of Vertical Curve

Perception and Brake Reaction Time(t) =

2.5sDesign speed (v) = 20 kmph

Stopping Sight Distance (SSD) = 0.278*V*t

$+ \frac{V^2}{254*f}$

Here, f= coefficient of longitudinal friction= 0.4 (NRRS, for 20 Kmph)

= 0.278*20*2.5+$\frac{20^2}{254*0.4}$ = 17.84m

Safe stopping sight distance = 20m for 20Kmph for Reaction time 2.5 sec

(From NRRS 2071) so adopted SSD = 20m

Intermediate Sight Distance (ISD) = 2*SSD = 2*20 = 40m

For single lane opposing vehicle occupies same lane in single lane road, so it should be designed for ISD.

Sample Calculation of Summit Curve

At PVI 0+271.764m & elevation 1380.384m,

Take Intermediate Sight distance (S)=40m

Grade in (N1) = -

0.71% grade out (N2)

= -5.45%

Grade change (N) =N1-N2 =-0.71% -(- 5.45)%=4.74%=0.0474

Case 1: L<S

L = 2S – $\frac{9.6}{N}$ = 2*40 – $\frac{9.6}{0.0474}$ = -122.53m

Case 2: L > S

L = $\frac{NS^2}{9.6}$ = $\frac{0.0474*40^2}{9.6}$ = 7.9m

Case 1 is satisfied i.e., L < S

Hence, adopted Length of Summit curve = 20m.
Also,
L = 2T, So, Tangent length (T) = $\frac{L}{2}$ =$\frac{20}{2}$ = 10m

Radius (R) = $\frac{2T}{N}$ = $\frac{2*10}{0.0474}$ = 421.94m

Elevation of Beginning of Vertical Curve (BVC)=Elevation of PVI + N1%* Tangent length

= 1380.384+ 0.0071*10 =1380.454m

Elevation of Ending of Vertical Curve (EVC)=Elevation of PVI - N2%* Tangent length

= 1382.384-0.0545*10 = 1379.839m

Chainage of BVC= Chainage of PVI - Tangent length

= 0+271.764 -10 = 0 + 261.764m

Chainage of EVC= Chainage of BVC + Curve length

= 0+261.764 + 20 = 0+281.764m

A) Sample Calculation of Valley Curve At

PVI 0+131.991m & elevation

1381.371m,Sight Distance (S) = 20m

Grade in (N1) = -9.97%

Grade out (N2) = -0.71%

Grade change (N) = -9.97% - (-0.71%) 9.26%= - 0.0926

Case 1 : When L > S

$$L = \frac{NS^2}{1.5+0.035S} = \frac{0.0926*20^2}{1.5+0.035*20} = 16.83m$$

Case 2 : When L < S

$$L = 2S - \frac{1.5+0.035S}{N} = 2*20 - \frac{1.5+0.035*20}{0.0926} = 16.24m$$

Case 2 is satisfied i.e. L < S

Hence, adopted Length of Valley curve = 20m.
Also,
L = 2T , So, Tangent length (T) = $\frac{L}{2} = \frac{20}{2}$ = 10m

Radius (R) = $\frac{2T}{N} = \frac{2*10}{0.0926}$ = 215.98m

Elevation of Beginning of Vertical Curve (BVC) = Elevation of PVI + N1% * Tangent Length

= 1381.371+0.0997*10

= 1382.368m

Elevation of Ending of Vertical Curve (EVC) = Elevation of PVI – N2% * Tangent Length

= 1381.371-0.0071*10

= 1381.301m

Chainage of BVC = Chainage of PVI – Tangent Length

= 0+131.991-10

= 0+121.991m

Chainage of EVC = Chainage of BVC 6 Curve Length

= 0+121.991 + 20

= 0+141.991m

3.3. Drainage Design

Runoff for Drainage design is estimated using a rational method.

$$Q = (1/3.6) * CiA$$

Where,

Q = runoff in m3 /s

C =Catchment Coefficient A=Area in Km^2

i= intensity of rainfall in mm/hr, governed by the time of concentration (tc) tc=0.01947K0.77

$K=(L^3/\Delta H)^{0.5}$

L= Length of travel of water

ΔH= Elevation difference

A) Hydrological analysis:

Table 3-1 Rainfall Data

Extreme rainfalll for suryabinayek													
Month	jan	feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
Month index	1	2	3	4	5	6	7	8	9	10	11	12	Extreme
2013	_	_	_	_	_	_	209	350	91	101	0	0	394.8
2014	2	14	0	0	30.8	172.2	365	279	68	122	0	9.8	364.9
2015	0	21	61	27	77	130.4	309	370	90	40.4	0	0	370.1
2016	1.4	11	3.4	3.2	136	317.5	363	153	232	65.3	0	0	363.3
2017	23.4	0	90	86	126	127.7	255	267	84	29.9	1.4	0	266.9
2018	8.2	1.8	50	125	141	197.9	393	378	52	4	0	0	392.9
2019	17.8	93	37	112	138	144.7	401	241	290	24.8	0	24	400.6
2020	69.1	20	28	75	224	315.8	300	291	205	3.6	1.2	0	315.8
2021	0	8.4	6.8	59	233	188.7	418	307	188	64.5	0	41	417.9
2022	3	49	17	15	255	302.8	330	226	234	78.7	0	0	329.9

3.3.1. IDF CURVE GENERATION:

Gumbels Method was implied for the generation of the Intensity Duration and Frequency Curve.

The calculation was done as below:

Peak value of precipitation was taken.

			Time in Min							
Year	Max Instantaneous Rainfall(x) of 24 hr		5	10	15	30	60	120	720	1440
2013	349.8		52.96906528	66.73684	76.39461	96.25117931	121.2689	152.7892233	277.6364	349.8
2014	364.9		55.25560869	69.6177	79.69238	100.4061044	126.5038	159.3847558	289.6213	364.9
2015	370.1		56.04302762	70.60979	80.82803	101.8369396	128.3065	161.656065	293.7486	370.1
2016	363.3		55.01332595	69.31245	79.34295	99.96584746	125.9491	158.6858914	288.3514	363.3
2017	266.9		40.41579052	50.92071	58.28966	73.44036523	92.52906	116.579313	211.8387	266.9
2018	392.9		59.49555674	74.9597	85.80744	108.1106013	136.2108	171.6148823	311.8449	392.9
2019	400.6		60.66154246	76.42875	87.48908	110.229338	138.8803	174.9781671	317.9564	400.6
2020	315.8		47.82055693	60.25013	68.96918	86.89571877	109.4817	137.9383554	250.6506	315.8
2021	417.9		63.28122465	79.72935	91.26732	114.9896164	144.8778	182.5346381	331.6874	417.9
2022	329.9		49.95567363	62.9402	72.04855	90.77548328	114.3699	144.0970977	261.8418	329.9
		Mean	54.09113725	68.15056	78.01292	98.29011938	123.8378	156.0258389	283.5178	357.21
		STD_DEVA	6.729275785	8.478356	9.705295	12.2279056	15.4062	19.41059022	35.27138	44.43915816

N	10
Yn	0.4952
Sn	0.9497

Gumbel Extreme Value

MEAN	357.21
SD	55.81
SKEW	-0.429

T	Yt	Kt	XT
1			
2	0.367	-0.1	351.63
5	1.5	1.06	416.366
10	2.25	1.85	460.4585
20	2.97	2.61	502.8741
50	3.902	3.59	557.5679
100	4.6	4.32	598.3092

Where,

Yt = reduced variate for given T

Yt = -ln(ln(T/T-1))

Yn = reduced mean, function of sample size N

Sn = reduced standard deviation, function of sample size N

Kt = frequency factor, (Yt-Yn)/Sn

X' = Mean Value of Precipitation

Sx = Standard Deviation of precipitation

Final Extreme Rainfall value by Gumbel's Method is given as:

Xt=X'+Kt*Sx

The values of Yn and Sn are the function of the sample size and arev calculated from the tables below:

Table Q2(a)

Reduced mean Y_n in Gumbel's Extreme Value Distribution
N = sample size

N	0	1	2	3	4	5	6	7	8	9
10	0.4952	0.4996	0.5035	0.5070	0.5100	0.5128	0.5157	0.5181	0.5202	0.5220
20	0.5236	0.5252	0.5268	0.5283	0.5296	0.5309	0.5320	0.5332	0.5343	0.5353
30	0.5362	0.5371	0.5380	0.5388	0.5396	0.5402	0.5410	0.5418	0.5424	0.5430
40	0.5436	0.5442	0.5448	0.5453	0.5458	0.5463	0.5468	0.5473	0.5477	0.5481
50	0.5485	0.5489	0.5493	0.5497	0.5501	0.5504	0.5508	0.5511	0.5515	0.5518
60	0.5521	0.5524	0.5527	0.5530	0.5533	0.5535	0.5538	0.5540	0.5543	0.5545
70	0.5548	0.5550	0.5552	0.5555	0.5557	0.5559	0.5561	0.5563	0.5565	0.5567
80	0.5569	0.5570	0.5572	0.5574	0.5576	0.5578	0.5580	0.5581	0.5583	0.5585
90	0.5586	0.5587	0.5589	0.5591	0.5592	0.5593	0.5595	0.5596	0.5598	0.5599
100	0.5600									

Table Q2(b)

Reduced mean Standard Deviation S_n in Gumbel's Extreme Value Distribution
N = sample size

N	0	1	2	3	4	5	6	7	8	9
10	0.9496	0.9676	0.9833	0.9971	1.0095	1.0206	1.0316	1.0411	1.0493	1.0565
20	1.0628	1.0696	1.0754	1.0811	1.0864	1.0915	1.0961	1.1004	1.1047	1.1086
30	1.1124	1.1159	1.1193	1.1226	1.1255	1.1285	1.1313	1.1339	1.1363	1.1388
40	1.1413	1.1436	1.1458	1.1480	1.1499	1.1519	1.1538	1.1557	1.1574	1.1590
50	1.1607	1.1623	1.1638	1.1658	1.1667	1.1681	1.1696	1.1708	1.1721	1.1734
60	1.1747	1.1759	1.1770	1.1782	1.1793	1.1803	1.1814	1.1824	1.1834	1.1844
70	1.1854	1.1863	1.1873	1.1881	1.1890	1.1898	1.1906	1.1915	1.1923	1.1930
80	1.1938	1.1945	1.1953	1.1959	1.1967	1.1973	1.1980	1.1987	1.1994	1.2001
90	1.2007	1.2013	1.2020	1.2026	1.2032	1.2038	1.2044	1.2049	1.2055	1.2060
100	1.2065									

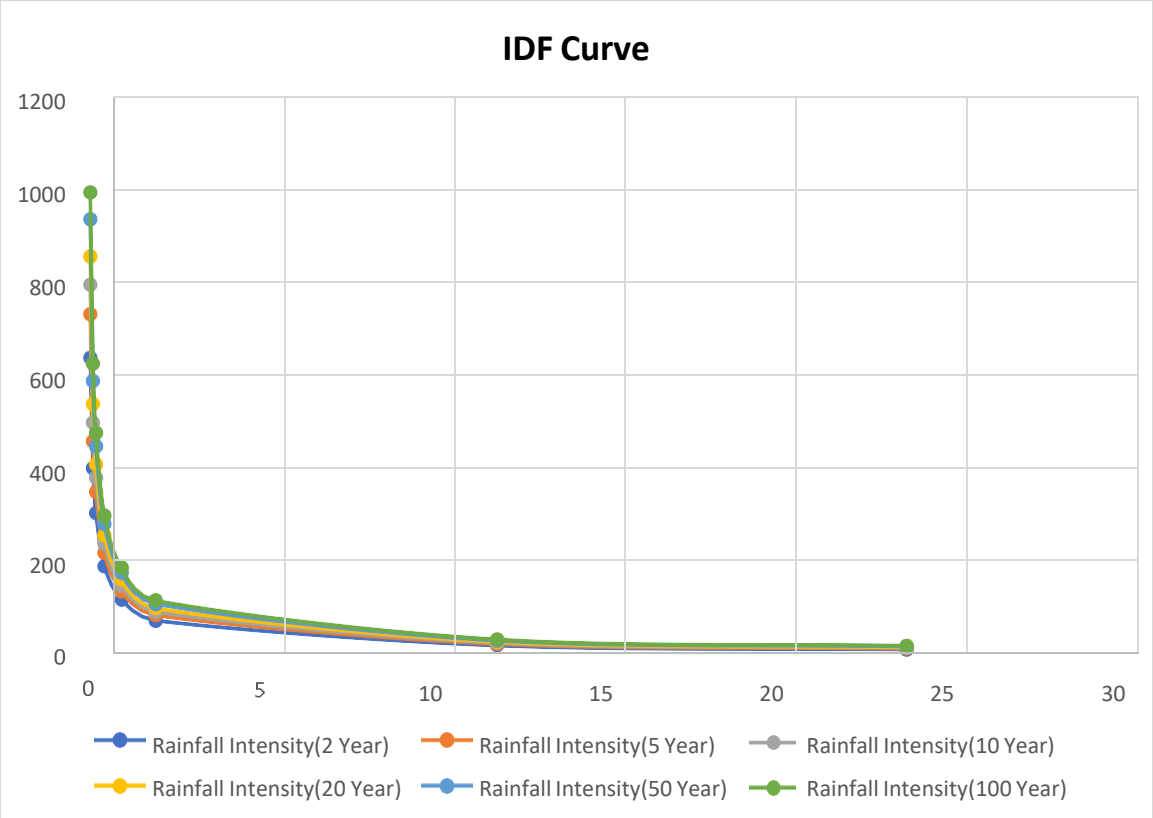
					Return Period(Year)											
					2		5		10		20		50		100	
Time(min)	Time(hr)	Mean	STD_DEVIA		Rainfall	Rainfall Intensity(2 Year)	Rainfall	Rainfall Intensity(5 Year)	Rainfall	Rainfall Intensity(10 Year)	Rainfall	Rainfall Intensity(20 Year)	Rainfall	Rainfall Intensity(50 Year)	Rainfall	Rainfall Intensity(100 Year)
5	0.083333	54.09114	6.729275785		53.41821	641.018516	61.22417	734.6900349	66.5403	798.4835694	71.65455	859.8545646	78.24924	938.9908478	83.16161	997.9393036
10	0.166667	68.15056	8.478856212		67.30273	403.8163609	77.13762	462.8257201	83.83552	503.0131285	90.27907	541.6744329	98.58786	591.5271674	104.7771	628.6623676
15	0.25	78.01292	9.705295109		77.04239	308.1695398	88.30053	353.2021291	95.96772	383.8708616	103.3437	413.3749587	112.8549	451.4197156	119.9398	479.7591773
30	0.5	98.29012	12.22799066		97.06733	194.1346576	111.2517	222.5033986	120.9117	241.8234895	130.205	260.409906	142.1883	284.376601	151.1147	302.2293432
60	1	123.8378	15.40619567		122.2972	122.2971708	140.1684	140.1683578	152.3393	152.3392524	164.048	164.0479611	179.146	179.1460328	190.3926	190.3925557
120	2	156.0258	19.41059022		154.0848	77.04238994	176.6011	88.30053227	191.9354	95.96771541	206.6875	103.3437397	225.7099	112.8549289	239.8796	119.9398794
720	12	283.5178	35.27113832		279.9806	23.33255221	320.9054	26.74211926	348.7698	29.06415198	375.5761	31.29800625	410.142	34.17850255	435.8901	36.32417836
1440	24	357.21	44.43915816		352.7661	14.69858684	404.3155	16.84647949	439.4224	18.30926844	473.1962	19.71650845	516.7466	21.53110741	549.1872	22.88279847

RAINFALL INTENSITY FOR VARIOUS RETURN PERIODS:

Time in hr	Rainfall Intensity(2 Year)	Rainfall Intensity(5 Year)	Rainfall Intensity(1 0 Year)	Rainfall Intensity(2 0 Year)	Rainfall Intensity(5 0 Year)	Rainfall Intensity(100 Year)
0.083333	641.0185	734.6900349	798.4836	859.8546	938.9908478	997.9393
0.166667	403.8164	462.8257201	503.0131	541.6744	591.5271674	628.6624
0.25	308.1696	353.2021291	383.8709	413.375	451.4197156	479.7592
0.5	194.1347	222.5033986	241.8235	260.4099	284.376601	302.2293
1	122.2972	140.1683578	152.3393	164.048	179.1460328	190.3926
2	77.04239	88.30053227	95.96772	103.3437	112.8549289	119.9398
12	23.33255	26.74211926	29.06415	31.29801	34.17850255	36.32418

24	14.69859	16.84647949	18.30927	19.71651	21.53110741	22.8828

IDF CURVE



3.3.2 Side Drain Design

The chainage with the highest catchment area is considered for the design of side drain.Upon using Google Earth Pro, we found the following characteristics of our catchment area:



(Source:Google Earth Pro)

The highest CA was found to be 17393 square metres with chainage from 0+800 to 0+1000.
Length of longest channel = 191 m.
Area of Catchment = 1.73 hectares.
Farthest point of catchment elevation=1510m.
Lowest point of catchment elevation=1441m.
Elevation difference=69m.

3.3.3. Calculation of discharge using rational method, Q= C*I*A/ 360

Where, A= catchment area in hectares,

I= rainfall intensity in mm/hr

C= Coefficient of runoff (assumed 0.4 in our case)

Using Kirpich Formula, time of

concentration(tc)=0.02741 hour and Longitudinal

Slope of Channel=(H/L)=0.361

Here, C =0.4 , I= 353.2 mm/hr and A=1.73

hectares

So, Q= (0.4*353.2*1.73)/360 = 0.678 m^3/s

For a return period of 20 years our drainage discharge is found to be 0.678 m^3/s, but we take Q= 1.1*0.678= 0.7458m^3/s considering factor of safety.

3.3.4. Determination of Channel Dimension:

After calculation of design discharge for drainage, we selected rectangular channel section considering side slope as 1:1 i.e. D/B=1 .

Channel dimension is determined by using manning's formula,

Q= $1/n * A * R^{2/3} * S^{1/2}$

Longitudinal slope of channel (S) = 7% = 0.07

Manning's rugosity coefficient (n) = 0.017 (for concrete finish with gravel on bottom)B= Bottom width of channel

D= Depth of channel

Here, B/D= 1, so B= D

Cross sectional area of channel (A)=

B*D= D^2Wetted Perimeter of channel

(P) = B + 2D= 3D

Hydraulic Radius (R) = (A/P) = D/3

Now. After putting all the values and solving equation (1) we get, D= 0.42 m but taken as 0.45m.So, B=0.45m.

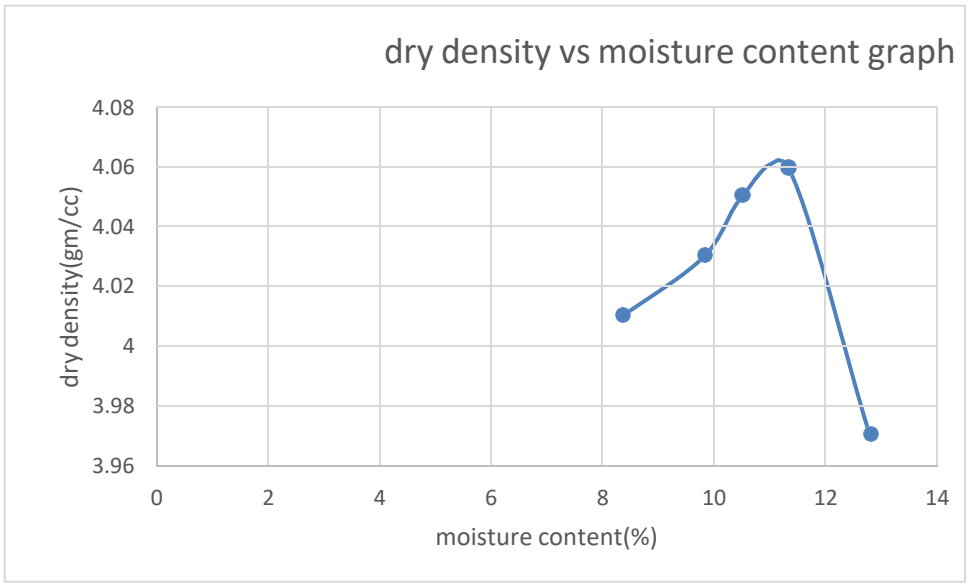
Hence, Dimension of side drainage =45cm * 45cm.

3.3.5. CBR Test Calculations

Observations:

Observation:	
Sample	Subgrade
Weight of hammer	4.86
Blows per layer	56
No. of layers	3
Weight of the mould	3140gm
volume of mould	944cc
Sample Calculation	
Test no.1	
Weight of soil+mould:	7240gm
wt. of soil	4100gm
Density of soil(γ)	(4100/944)=4.34gm/cc
For water content:	
Wt. of wet soil (w1)	24.95gm
wt. dry soil(w2)	24.15gm
wt. of water (w3)	0.8
water content(w)	(w3/w2)*100=3.31%
dry density, (γd)	γ/(1+w)=4.26gm/cc

Test No.	1	2	3	4
Wt.Of Soil+Mould(gm)	7240	7320	7360	7410
Wt . of Mould(gm)	3140	3140	3140	3140
Wt Of wet soil(gm)	4100	4180	4220	4270
Wet Density(gm/cc)	4.34	4.43	4.47	4.52
Water Content(%)	8.35	9.82	10.5	11.33
Dry Density(gm/cc)	4.01	4.03	4.05	4.06


dry density vs moisture content graph
4.08
4.06
4.04
4.02
4
3.98
3.96
dry density(gm/cc)
0
2
4
6
8
10
12
14
moisture content(%)

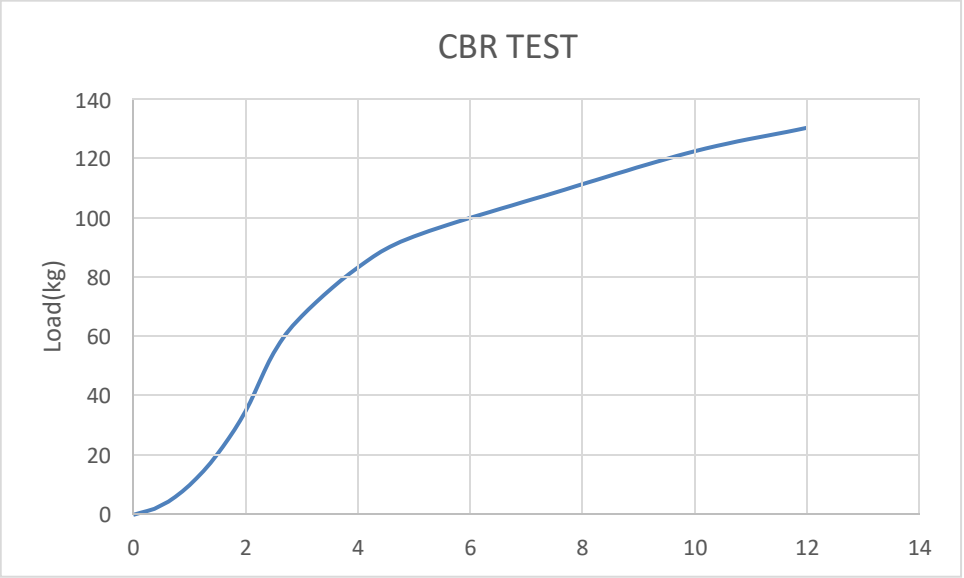
CBR TESTING OF SAMPLES

Proving ring calibration factor is 93.5 divisions

=0.3 KN Therefore, 1 division = 0.327 kg

SAMPLE NO. 1

SN	Penetration (mm)	Proving Ring Division	Load (kg)	Corrected load(kg)
1	0	0	0	
2	0.5	9	3.1	
3	1	30	10	
4	1.5	62	20.5	
5	2	107	35	
6	2.5	166	54.6	70
7	3	204	67	
8	4	255	83.4	
9	5	287	93.9	99.5
11	7.5	332	108.6	
14	10	374	122.6	
15	12	399	130.5	



From above graph,

Load corresponding to 2.5 mm

penetration =70kg Therefore,

$CBR_{2.5\ mm} = (70 \times 100) / 1370$

$= 5.11\%$

Load corresponding to 5 mm penetration =95kg Therefore,

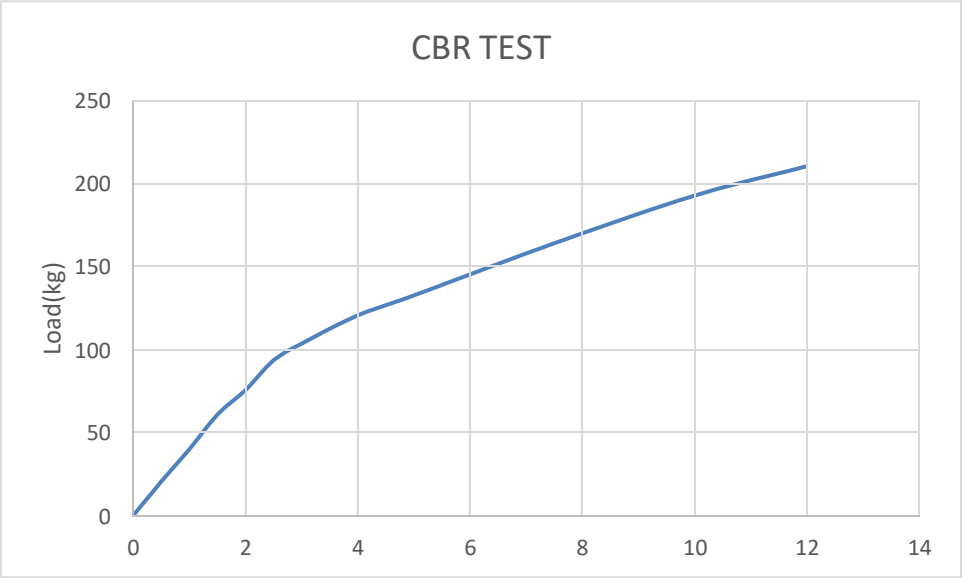
$CBR_{5\ mm} = (99.5 \times 100) / 2055$

$=4.83\%$

As, $CBR_{5\ mm} < CBR_{2.5\ mm}$, the CBR value of the given soil sample = 5.11%.

SAMPLE NO. 2

SN	Penetration (mm)	Proving Ring Division	Load (kg)
1	0	0	0
2	0.5	62	20.5
3	1	122	39.9
4	1.5	185	60.8
5	2	230	75.5
6	2.5	285	93.5
7	3	316	103.5
8	4	368	120.5
9	5	405	132.5
11	7.5	501	163.9
14	10	588	192.6
15	12	643	210.3



From above graph, Load corresponding to 2.5 mm penetration = 88.5kg Therefore, $CBR_{2.5\ mm}$ = (88.5×100) / 1370 = 6.46%

Load corresponding to 5mm penetration =139.5 kg Therefore,

$CBR_{5\ mm}$ = (139.5×100) / 2055

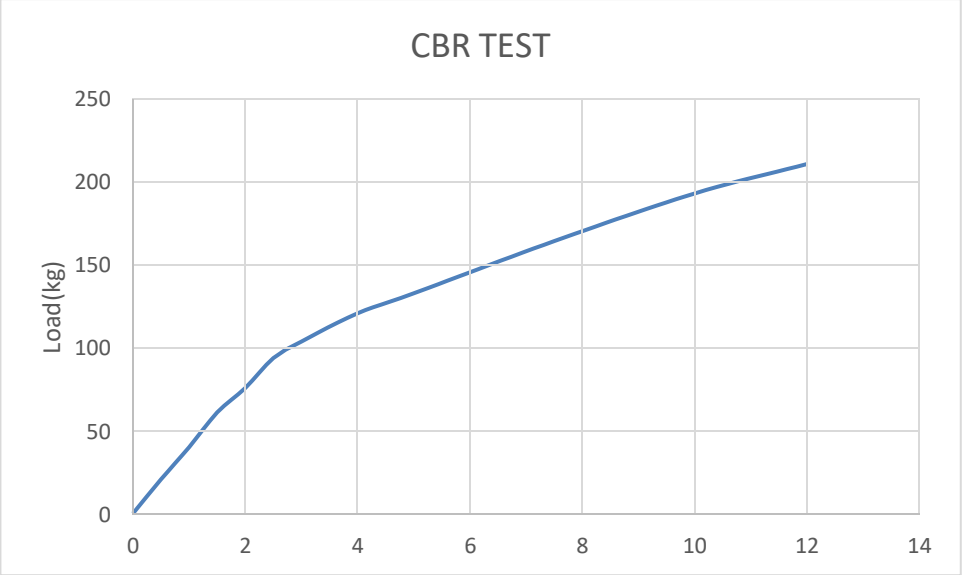
=6.79%

As, $CBR_{5\ mm}$> $CBR_{2.5\ mm}$, the CBR value of the given soil sample = 6.79%.

SAMPLE NO. 3

SN	Penetration (mm)	Proving Ring Division	Load (kg)	Corrected load(kg)
1	0	0	0	
2	0.5	62	20.5	
3	1	122	39.9	
4	1.5	185	60.8	
5	2	230	75.5	
6	2.5	285	93.5	130
7	3	316	103.5	
8	4	368	120.5	

9	5	405	132.5	170
11	7.5	501	163.9	
14	10	588	192.6	
15	12	643	210.3	



From above graph,

Load corresponding to 2.5 mm penetration = 130kg Therefore,

CBR2.5 mm = (130 ×100) / 1370

= 9.49%

Load corresponding to 5 mm penetration =170 kg Therefore,

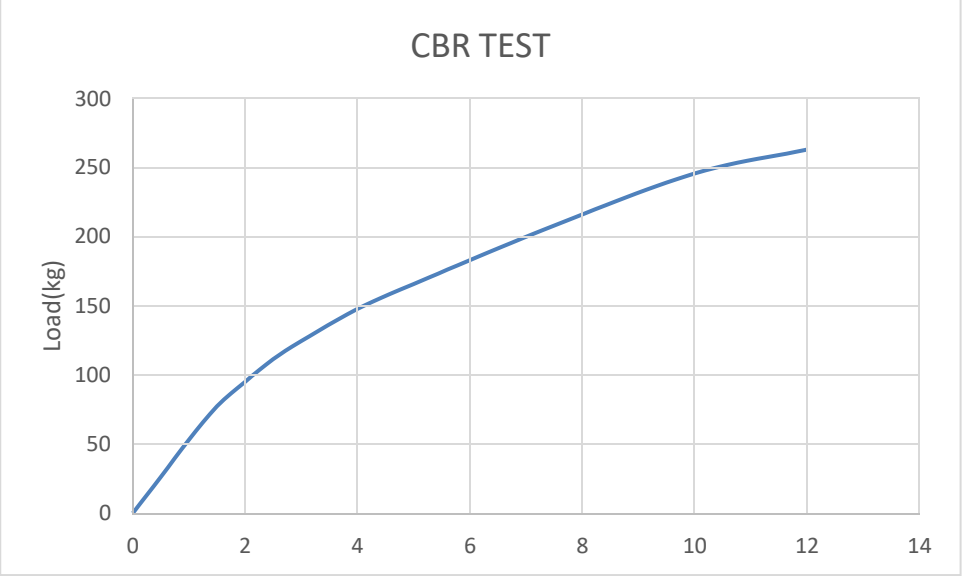
CBR5 mm = (170×100) / 2055

=8.27%

As, CBR5 mm < CBR2.5 mm, the CBR value of the given soil sample = 9.49%

SAMPLE NO. 4

SN	Penetration (mm)	Proving Ring Division	Load (kg)
1	0	0	0
2	0.5	80	26.2
3	1	162	53
4	1.5	236	77.2
5	2	290	94.9
6	2.5	340	111.2
7	3	380	124.3
8	4	451	147.5
9	5	506	165.5
11	7.5	635	207.7
14	10	750	245.3
15	12	803	262.6



From above graph,

Load corresponding to 2.5 mm penetration = 112.5kg Therefore,

CBR2.5 mm = (112.5×100) / 1370

= 8.212%

Load corresponding to 5 mm penetration =162.5kg Therefore,

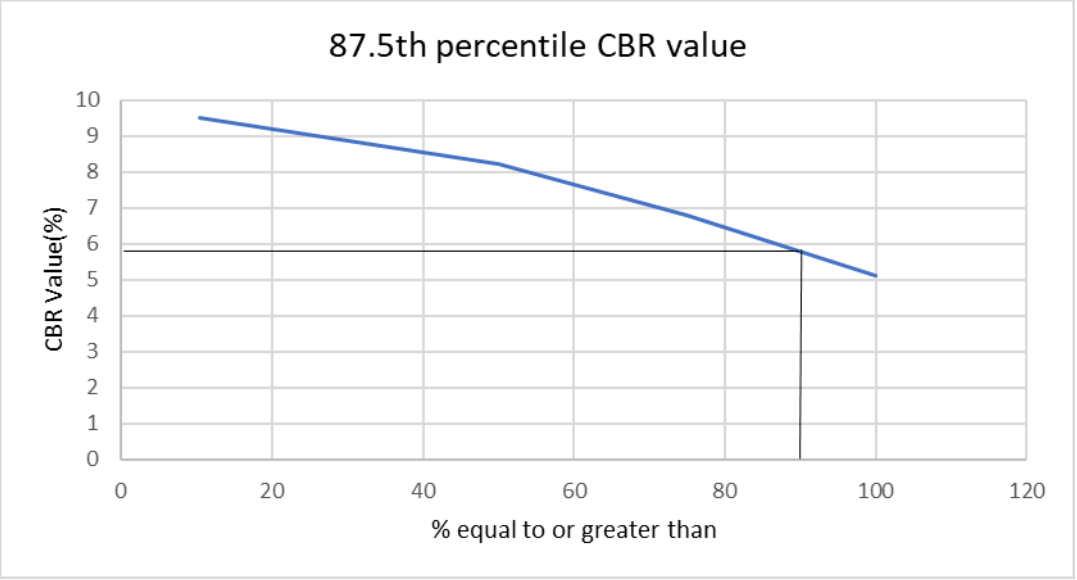
CBR5 mm = (162.5×100) / 2055

=7.9%

As, CBR5 mm<CBR2.5 mm, the CBR value ofthe given soil sample = 8.21%

Calculation of 87.5th percentile of CBR

SN	CBR	CBR in descending order	Rank	%equal to or greater than
1	5.11	9.49	1	25
2	6.79	8.21	2	50
3	9.49	6.79	3	75
4	8.21	5.11	4	100



From graph,

87.5th percentile CBR=5.8%

For conventional design adopting safety considerations, we have adopted the CBR value as 5%.

3.4. Pavement Design

3.4.1. Steps for Pavement Design

a) The number of commercial vehicles expected to use the roadway on the day it is opened was determined.

b) The traffic was projected over the lifetime of roadway.

c) The number of commercial vehicles was converted to cumulative equivalent standard axle load of 8160 kg to be used for design.

d) The strength of subgrade soil over which the road is to be built is determined i.e. the 87.5 percentile CBR value was determined.

e) Flexible Pavement Design Guidelines recommended by DOR, IRC recommendation, Road Note 31 (Catalogue Method) were used to determine the thickness of different pavement layers.

f) And the most economical combination of pavement materials was adopted which will give satisfactory service over the design life.

3.4.2 Calculation of Design Traffic

The local concerned authorities were consulted for the prediction of expected traffic. The road is assumed to be completed at the end of 2026 and comes in operation. And its design life is assumed to be 15 years (from DOR pavement guidelines, 2014, for roads other than national highways, urban road, and expressways). Assuming the growth rate of 7.5% (from DOR pavement guidelines, 2014, 7.5% growth rate of commercial vehicle is assumed in absence of detail traffic study), the expected traffic at the opening of road at the end of construction period (2025) was calculated using the formula below:

$A = P * (1 + r)\, n$ Equation 3-1

Where, A= number of heavy vehicles per day for design (laden weight > 3 tonnes)
P= number of heavy vehicles per day least count r=Growth Rate (7.5%)
n=number of years between the last count and the year of completion of construction.

The VDF was adopted as recommended in "Pavement Design Guidelines (Flexible Pavement)", Planning, Monitoring and Evaluation Unit, DOR. The Cumulative number of traffic for the period of 15 years was calculated using the formula below:

$Ns = 365 * A * VDF * LDF * ((1+r)n - 1)$ Equation 3-2

We take 20% of this traffic from the main road as diverted to calculate the estimated traffic of our designed road. The diverted and generated traffic are taken 20 % each as this road section can act as shortcut to Pathivara temple and other places of Suryabinayek decreasing the travel cost. Also the extensive agriculture farm and animal husbandry lies in the vicinity of the road alignment. So the percentage taken can be justified.

Rate of traffic growth = 7.5 %

Base period=3 years

Design life=15 years

A) Sample Calculation for Tractor

Design Traffic at the end of construction period in terms of standard axles,

$A=P*(1+r)^n$, P=No of heavy veh/day least count

$=14*(1+7.5\%)^3 = 17.39 \sim 18$

Cumulative number of standard axles at the end of design period,

$Ns=365*A*VDF*LDF*((1+r)^r-1)/r$

$=365*18*1*((1+0.075)15-1)/0.075$

= 171597.6 esal

20% of the traffic=34319.5 esal

In a similar way, the cumulative number of standard axles is calculated for different vehicle types. And the design traffic is determined.

Table 3- 1 Vehicle types and their total design traffic

Vehicle Type	Traffic Data in 2023	Traffic Data up to 2026	Data In Whole Number	VDF	CSA (In esal)	Diverted Traffic(20%) (in esal)	Generated Traffic(20%) (in easl)	Total Design Traffic (in easl)
Two-axle Truck	11	13.66	14	4.75	633958.0073	126791.60	126791.60	887541.21
Bus	8	9.94	10	0.5	47666.02	9533.20	9533.20	66732.43
Minibus	6	7.45	8	0.35	26692.97	5338.59	5338.59	37370.16
Tractor	15	18.63	19	1	181130.86	36226.17	36226.17	253583.20
Sum								1245227
								1.245msa

3.3.3. Calculation of Pavement Thickness

For cumulative traffic of 1.245 msa and CBR value of 5%

Using Road Note 31:

Traffic classes (10⁶ esa)	
T1 =	< 0.3
T2 =	0.3 - 0.7
T3 =	0.7 - 1.5
T4 =	1.5 - 3.0
T5 =	3.0 - 6.0
T6 =	6.0 - 10
T7 =	10 - 17
T8 =	17 - 30

Subgrade strength classes (CBR%)	
S1 =	2
S2 =	3 , 4
S3 =	5 - 7
S4 =	8 - 14
S5 =	15 - 29
S6 =	30+

Figure: Traffic classes and Subgrade strength classes according to Road Note 31

Traffic class = T3

Subgrade strength in CBR=S3

CHART 1 GRANULAR ROADBASE / SURFACE DRESSING

	T1	T2	T3	T4	T5	T6	T7	T8
S1	SD 150 / 175 / 300	SD 150 / 225* / 300	SD 200 / 200 / 300	SD 200 / 250* / 300	SD 200 / 300* / 300	SD 225 / 325* / 300		
S2	SD 150 / 150 / 200	SD 150 / 200 / 200	SD 200 / 175 / 200	SD 200 / 225* / 200	SD 200 / 275* / 200	SD 225 / 300* / 200		
S3	SD 150 / 200	SD 150 / 250	SD 200 / 225	SD 200 / 275*	SD 200 / 325*	SD 225 / 350*		
S4	SD 150 / 125	SD 150 / 175	SD 200 / 150	SD 200 / 200	SD 200 / 250	SD 225 / 275		
S5	SD 150 / 100	SD 150 / 100	SD 175 / 100	SD 200 / 125	SD 225 / 150	SD 250 / 175		
S6	SD 150	SD 150	SD 175	SD 200	SD 225	SD 250		

Fig 3-4: Thickness as per Road Note 31

From design Catalogue,

Granular subbase GS =225mm

Granular road base, GB1- GB3 =200mm

Surface Course- Surface Dressing, SD

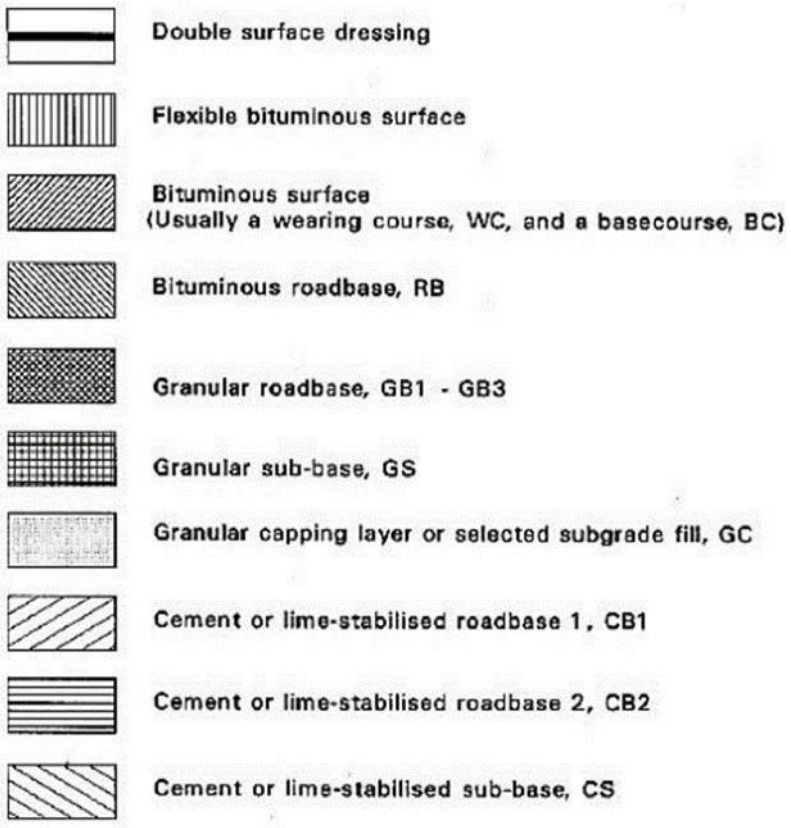


Fig 3-5: Surface Dressing Types as per Road Note 31

3.4.4.DOR Flexible Pavement design guidelines (2014)

This is the guideline provided by DOR for the design of flexible pavements in Nepal. The method is similar to IRC (2001). It provides tables and graphs to design the pavement thickness. It designs pavement for traffic greater than 1 msa.

NRS (2070) mentions the instrument to be used for the CBR test with standard dimensions. The soil samples are tested with a minimum of seven samples. The load penetration curves are drawn. The ratio of load and standard loads are taken at 2.5 mm and 5mm values. The CBR value should be less at a 5mm penetration value. Otherwise, the test is repeated. Finally, if the

CBR value at 5 mm penetration exceeds the penetration at 2.5 mm the CBR is taken to be 5 mm value.

Flexible Pavement Design Guideline (2nd Revision, 2021)

Design Traffic, msa	GBS, mm	Base (WMM), mm	DBM, mm	AC, mm
5	200	250	0	40
10	250	250	0	40
20	250	250	0	60
30	300	300	75	40
40	300	300	90	40
50	300	300	110	40
70	300	300	150	40
80	300	300	150	50
90	300	300	150	60
100	300	300	150	60
110	300	300	150	65
120	300	300	150	75
130	300	300	150	80
140	300	300	150	85
150	300	300	150	90
160	300	300	150	95
170	300	300	150	95
180	300	300	150	100
190	300	300	150	105
200	300	300	150	110
250	300	300	150	120

From Table we obtained the thickness values for 1.245 msa

Granular Sub-base=200mm

Wet mix macadam Base=250mm

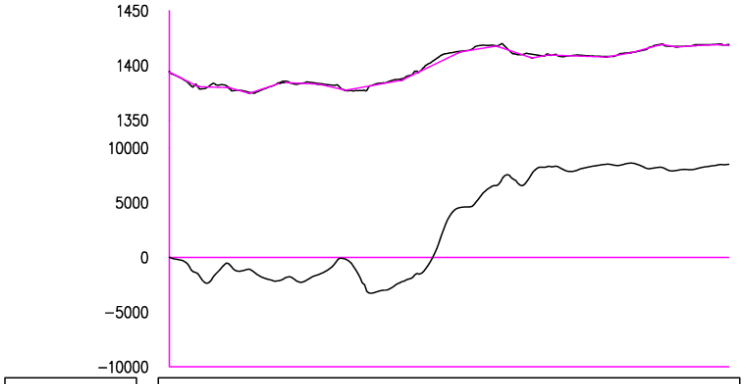
Asphalt concrete=50mm



Fig3.2: Pavement Design

3.5. Mass Haul Diagram:

The Mass Haul Diagram is a graphical representation used in civil engineering and construction projects, particularly in road construction and earthwork operations. It provides a visual depiction of the movement of earth materials (such as soil, rock, or other excavated materials) across a project site during construction. The diagram helps in planning and optimizing the transportation of earth materials, minimizing costs, and ensuring efficient utilization of construction resources.



Chainage	0+000	0+500	1+000	1+500	2+000	2+500 2+551.2
Cut Volume	19.285	57.645	69.751	68.420	14.883	35.730 2.798
Fill Volume	0.231	27.723	7.727	0.190	1.595	1.440 0.010
Volume Difference	19.054	29.922	62.024	68.230	13.288	34.290 2.788

4. Estimating and Costing

4.1. Engineering cost estimate

It's a technique where the system being costed is broken down into lower-level components (such as parts or assemblies), each of which is costed separately for direct labor, direct material, and other costs. And this project cost estimate is prepared only after fixing the alignment, detailed survey, and detailed engineering design. Engineering cost estimate is done according to the prevailing practice of Nepal with standard rate analysis of government following the district rates and other sources of material, labor, and equipment rates. We can also estimate the quantity from standard drawings.

4.2. Quantity Estimation

Quantity estimation was carried out after the detailed engineering design of the road section. Its purpose is to give the client a complete list of all the quantities required for the project.

Abstract of Quantity		
Description	**Quantity**	**Unit**
Drain	1785.835	M3
Pavement	716.8066	M3
Base	3584.033	M3
Sub Base	2867.226	M3
ShoulderLayer-2	956.6974	M3
ShoulderLayer-3	956.6974	M3
Shoulder	6122.863	L
Cut	17924.79	M3
Fill	11257.82	M3
Drain Cut	1577.525	M3
Structure Cut	954.9308	M3
Back Fill	272.2653	M3

Gabion Retaining Wall	770	M3
Masonry Breast Wall	480.9264	M3
Pipe Culvert	5	Num

4.3. Construction Cost

The total construction cost is calculated in the basis of rate analysis and detailed quantity estimate. The total project cost of the Ganesh temple to Gundu Pathivara Road is NRs. 171,713,381.81including VAT and contingencies and cost per km is Rs. 57,485,587.9

Table 4-1 Total Project Cost

Item No	Description	Amount	%
1	PIPE DRAIN, PIPE CULVERTS & CONCRETE CHANNELS	4,15,386.52	0.33
2	EARTH WORK	36,79,851.67	2.94
3	SUB GRADE	53,16,988.12	4.24
4	SUB BASE, BASE, HARD SHOULDER & GRAVEL WEARING	4,47,84,364.57	35.73
5	BITUMINOUS SURFACE AND BASE COURSE	2,64,71,030.12	21.12
6	CONCRETE FOR STRUCTURES	66,34,397.86	5.29
7	RIVER TRAINING & PROTECTION WORKS	72,30,013.96	5.77
8	MASONRY FOR STRUCTURES	3,08,06,202.08	24.58
A	**Total Estimated Cost**	**12,53,38,234.89**	**100.00**
B	**Work Charge Establishment Cost @ 2% of (A)**	**25,06,764.70**	
C	**Miscellaneous Minor Cost @ 2% of (A)**	**25,06,764.70**	
D	**VAT @ 13% (A)**	**1,62,93,970.54**	
E	**Total Estimated Cost with Contingencies (4%) and VAT (A+B+C+D)**	**14,66,45,734.83**	
F	**Physical Contingencies @ 10% of (A)**	**1,25,33,823.49**	
G	**Price Contingencies @ 10% of (A)**	**1,25,33,823.49**	
H	**Grand Total Estimated Cost with Contingencies and VAT (E+F+G)**	**17,17,13,381.81**	
I	**Total Length of Road (km)**	**2.55**	
J	**Cost per km with Contingencies (4%) and VAT (Excluding Physical and Price Contingencies) (E/I)**	**5,74,85,587.94**	

5. Conclusion

Detailed Engineering Design, Cost Estimation of Ganesh temple to Gundu Pathivara has been completed. The road segment of 2.55km was designed using SW Road V2 software. The total cost of construction of the road segment has estimated to be Rs 17,17,13,381.81 and Rs 5,74,85,587.94 per km. The cost estimated seems to be very costly due to the fact-

• The Nepal Rural Road Standards Norms were followed for the design of effective, safe, and smooth traffic movement.

• Special care was taken so that alignment did not cross cultivable land and followed the existing trail as closely as possible.

• Pavement design was done using DOR Flexible Pavement design Guidelines (2014).

• Longitudinal and cross-drainage structures such as pipe culverts, drains, etc were designed using relevant formulas, whenever necessary.

•SW Software is used for quantity and cost estimation.

The designed road reduces the existing high grade of the road, which will provide comfort to the passengers while travelling. The designed side drain as well as pipe culvert will drain off the surface water efficiently which will protect the pavement ensuring long life of road. The road connects Ganesh temple to Gundu pathivara; which is most likely to be used by local as well as travelers during the visit to Pathivara temple as well as quite famous Gundu Dummy plane. So, the road is likely to come in handy in future. Regular repair and maintenance of the road is recommended to ensure the prolongation of life of road.

6. List of Annex

- Annex I Design of Horizontal curve
- Annex II: Design of Vertical Curve
- Annex III: Estimation of earthwork
- Annex IV: Drain Quantity Estimate
- Annex V: Retaining Wall Quantity Estimate
- Annex VI: Masonry Breastplate Wall Quantity Estimate
- Annex VII: Summary of Rates
- Annex VIII: Rate Analysis
- Annex IX: Abstract of

Annex-I Calculation of Horizontal Curve

Starting Station (m)	Ending Station (m)	IP Station (m)	Delta angle	Tangent length	Length Of Curve (m)	Radius (m)	Transition length (m)	Extra Widening	Super elevation
0+000	0+000	0+000	0	0	0	0	0	0	0
0+007.321	0+164.528	0+085.924	97.90541646	105.658	157.207	92	0	0.9	0.0192
0+190.24	0+343.16	0+266.7	67.39744405	86.695	152.92	130	0	0.6	0.0136
0+364.144	0+449.722	0+406.933	35.27524886	44.194	85.578	139	0	0.6	0.0127
0+469.627	0+561.264	0+515.445	69.08464954	52.316	91.637	76	0	0.9	0.0233
0+563.698	0+658.527	0+611.113	77.61870613	56.3	94.829	70	0	0.9	0.0253
0+779.017	0+846.26	0+812.638	66.42591661	37.973	67.242	58	0	1.2	0.034
0+891.232	0+944.748	0+917.99	13.10355662	26.875	53.516	234	0	0.6	0.0075
0+961.377	1+051.624	1+006.5	13.05748355	45.32	90.247	396	0	0	0.0047
1+102.072	1+155.815	1+128.944	128.3020645	49.538	53.743	24	16.67	1.5	0.0074
1+166.291	1+335.686	1+250.989	8.839352549	84.866	169.395	1098	0	0	0.00161
1+466.401	1+510.229	1+488.315	41.16636544	22.908	43.828	61	0	0.9	0.029

1+535.903	1+571.348	1+553.626	96.70546712	23.614	35.444	21	19.047	1.5	0.084
1+572.201	1+667.792	1+619.997	78.2425934	56.931	95.591	70	0	0.9	0.0253
1+691.109	1+832.862	1+761.986	16.99133492	71.401	141.753	478	0	0	0.0037
1+887.579	2+113.31	2+000.444	38.72288964	117.367	225.731	334	0	0	0.0053
2+163.172	2+304.458	2+233.815	49.06087516	75.301	141.285	165	0	0.6	0.0107
2+304.875	2+465.714	2+385.295	21.63237639	81.389	160.839	426	0	0	0.00415
2+551.193	2+551.193	2+551.193	0	0	0	0	0	0	0

Annex-II Calculation of Vertical Curve

VERTICAL CURVE DATA TABLE									
VIP No.	VIP		Length (m)	BVC		MVC		EVC	
	Chainage (m)	Elevation (m)		Chainage (m)	Elevation (m)	Chainage (m)	Elevation (m)	Chainage (m)	Elevation (m)
1	0+000	1394.532	0	0+000	1394.532	0+000	1394.532	0+000	1394.532
2	0+131.991	1381.371	20	0+121.991	1382.368	0+131.991	1381.834	0+141.991	1381.301
3	0+271.764	1380.384	20	0+261.764	1380.454	0+271.764	1380.146	0+281.764	1379.839
4	0+364.289	1375.343	20	0+354.289	1375.888	0+364.289	1375.961	0+374.289	1376.034
5	0+513.364	1385.648	20	0+503.364	1384.957	0+513.364	1385.218	0+523.364	1385.479
6	0+678.544	1382.865	20	0+668.544	1383.033	0+678.544	1382.6	0+688.544	1382.166
7	0+801.285	1374.287	20	0+791.285	1374.986	0+801.285	1374.703	0+811.285	1374.419
8	0+856.739	1375.022	20	0+846.739	1374.89	0+856.739	1375.306	0+866.739	1375.721
9	1+120.405	1393.45	20	1+110.405	1392.751	1+120.405	1393.6	1+130.405	1394.448
10	1+305.029	1411.888	20	1+295.029	1410.889	1+305.029	1411.658	1+315.029	1412.426
11	1+421.548	1418.157	20	1+411.548	1417.619	1+421.548	1417.914	1+431.548	1418.21

12	1+490.835	1418.521	20	1+480.835	1418.468	1+490.835	1418.159	1+500.835	1417.85
13	1+662.816	1406.99	20	1+652.816	1407.661	1+662.816	1407.493	1+672.816	1407.325
14	1+759.874	1410.237	20	1+749.874	1409.903	1+759.874	1410.031	1+769.874	1410.158
15	1+993.024	1408.393	20	1+983.024	1408.472	1+993.024	1408.577	2+003.024	1408.681
16	2+108.077	1411.7	20	2+098.077	1411.413	2+108.077	1411.833	2+118.077	1412.254
17	2+252.133	1419.677	20	2+242.133	1419.124	2+252.133	1419.226	2+262.133	1419.328
18	2+313.876	1417.52	20	2+303.876	1417.87	2+313.876	1417.772	2+323.876	1417.674
19	2+403.021	1418.89	20	2+393.021	1418.736	2+403.021	1418.852	2+413.021	1418.968
20	2+516.417	1419.775	20	2+506.417	1419.697	2+516.417	1419.654	2+526.417	1419.611
21	2+551.193	1419.207	0	2+551.193	1419.207	2+551.193	1419.207	2+551.193	1419.207

Annex-III Estimate of Earthwork

Chainage (m)	Mean Distance (m)	AREA (m2)					VOLUME (m3)				
		Cut	Fill	Structure Cut	Drain Cut	BackFill	Cut	Fill	Structure Cut	Drain Cut	BackFill
0+000	5	2.986	0.046	0	0.871	0	14.931	0.231	0	4.354	0
0+010	10	0	8.183	0	0	0	0	81.825	0	0	0
0+020	10	0	5.985	0	0	0	0	59.851	0	0	0
0+030	10	0	3.953	0	0.134	0	0	39.526	0	1.337	0
0+040	10	0	3.533	0	0.119	0	0	35.335	0	1.192	0
0+050	10	0	4.037	0	0.03	0	0	40.367	0	0.302	0
0+060	10	0	5.637	0	0	0	0	56.371	0	0	0
0+070	10	0	9.16	0	0	0	0	91.601	0	0	0
0+080	10	0	15.974	0	0	0	0	159.738	0	0	0
0+090	10	0	26.391	0	0	0	0	263.909	0	0	0
0+100	10	0.354	28.282	2.486	0	0.049	3.544	282.821	24.864	0	0.487
0+110	10	0.217	18.371	3.137	0.2	0.118	2.166	183.709	31.368	2	1.18
0+120	10	3.771	11.991	0	0.385	0	37.706	119.907	0	3.853	0
0+130	10	0	9.677	0	0.1	0	0	96.767	0	0.998	0
0+140	10	0	29.187	0	0	0	0	291.872	0	0	0
0+150	10	0	31.764	0	0	0	0	317.641	0	0	0
0+160	10	0	19.943	0	0	0	0	199.43	0	0	0
0+170	10	0	6.479	0	0	0	0	64.786	0	0	0
0+180	10	4.665	0.06	0	0.915	0	46.648	0.599	0	9.151	0

0+190	10	18.949	0	0	1.105	0	189.489	0	0	11.05	0
0+200	10	31.116	0	0	1.105	0	311.161	0	0	11.05	0
0+210	10	23.408	0	0	1.105	0	234.077	0	0	11.05	0
0+220	10	18.042	0	0	1.105	0	180.418	0	0	11.05	0
0+230	10	19.223	0	0	1.105	0	192.231	0	0	11.05	0
0+240	10	22.581	0	0	1.105	0	225.808	0	0	11.05	0
0+250	10	21.188	0	0	1.105	0	211.878	0	0	11.05	0
0+260	10	14.447	0.198	0	0.91	0	144.474	1.981	0	9.098	0
0+270	10	0.007	4.484	0	0.301	0	0.074	44.842	0	3.013	0
0+280	10	0	21.203	0	0	0	0	212.031	0	0	0
0+290	10	0	23.58	0	0	0	0	235.804	0	0	0
0+300	10	0	15.852	0	0.045	0	0	158.524	0	0.447	0
0+310	10	0.596	8.886	0	0.553	0	5.955	88.86	0	5.525	0
0+320	10	2.761	4.372	0	0.553	0	27.607	43.721	0	5.525	0
0+330	10	4.874	2.108	0	0.553	0	48.742	21.079	0	5.525	0
0+340	10	6.021	1.309	0	0.553	0	60.208	13.086	0	5.525	0
0+350	10	5.54	0.984	0	0.553	0	55.403	9.844	0	5.525	0
0+360	10	4.847	1.342	0	0.553	0	48.468	13.422	0	5.525	0
0+370	10	1.795	5.814	0	0.553	0	17.946	58.136	0	5.525	0
0+380	10	0	12.249	0	0.123	0	0	122.485	0	1.228	0
0+390	10	0	15.624	0	0	0	0	156.236	0	0	0
0+400	10	0	15.176	0	0	0	0	151.759	0	0	0
0+410	10	0	13.193	0	0	0	0	131.926	0	0	0

0+420	10	0	10.489	0	0	0	0	104.885	0	0	0
0+430	10	0	8.41	0	0	0	0	84.103	0	0	0
0+440	10	0	6.951	0	0.017	0	0	69.514	0	0.165	0
0+450	10	0	6.153	0	0.067	0	0	61.526	0	0.668	0
0+460	10	0	5.576	0	0.081	0	0	55.757	0	0.813	0
0+470	10	0	6.152	0	0.15	0	0	61.518	0	1.495	0
0+480	10	0.021	5.123	0	0.222	0	0.21	51.226	0	2.218	0
0+490	10	2.981	2.256	0	0.553	0	29.813	22.564	0	5.525	0
0+500	10	5.212	2.772	0	0.553	0	52.12	27.723	0	5.525	0
0+510	10	6.432	0.572	0	0.634	0	64.315	5.72	0	6.336	0
0+520	10	8.684	1.19	0	0.657	0	86.837	11.903	0	6.565	0
0+530	10	11.483	0.353	0	0.889	0	114.833	3.526	0	8.891	0
0+540	10	7.404	1.166	0	0.553	0	74.035	11.656	0	5.525	0
0+550	10	2.304	0.932	0	0.819	0	23.042	9.322	0	8.19	0
0+560	10	0	9.296	0	0.208	0	0.002	92.956	0	2.078	0
0+570	10	0	15.618	0	0.002	0	0	156.181	0	0.016	0
0+580	10	0	13.905	0	0	0	0	139.049	0	0	0
0+590	10	0	9.057	0	0	0	0	90.572	0	0	0
0+600	10	0	3.458	0	0.195	0	0	34.575	0	1.947	0
0+610	10	1.889	0.1	0	0.783	0	18.891	0.998	0	7.833	0
0+620	10	7.854	0	0	1.105	0	78.536	0	0	11.05	0
0+630	10	11.85	0	0	1.105	0	118.503	0	0	11.05	0
0+640	10	11.259	0	0	1.105	0	112.593	0	0	11.05	0
0+650	10	9.899	0	0	1.105	0	98.989	0	0	11.05	0
0+660	10	7.939	0	0	1.105	0	79.391	0	0	11.05	0
0+670	10	6.145	0	0	1.105	0	61.45	0	0	11.05	0
0+680	10	5.408	0	0	1.105	0	54.083	0	0	11.05	0

0+690	10	6.221	0	0	1.105	0	62.208	0	0	11.05	0
0+700	10	7.795	0	0	1.105	0	77.949	0	0	11.05	0
0+710	10	9.456	0	0	1.105	0	94.564	0	0	11.05	0
0+720	10	11.205	0	0	1.105	0	112.053	0	0	11.05	0
0+730	10	13.042	0	0	1.105	0	130.418	0	0	11.05	0
0+740	10	14.966	0	0	1.105	0	149.657	0	0	11.05	0
0+750	10	17.6	0	0	1.105	0	176	0	0	11.05	0
0+760	10	24.977	0	2.435	1.084	0.731	249.77	0	24.345	10.839	7.312
0+770	10	24.338	0	0	1.105	0	243.38	0	0	11.05	0
0+780	10	7.495	0.038	0	0.927	0	74.952	0.379	0	9.265	0
0+790	10	0.067	2.257	0	0.303	0	0.673	22.574	0	3.032	0
0+800	10	0.037	3.478	0	0.211	0	0.366	34.781	0	2.109	0
0+810	10	0	9.711	0	0	0	0	97.109	0	0	0
0+820	10	0	13.705	0	0	0	0	137.047	0	0	0
0+830	10	0	18.974	0	0	0	0	189.742	0	0	0
0+840	10	0	30.193	0	0	0	0	301.935	0	0	0
0+850	10	0	28.136	0	0	0	0	281.364	0	0	0
0+860	10	0	35.664	0	0	0	0	356.645	0	0	0
0+870	10	0	36.882	0	0	0	0	368.821	0	0	0
0+880	10	0.924	22.653	8.791	0	0.676	9.241	226.534	87.912	0	6.757
0+890	10	0.647	25.851	7.166	0	0.358	6.469	258.514	71.661	0	3.578
0+900	10	0	59.421	0	0	0	0	594.211	0	0	0
0+910	10	0	14.531	0	0	0	0	145.314	0	0	0
0+920	10	0	3.548	0	0	0	0	35.477	0	0	0
0+930	10	2.66	0.13	0	0.663	0	26.604	1.302	0	6.634	0
0+940	10	5.342	0.066	0	0.889	0	53.419	0.663	0	8.891	0
0+950	10	5.834	0.07	0	1.027	0	58.338	0.697	0	10.273	0

0+960	10	4.874	0.014	0	0.985	0	48.742	0.14	0	9.848	0
0+970	10	3.256	0.198	0	0.729	0	32.556	1.981	0	7.287	0
0+980	10	2.036	0.437	0	0.586	0	20.356	4.369	0	5.864	0
0+990	10	3.397	2.33	0	0.553	0	33.973	23.299	0	5.525	0
1+000	10	6.411	0.773	0	0.564	0	64.11	7.727	0	5.641	0
1+010	10	9.547	0.222	0	0.826	0	95.472	2.222	0	8.262	0
1+020	10	12.383	0.009	0	1.089	0	123.826	0.087	0	10.893	0
1+030	10	13.847	0.019	0	1.077	0	138.474	0.194	0	10.774	0
1+040	10	11.273	0.16	0	0.897	0	112.727	1.602	0	8.969	0
1+050	10	9.449	0.386	0	0.729	0	94.486	3.864	0	7.29	0
1+060	10	8.947	0.473	0	0.687	0	89.473	4.729	0	6.865	0
1+070	10	5.68	0.576	0	0.574	0	56.805	5.757	0	5.74	0
1+080	10	10.343	0.241	0	0.859	0	103.433	2.405	0	8.59	0
1+090	10	9.514	1.232	0	0.553	0	95.135	12.32	0	5.528	0
1+100	10	7.626	4.393	0	0.553	0	76.261	43.929	0	5.525	0
1+110	10	10.277	0.933	0	0.553	0	102.77	9.325	0	5.525	0
1+120	10	25.648	0	0	1.105	0	256.482	0	0	11.05	0
1+130	10	14.109	5.192	0	0.553	0	141.091	51.916	0	5.525	0
1+140	10	0.841	6.879	0	0.553	0	8.409	68.788	0	5.525	0
1+150	10	6.613	0.072	0	0.879	0	66.13	0.721	0	8.793	0
1+160	10	17.217	0	0	1.105	0	172.175	0	0	11.05	0
1+170	10	25.507	0	0	1.105	0	255.073	0	0	11.05	0

1+180	10	28.621	0	0	1.105	0	286.211	0	0	11.05	0
1+190	10	29.601	0	0	1.105	0	296.007	0	0	11.05	0
1+200	10	34.321	0	0	1.105	0	343.211	0	0	11.05	0
1+210	10	39.584	0	0	1.105	0	395.837	0	0	11.05	0
1+220	10	41.633	0	7.8	1.063	2.868	416.325	0	77.997	10.627	28.681
1+230	10	45.512	0	11.48	1.063	4.722	455.12	0	114.804	10.627	47.224
1+240	10	50.754	0	11.48	1.063	4.722	507.54	0	114.804	10.627	47.224
1+250	10	49.298	0	11.48	1.063	4.722	492.98	0	114.804	10.627	47.224
1+260	10	43.872	0	11.48	1.063	4.722	438.719	0	114.804	10.627	47.224
1+270	10	39.093	0	4.869	1.063	1.462	390.932	0	48.691	10.627	14.624
1+280	10	31.871	0	4.869	1.063	1.462	318.707	0	48.691	10.627	14.624
1+290	10	26.171	0	0	1.105	0	261.71	0	0	11.05	0
1+300	10	19.342	0	0	1.105	0	193.422	0	0	11.05	0
1+310	10	12.72	0	0	1.105	0	127.204	0	0	11.05	0
1+320	10	6.472	0	0	1.105	0	64.719	0	0	11.05	0
1+330	10	2.703	0.038	0	0.933	0	27.031	0.375	0	9.333	0
1+340	10	1.552	0.117	0	0.723	0	15.515	1.174	0	7.231	0
1+350	10	0.887	0.201	0	0.556	0	8.868	2.01	0	5.557	0
1+360	10	0.32	0.385	0	0.398	0	3.196	3.852	0	3.977	0
1+370	10	0.145	0.424	0	0.467	0	1.452	4.237	0	4.673	0
1+380	10	4.539	0.03	0	0.969	0	45.394	0.303	0	9.69	0

1+390	10	15.215	0	0	1.105	0	152.154	0	0	11.05	0
1+400	10	23.934	0	0	1.105	0	239.337	0	0	11.05	0
1+410	10	25.204	0	0	1.105	0	252.038	0	0	11.05	0
1+420	10	25.268	0	0	1.105	0	252.677	0	0	11.05	0
1+430	10	23.566	0	0	1.105	0	235.66	0	0	11.05	0
1+440	10	19.396	0	0	1.105	0	193.956	0	0	11.05	0
1+450	10	14.845	0	0	1.105	0	148.452	0	0	11.05	0
1+460	10	12.728	0	0	1.105	0	127.281	0	0	11.05	0
1+470	10	12.098	0	0	1.105	0	120.982	0	0	11.05	0
1+480	10	6.568	0.077	0	0.888	0	65.68	0.771	0	8.876	0
1+490	10	1.183	2.016	0	0.553	0	11.826	20.161	0	5.525	0
1+500	10	5.847	0.019	0	0.995	0	58.469	0.19	0	9.951	0
1+510	10	25.197	0	0	1.105	0	251.966	0	0	11.05	0
1+520	10	35.809	0	0	1.105	0	358.092	0	0	11.05	0
1+530	10	19.408	0	0	1.105	0	194.077	0	0	11.05	0
1+540	10	8.468	0	0	1.105	0	84.678	0	0	11.05	0
1+550	10	0	4.333	0	0.055	0	0	43.326	0	0.551	0
1+560	10	0	23.545	0	0	0	0	235.452	0	0	0
1+570	10	0.124	18.427	3.334	0	0.207	1.24	184.266	33.337	0	2.075
1+580	10	0.315	16.06	4.685	0	0.405	3.153	160.596	46.849	0	4.051

1+590	10	0	25.835	0	0	0	0	258.351	0	0	0
1+600	10	0	16.763	0	0.223	0	0.004	167.627	0	2.227	0
1+610	10	2.119	6.24	0	0.553	0	21.186	62.403	0	5.525	0
1+620	10	11.292	0.228	0	0.799	0	112.916	2.275	0	7.987	0
1+630	10	23.297	0	0	1.105	0	232.966	0	0	11.05	0
1+640	10	27.521	0	0	1.105	0	275.212	0	0	11.05	0
1+650	10	31.161	0	0	1.105	0	311.607	0	0	11.05	0
1+660	10	28.957	0	0	1.105	0	289.57	0	0	11.05	0
1+670	10	19.884	0	0	1.105	0	198.842	0	0	11.05	0
1+680	10	12.379	0	0	1.105	0	123.795	0	0	11.05	0
1+690	10	6.143	0.031	0	0.992	0	61.433	0.309	0	9.92	0
1+700	10	0.945	0.916	0	0.553	0	9.452	9.165	0	5.525	0
1+710	10	0.059	3.024	0	0.352	0	0.593	30.238	0	3.52	0
1+720	10	4.981	0.63	0	0.6	0	49.805	6.297	0	5.996	0
1+730	10	4.245	0.019	0	1.005	0	42.451	0.186	0	10.05	0
1+740	10	0	3.884	0	0.18	0	0	38.839	0	1.798	0
1+750	10	0.843	0.522	0	0.312	0	8.427	5.223	0	3.12	0
1+760	10	3.879	0.584	0	0.621	0	38.792	5.835	0	6.206	0
1+770	10	1.566	3.889	0	0.553	0	15.663	38.889	0	5.525	0
1+780	10	0	11.667	0	0	0	0	116.667	0	0	0
1+790	10	0	11.375	0	0	0	0	113.755	0	0	0
1+800	10	0	10.258	0	0	0	0	102.583	0	0	0
1+810	10	0	7.589	0	0	0	0	75.895	0	0	0
1+820	10	0	4.412	0	0.118	0	0	44.117	0	1.182	0
1+830	10	0.265	1.807	0	0.368	0	2.646	18.067	0	3.677	0

1+840	10	1.349	0.376	0	0.598	0	13.495	3.762	0	5.983	0
1+850	10	3.474	0.109	0	0.834	0	34.744	1.088	0	8.342	0
1+860	10	5.532	0.005	0	1.07	0	55.317	0.052	0	10.701	0
1+870	10	5.816	0	0	1.105	0	58.16	0	0	11.05	0
1+880	10	5.096	0	0	1.105	0	50.956	0	0	11.05	0
1+890	10	4.387	0	0	1.105	0	43.869	0	0	11.05	0
1+900	10	3.804	0.005	0	1.053	0	38.039	0.047	0	10.531	0
1+910	10	3.321	0.016	0	0.999	0	33.214	0.158	0	9.986	0
1+920	10	2.927	0.03	0	0.954	0	29.268	0.299	0	9.537	0
1+930	10	2.62	0.044	0	0.919	0	26.198	0.441	0	9.185	0
1+940	10	2.4	0.056	0	0.893	0	24.004	0.561	0	8.931	0
1+950	10	2.269	0.064	0	0.877	0	22.686	0.642	0	8.775	0
1+960	10	2.225	0.067	0	0.872	0	22.246	0.674	0	8.717	0
1+970	10	2.269	0.065	0	0.876	0	22.687	0.651	0	8.76	0
1+980	10	2.402	0.058	0	0.89	0	24.017	0.578	0	8.902	0
1+990	10	2.247	0.051	0	0.891	0	22.467	0.515	0	8.91	0
2+000	10	0.894	0.159	0	0.594	0	8.941	1.595	0	5.941	0
2+010	10	0	2.061	0	0.104	0	0	20.613	0	1.035	0
2+020	10	0	5.43	0	0	0	0	54.303	0	0	0
2+030	10	0.642	5.03	0	0.553	0	6.423	50.297	0	5.525	0
2+040	10	1.954	5.404	0	0.553	0	19.542	54.035	0	5.525	0
2+050	10	3.416	3.64	0	0.553	0	34.16	36.399	0	5.525	0
2+060	10	4.887	1.285	0	0.564	0	48.865	12.852	0	5.642	0
2+070	10	4.712	0	0	1.105	0	47.119	0	0	11.05	0
2+080	10	3.97	0.002	0	1.079	0	39.702	0.019	0	10.787	0
2+090	10	3.355	0.022	0	0.979	0	33.546	0.224	0	9.788	0
2+100	10	2.224	0.081	0	0.852	0	22.244	0.807	0	8.515	0
2+110	10	0.48	0.352	0	0.44	0	4.801	3.523	0	4.401	0
2+120	10	0	3.06	0	0.058	0	0	30.601	0	0.577	0
2+130	10	0	5.683	0	0	0	0	56.828	0	0	0

2+140	10	0	7.128	0	0	0	0	71.279	0	0	0
2+150	10	0	8.322	0	0	0	0	83.222	0	0	0
2+160	10	0	9.692	0	0	0	0	96.921	0	0	0
2+170	10	0	10.725	0	0	0	0	107.251	0	0	0
2+180	10	0.318	6.98	0	0.446	0	3.181	69.799	0	4.464	0
2+190	10	0.993	2.503	0	0.381	0	9.935	25.035	0	3.81	0
2+200	10	2.384	0.237	0	0.718	0	23.84	2.371	0	7.182	0
2+210	10	3.1	0.178	0	0.695	0	31.001	1.783	0	6.953	0
2+220	10	3.021	0.213	0	0.678	0	30.214	2.127	0	6.778	0
2+230	10	2.016	0.202	0	0.805	0	20.165	2.02	0	8.048	0
2+240	10	1.218	0.205	0	0.573	0	12.179	2.052	0	5.732	0
2+250	10	1.403	5.139	0	0.389	0	14.027	51.389	0	3.89	0
2+260	10	0.822	10.425	0	0.553	0	8.222	104.255	0	5.525	0
2+270	10	0	10.91	0	0	0	0	109.103	0	0	0
2+280	10	0	7.528	0	0	0	0	75.279	0	0	0
2+290	10	0	3.442	0	0.063	0	0	34.418	0	0.632	0
2+300	10	0.265	0.258	0	0.498	0	2.649	2.577	0	4.977	0
2+310	10	2.296	0.007	0	0.988	0	22.961	0.067	0	9.88	0
2+320	10	2.914	0.001	0	1.059	0	29.138	0.012	0	10.588	0
2+330	10	2.152	0.028	0	0.878	0	21.52	0.283	0	8.784	0
2+340	10	1.304	0.098	0	0.673	0	13.044	0.978	0	6.731	0
2+350	10	0.507	0.212	0	0.465	0	5.067	2.118	0	4.648	0
2+360	10	0.001	0.623	0	0.26	0	0.005	6.229	0	2.597	0
2+370	10	0	1.554	0	0.067	0	0	15.541	0	0.674	0
2+380	10	0.051	1.792	0	0.268	0	0.51	17.92	0	2.684	0
2+390	10	2.64	0.597	0	0.555	0	26.403	5.975	0	5.553	0
2+400	10	4.313	0.008	0	1.049	0	43.126	0.085	0	10.49	0
2+410	10	4.261	0	0	1.105	0	42.608	0	0	11.05	0
2+420	10	3.637	0	0	1.105	0	36.367	0	0	11.047	0
2+430	10	3.137	0.004	0	1.049	0	31.366	0.039	0	10.487	0

2+440	10	2.692	0.014	0	0.993	0	26.925	0.144	0	9.927	0
2+450	10	2.303	0.031	0	0.939	0	23.032	0.306	0	9.392	0
2+460	10	2	0.052	0	0.866	0	19.996	0.517	0	8.662	0
2+470	10	1.874	0.077	0	0.839	0	18.736	0.772	0	8.39	0
2+480	10	2.056	0.108	0	0.799	0	20.561	1.076	0	7.988	0
2+490	10	2.392	0.126	0	0.782	0	23.922	1.262	0	7.819	0
2+500	10	2.803	0.144	0	0.77	0	28.029	1.44	0	7.702	0
2+510	10	3.119	0.098	0	0.854	0	31.192	0.983	0	8.539	0
2+520	10	1.342	0.91	0	0.467	0	13.424	9.102	0	4.669	0
2+530	10	0	2.076	0	0.073	0	0	20.76	0	0.728	0
2+540	10	0.584	0.24	0	0.465	0	5.839	2.404	0	4.654	0
2+550	5.5965	3.382	0.025	0	0.977	0	18.925	0.139	0	5.465	0
2+551.193	0.5965	3.689	0.017	0	1.002	0	2.201	0.01	0	0.598	0

Annex-IV Drain Quantity Estimate

Structure (Drain) Quantity											
Chainage (m)	**Length (m)**	**Left**					**Right**				
		Drain	**PCC Capping M10 M3**	**RRM (1 in 6 C/S Ratio) M3**	**PCC Base M10 M3**	**Stone Soling M3**	**Drain**	**PCC Capping M10 M3**	**RRM (1 in 6 C/S Ratio) M3**	**PCC Base M10 M3**	**Stone Soling M3**
0+000	5	Drain	0	0.9	0.425	0.425	Drain	0	0.9	0.425	0
0+010	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
0+020	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
0+030	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
0+040	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
0+050	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
0+060	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
0+070	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
0+080	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
0+090	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
0+100	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
0+110	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
0+120	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
0+130	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
0+140	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
0+150	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
0+160	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
0+170	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
0+180	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
0+190	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
0+200	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1

0+210	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
0+220	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
0+230	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
0+240	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
0+250	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
0+260	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
0+270	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
0+280	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
0+290	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
0+300	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
0+310	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
0+320	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
0+330	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
0+340	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
0+350	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
0+360	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
0+370	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
0+380	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
0+390	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
0+400	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
0+410	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
0+420	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
0+430	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
0+440	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
0+450	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
0+460	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
0+470	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
0+480	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
0+490	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
0+500	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1

0+510	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
0+520	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
0+530	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
0+540	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
0+550	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
0+560	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
0+570	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
0+580	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
0+590	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
0+600	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
0+610	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
0+620	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
0+630	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
0+640	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
0+650	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
0+660	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
0+670	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
0+680	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
0+690	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
0+700	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
0+710	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
0+720	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
0+730	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
0+740	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
0+750	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
0+760	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
0+770	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
0+780	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
0+790	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
0+800	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1

0+810	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
0+820	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
0+830	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
0+840	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
0+850	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
0+860	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
0+870	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
0+880	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
0+890	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
0+900	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
0+910	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
0+920	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
0+930	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
0+940	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
0+950	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
0+960	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
0+970	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
0+980	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
0+990	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
1+000	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
1+010	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
1+020	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
1+030	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
1+040	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
1+050	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
1+060	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
1+070	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
1+080	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
1+090	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
1+100	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1

1+110	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
1+120	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
1+130	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
1+140	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
1+150	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
1+160	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
1+170	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
1+180	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
1+190	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
1+200	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
1+210	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
1+220	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
1+230	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
1+240	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
1+250	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
1+260	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
1+270	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
1+280	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
1+290	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
1+300	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
1+310	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
1+320	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
1+330	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
1+340	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
1+350	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
1+360	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
1+370	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
1+380	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
1+390	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
1+400	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1

1+410	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
1+420	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
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1+490	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
1+500	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
1+510	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
1+520	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
1+530	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
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1+570	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
1+580	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
1+590	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
1+600	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
1+610	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
1+620	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
1+630	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
1+640	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
1+650	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
1+660	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
1+670	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
1+680	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
1+690	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
1+700	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1

1+710	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
1+720	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
1+730	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
1+740	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
1+750	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
1+760	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
1+770	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
1+780	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
1+790	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
1+800	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
1+810	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
1+820	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
1+830	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
1+840	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
1+850	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
1+860	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
1+870	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
1+880	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
1+890	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
1+900	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
1+910	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
1+920	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
1+930	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
1+940	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
1+950	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
1+960	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
1+970	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
1+980	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
1+990	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
2+000	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1

2+010	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
2+020	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
2+030	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
2+040	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
2+050	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
2+060	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
2+070	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
2+080	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
2+090	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
2+100	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
2+110	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
2+120	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
2+130	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
2+140	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
2+150	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
2+160	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
2+170	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
2+180	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
2+190	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
2+200	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
2+210	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
2+220	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
2+230	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
2+240	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
2+250	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
2+260	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
2+270	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
2+280	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
2+290	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
2+300	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1

2+310	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
2+320	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
2+330	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
2+340	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
2+350	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
2+360	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
2+370	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
2+380	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
2+390	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
2+400	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
2+410	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
2+420	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
2+430	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
2+440	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
2+450	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
2+460	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
2+470	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
2+480	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
2+490	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
2+500	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
2+510	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
2+520	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
2+530	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
2+540	10	Drain	0	1.8	0.85	0.85	Drain	0	1.8	0.85	1
2+550	5.5965	Drain	0	1.007	0.476	0.476	Drain	0	1.007	0.476	0
2+551.193	0.5965	Drain	0	0.107	0.051	0.051	Drain	0	0.107	0.051	0

Annex- VI Retaining Wall Quantity Estimate

Structure (Gabion Retaining Wall) Quantity									
Chainage (m)	Length (m)	Left				Right			
		Gabion Retaining Wall	Height M	Gabion Works	Geotextile M2	Gabion Retaining Wall	Height M	Gabion Works M3	Geotextile M2
0+100	10					GabionRetainingWallDyn	4	70	52
0+110	10					GabionRetainingWallDyn	5	100	66
0+880	10	GabionRetainingWallDyn	5	100	66	GabionRetainingWallDyn	5	100	66
0+890	10	GabionRetainingWallDyn	5	100	66	GabionRetainingWallDyn	5	100	66
1+570	10	GabionRetainingWallDyn	5	100	66				
1+580	10	GabionRetainingWallDyn	5	100	66				

Annex-VI Masonry Breast Wall Quantity Estimate

Structure (MasonryBreastWall) Quantity													
Chainage (m)	**Length (m)**	**Left**						**Right**					
		MasonryBreastWall	**Height M**	**PCC Capping M10 M3**	**RRM (1 in 6 C/S Ratio) M3**	**PCC Base M10 M3**	**Stone Soling M3**	**MasonryBreastWall**	**Height M**	**PCC Capping M10 M3**	**RRM (1 in 6 C/S Ratio) M3**	**PCC Base M10 M3**	**Stone Soling M3**
0+760	10	MasonryBreastWallA	2	0.25	17.358	1.206	1.809						
1+220	10	MasonryBreastWallA	2.5	0.25	25.467	1.507	2.261	MasonryBreastWallA	2.5	0.25	25.467	1.507	2.261
1+230	10	MasonryBreastWallA	3	0.25	35.082	1.809	2.713	MasonryBreastWallA	3	0.25	35.082	1.809	2.713
1+240	10	MasonryBreastWallA	3	0.25	35.082	1.809	2.713	MasonryBreastWallA	3	0.25	35.082	1.809	2.713
1+250	10	MasonryBreastWallA	3	0.25	35.082	1.809	2.713	MasonryBreastWallA	3	0.25	35.082	1.809	2.713
1+260	10	MasonryBreastWallA	3	0.25	35.082	1.809	2.713	MasonryBreastWallA	3	0.25	35.082	1.809	2.713
1+270	10	MasonryBreastWallA	2	0.25	17.358	1.206	1.809	MasonryBreastWallA	2	0.25	17.358	1.206	1.809
1+280	10	MasonryBreastWallA	2	0.25	17.358	1.206	1.809	MasonryBreastWallA	2	0.25	17.358	1.206	1.809

Annex-VII Summary of Rates

S.No	Item Description	Unit	Quantity	Rate	Amount
A	**Labour**				
A.1	Unskilled	md	13,454.42	900.00	1,21,08,978.71
A.2	Skilled	md	3,903.97	1,200.00	46,84,760.00
	Total Labour Cost				**1,67,93,738.71**
B	**Material**				
B.1	Aggregate 5-10mm	cum	244.09	4,820.19	11,76,558.55
B.2	Aggregate10-20mm	cum	371.39	5,081.74	18,87,300.13
B.3	Aggregate 20-40mm	cum	437.36	4,969.65	21,73,522.94
B.4	Aggregates	cum	5,494.28	4,521.29	2,48,41,261.88
B.5	Binding wire	kg	533.35	106.48	56,791.99
B.6	Bitumen	t	109.11	1,15,790.15	1,26,34,204.02
B.7	Bitumen MC30	t	19.98	1,19,000.79	23,77,542.01
B.8	Boulders/stones	cum	896.50	4,059.80	36,39,610.70
B.9	Cement	t	286.67	14,481.66	41,51,505.29
B.10	Coarse sand	cum	218.68	4,617.19	10,09,685.12
B.11	Geotextile	sqm	616.80	0.73	450.24
B.12	Mesh wire/10SWG	kg	9,936.75	102.62	10,19,708.39
B.13	RCC Pipe 900mm NP3	m	50	6,384.06	3,19,202.88
B.14	Sand	cum	720.47	4,544.58	32,74,246.93
B.15	Selvedge wire/8SWG	kg	1,206.13	102.62	1,23,772.44
B.16	Water	KL	5,466.64	377.63	20,64,366.40
B.17	Stone	cum	2,678.46	4,770.94	1,27,78,773.25
B.18	Sub base Material S1 type or S2 type	cum	4,894.62	2,017.62	98,75,488.69
B.19	Stone Screening 13.2 mm	cum	726.52	3,701.09	26,88,904.06
B.20	Aggregate Below 5.6 mm	cum	424.45	4,521.29	19,19,083.35

B.21	Filler	t	21.24	0.01	0.21
	Total Material Cost				**8,80,11,979.49**
C	**Equipment**				
C.1	Boiler	hr	232.49	400.00	92,994.14
C.2	Concrete Mixer	hr	194.38	1,000.00	1,94,381.84
C.3	Generator (< 2 KVA)	hr	471.90	150.00	70,785.33
C.4	Pneumatic Tyred Roller	hr	45.03	1,200.00	54,041.96
C.5	Tractor	hr	327.32	300.00	98,196.29
C.6	Vibrator Roller	hr	508.28	400.00	2,03,312.38
C.7	Motor Grader	hr	377.31	1,600.00	6,03,701.94
C.8	Hydraulic Excavator	hr	388.29	1,600.00	6,21,265.14
C.9	Dozer	hr	225.16	2,100.00	4,72,828.44
C.10	Tractor with ripper attachment	hr	27.24	2,200.00	59,937.64
C.11	Smooth Wheel Roller	hr	90.07	500.00	45,034.97
C.12	Paver Finisher	hr	45.03	2,900.00	1,30,601.41
C.13	Bitumen Sprayer	hr	174.36	1,300.00	2,26,673.23
C.14	Batch mix HMP	hr	45.03	750.00	33,776.23
	Total Equipment Cost				**29,07,530.94**
D	**Total Labor Material & Equipment Cost**				**10,77,13,249.14**
E	**Miscellaneous Cost**				**1,76,25,113.42**
	Total Civil Cost				**12,53,38,362.55**

Annex-VIII Rate Analysis

SN	Item Code	Description of Work	Quantity	Unit	Kind	Description	Analysis Quantity	Unit	Rate	Amount	Total	Overhead (15%)	Grand Total	Item Unit Rate
1	07.02(D)	Laying RCC 900mm internal dia. Pipe	12.50	m	Labour	Skilled	1	md	1,200.00	1,200.00	90,301.42	13,545.21	1,03,846.63	8,307.73
						Unskilled	8	md	900.00	7,200.00				
					Material	Sand	0.12	cum	4,544.58	545.35				
						Cement	0.09	t	14,481.66	1,303.35				
						RCC Pipe 900mm NP3	12.50	m	6,384.06	79,800.72				
					Miscellaneous	Add 3% of Labour for Tools & Plant				252.00				
2	09.01(I-B)	Roadway Excavation in all types of soil by mechanical means.	360	cum	Labour	Skilled	1	md	1,200.00	1,200.00	13,500.00	2,025.00	15,525.00	43.13
						Unskilled	3	md	900.00	2,700.00				
					Equipment	Hydraulic Excavator	6	hr	1,600.00	9,600.00				

3	09.04(I-B-i)	Ordinary Soil Depth upto 3m By Mechanical Means	240	cum	Labour	Skilled	1	md	1,200.00	1,200.00	13,500.00	2,025.00	15,525.00	64.69
						Unskilled	3	md	900.00	2,700.00				
					Equipment	Hydraulic Excavator	6	hr	1,600.00	9,600.00				
4	09.09(B)	Construction of Embankment with Material Deposited from Roadway Cutting(With Machine)	300	cum	Labour	Skilled	1	md	1,200.00	1,200.00	61,989.36	9,298.40	71,287.76	237.63
						Unskilled	10	md	900.00	9,000.00				
					Material	Water	72	KL	377.63	27,189.36				
					Equipment	Dozer	6	hr	2,100.00	12,600.00				
						Motor Grader	6	hr	1,600.00	9,600.00				
						Vibrator Roller	6	hr	400.00	2,400.00				
5	10.04(II)	Compacting original ground supporting embankment	600	cum	Labour	Skilled	1	md	1,200.00	1,200.00	31,863.12	4,779.47	36,642.59	61.07

						Unskilled	4	md	900.00	3,600.00				
					Material	Water	24	KL	377.63	9,063.12				
					Equipment	Tractor with ripper attachment	6	hr	2,200.00	13,200.00				
						Vibrator Roller	12	hr	400.00	4,800.00				
6	10.08	Laying of hand pack Stone soling	5	cu m	Labour	Skilled	6	md	1,200.00	7,200.00	46,625.61	6,993.84	53,619.45	10,723.89
						Unskilled	12	md	900.00	10,800.00				
					Material	Stone	6	cu m	4,770.94	28,625.61				
7	12.01(A)	Providing and laying Granular Sub-Base Material By Mechanical means	300	cu m	Labour	Skilled	2	md	1,200.00	2,400.00	8,12,763.42	1,21,914.51	9,34,677.93	3,115.59
						Unskilled	12	md	900.00	10,800.00				
					Material	Sub base Material S1 type or S2 type	384	cu m	2,017.62	7,74,766.08				
						Water	18	KL	377.63	6,797.34				
					Equipment	Motor Grader	6	hr	1,600.00	9,600.00				
						Vibrator Roller	12	hr	400.00	4,800.00				
						Tractor	12	hr	300.00	3,600.00				

8	12.06(B)	Water Bound Macadam by Mechanical Means	360	cum	Labour	Skilled	3	md	1,200.00	3,600.00	22,66,136.52	3,39,920.48	26,06,056.99	7,239.05
						Unskilled	15	md	900.00	13,500.00				
					Material	Aggregates	435.60	cum	4,521.29	19,69,475.01				
						Stone Screening 13.2 mm	57.60	cum	3,701.09	2,13,182.78				
						Water	144	KL	377.63	54,378.72				
					Equipment	Vibrator Roller	6	hr	400.00	2,400.00				
						Motor Grader	6	hr	1,600.00	9,600.00				
9	13.01(B)	Prime Coat, with MC 30 / 70 by Manual Means	1000	lit	Labour	Skilled	3	md	1,200.00	3,600.00	2,42,277.17	36,341.58	2,78,618.74	278.62
						Unskilled	100	md	900.00	90,000.00				
					Material	Bitumen MC30	1.10	t	1,19,000.79	1,30,900.87				
						Water	10	KL	377.63	3,776.30				
					Equipment	Boiler	8	hr	400.00	3,200.00				
						Generator (< 2 KVA)	8	hr	150.00	1,200.00				
						Tractor	6	hr	300.00	1,800.00				
						Bitumen Sprayer	6	hr	1,300.00	7,800.00				

10	13.02(B)	Tack coat with Bitumen on prepared surfaces By Manual Means	1000	lit	Labour	Skilled	3	md	1,200.00	3,600.00	1,80,969.17	27,145.37	2,08,114.54	208.11
						Unskilled	40	md	900.00	36,000.00				
					Material	Bitumen	1.10	t	1,15,790.15	1,27,369.17				
					Equipment	Boiler	8	hr	400.00	3,200.00				
						Generator (< 2 KVA)	8	hr	150.00	1,200.00				
						Tractor	6	hr	300.00	1,800.00				
						Bitumen Sprayer	6	hr	1,300.00	7,800.00				
11	13.06	Bituminous Concrete / Asphalt Concrete	95.50	cum	Labour	Unskilled	15	md	900.00	13,500.00	22,17,700.94	3,32,655.14	25,50,356.08	26,705.30
						Skilled	5	md	1,200.00	6,000.00				
					Material	Bitumen	12.94	t	1,15,790.15	14,98,324.54				
						Aggregate10-20mm	49.48	cum	5,081.74	2,51,444.62				
						Aggregate 5-10mm	32.52	cum	4,820.19	1,56,752.66				
						Aggregate Below 5.6 mm	56.55	cum	4,521.29	2,55,679.09				
						Filler	2.83	t	0.01	0.03				

					Equipment	Paver Finisher	6	hr	2,900.00	17,400.00				
						Generator (< 2 KVA)	6	hr	150.00	900.00				
						Smooth Wheel Roller	12	hr	500.00	6,000.00				
						Pneumatic Tyred Roller	6	hr	1,200.00	7,200.00				
						Batch mix HMP	6	hr	750.00	4,500.00				
12	20.01	PCC M10(1:3:6)	15	cum	Labour	Skilled	2	md	1,200.00	2,400.00	1,78,073.33	26,711.00	2,04,784.33	13,652.29
						Unskilled	22	md	900.00	19,800.00				
					Material	Aggregate 20-40mm	13.50	cum	4,969.65	67,090.31				
						Coarse sand	6.75	cum	4,617.19	31,166.03				
						Cement	3.45	t	14,481.66	49,961.73				
						Water	2	KL	377.63	755.26				
					Equipment	Concrete Mixer	6	hr	1,000.00	6,000.00				
						Generator (< 2 KVA)	6	hr	150.00	900.00				
13	24.01(A-ii)	Gabion Box size 2 X 1 X 1 m (11 sqm)	6	cum	Labour	Skilled	3	md	1,200.00	3,600.00	45,456.04	6,818.41	52,274.44	8,712.41
						Unskilled	7	md	900.00	6,300.00				
					Material	Mesh wire/10SWG	72.45	kg	102.62	7,434.81				
						Selvedge wire/8SWG	8.88	kg	102.62	911.26				

						Binding wire	3.90	kg	106.48	415.28			
						Boulders/stones	6.60	cum	4,059.80	26,794.68			
14	24.01(B-ii)	Gabion Box size 2 X 1 X 1 m (11 sqm)	6	cum	Labour	Skilled	3	md	1,200.00	3,600.00	47,714.52	7,157.18	54,871.70	9,145.28
						Unskilled	8	md	900.00	7,200.00				
					Material	Mesh wire/10SWG	85.20	kg	102.62	8,743.22				
						Selvedge wire/8SWG	8.87	kg	102.62	910.24				
						Binding wire	4.38	kg	106.48	466.39				
						Boulders/stones	6.60	cum	4,059.80	26,794.68				
15	24.05	Laying and fixing of Geo-Textile all complete as per specification.	300	sqm	Labour	Skilled	1	md	1,200.00	1,200.00	3,262.79	489.42	3,752.21	12.51
						Unskilled	2	md	900.00	1,800.00				
					Material	Geotextile	360	sqm	0.73	262.79				
16	24.16	Providing and laying Filter material underneath pitching in slopes	1	cum	Labour	Unskilled	3	md	900.00	2,700.00	9,625.12	1,443.77	11,068.89	11,068.89

						Skilled	1	md	1,200.00	1,200.00				
					Material	Stone	1.20	cum	4,770.94	5,725.12				
17	26.06(A)	Random Rubble Masonry in Cement mortar (1:6)	5	cum	Labour	Skilled	7	md	1,200.00	8,400.00	73,667.97	11,050.20	84,718.17	16,943.63
						Unskilled	20	md	900.00	18,000.00				
					Material	Water	1	KL	377.63	377.63				
						Cement	0.48	t	14,481.66	6,951.20				
						Sand	1.98	cum	4,544.58	8,998.27				
						Stone	5.75	cum	4,770.94	27,432.88				
					Miscellaneous	Add 5% of Labour+Material for scaffolding, protection etc.				3,508.00				

Annex-IX Abstract of Cost

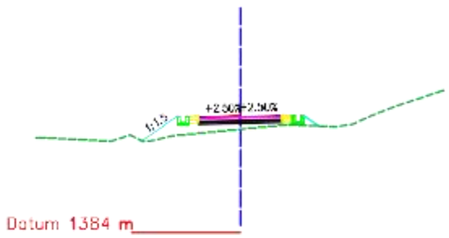
SN	Activity No.	Description of Work	Unit	Quantity	Rate	Amount(NRs)
7		**PIPE DRAIN, PIPE CULVERTS & CONCRETE CHANNELS**				
7.01	07.02(D)	Providing and Laying Reinforced cement concrete 900mm NP3 Flush jointed pipe for culverts including fixing with cement mortar 1:2 as per Drawing and Technical Specifications. [Laying RCC 900mm internal dia. Pipe]	m	50.00	8,307.73	4,15,386.52
Sub Total						**4,15,386.52**
9		**EARTH WORK**				
9.01	09.01(I-B)	Road way Excavation in all types of soil by mechanical means as per Drawing and technical specifications including removal of stumps and other deleterious matter, all lifts and lead as per Drawing and instruction of the Engineer. [Roadway Excavation in all types of soil by mechanical means.]	cum	17,924.80	43.13	7,73,006.78
9.02	09.04(I-B-i)	Earth work in excavation of foundation of structures, including construction of shoring and bracing, removal of stumps and other deleterious matter and backfilling with approved Material as per Drawing and Technical Specifications. [Ordinary Soil Depth upto 3m By Mechanical Means]	cum	3,581.76	64.69	2,31,695.26
9.03	09.09(B)	Providing, laying, spreading and compacting embankment with roadway cutting material and compact to the required density as per Drawing and Technical Specifications. (With machine) [Construction of Embankment with Material Deposited from Roadway Cutting(With Machine)]	cum	11,257.82	237.63	26,75,149.62
Sub Total						**36,79,851.67**
10		**SUB GRADE**				
10.01	10.04(II)	Loosening, leveling and Compacting original ground supporting embankment to facilitate placement of first layer of embankment, scarified to a depth of 150 mm, mixed with water at OMC and then compacted by rolling so as to achieve dry density as per Drawing and Technical Specifications. [Compacting original ground supporting embankment]	cum	2,724.44	61.07	1,66,384.04

10.02	10.08	Providing and laying of hand pack Stone soling with 150 to 200 mm thick stones and packing with smaller stone on prepared surface as per Drawing and Technical Specifications. [Laying of hand pack Stone soling]	cum	480.29	10,723.89	51,50,604.08
Sub Total						**53,16,988.12**
12		**SUB BASE, BASE, HARD SHOULDER & GRAVEL WEARING**				
12.01	12.01(A)	Providing and laying granular sub-base on prepared surface, mixing at OMC, and compacting to achieve the desired density, complete as per Drawing and Technical Specifications. [Providing and laying Granular Sub-Base Material By Mechanical means]	cum	3,823.92	3,115.59	1,19,13,791.27
12.02	12.06(B)	Providing, laying, spreading and compacting Water bound macadam including brooming requisite type of screening/ binding Materials to fill up the interstices of coarse aggregate, watering and compacting to the required density as per Drawing and Technical Specifications. [Water Bound Macadam by Mechanical Means]	cum	4,540.73	7,239.05	3,28,70,573.30
Sub Total						**4,47,84,364.57**
13		**BITUMINOUS SURFACE AND BASE COURSE**				
13.01	13.01(B)	Providing and applying prime coat with Hot Bitumen (including cutter) on prepared surface of granular base including cleaning of road surface and spraying by manual means as per Technical Specification . [Prime Coat, with MC 30 / 70 by Manual Means]	lit	18,162.92	278.62	50,60,530.52
13.02	13.02(B)	Providing and applying tack coat with hot Bitumen at the specified rate the prepared surfaces including cleaning as per Technical Speciation . [Tack coat with Bitumen on prepared surfaces By Manual Means]	lit	10,897.75	208.11	22,67,980.85
13.03	13.06	Providing and laying Bituminous concrete/ Asphalt concrete using crushed aggregates of specified grading, premixed with bituminous binder and filler as per Drawing and Technical Specifications [Bituminous Concrete / Asphalt Concrete]	cum	716.81	26,705.30	1,91,42,518.75
Sub Total						**2,64,71,030.12**
20		**CONCRETE FOR STRUCTURES**				
20.01	20.01	Providing and laying of Plain Cement Concrete M 10 (or 1:3:6 for nominal mix) in Foundation complete as per Drawing and Technical Specifications. [PCC M10(1:3:6)]	cum	485.95	13,652.29	66,34,397.86
Sub Total						**66,34,397.86**

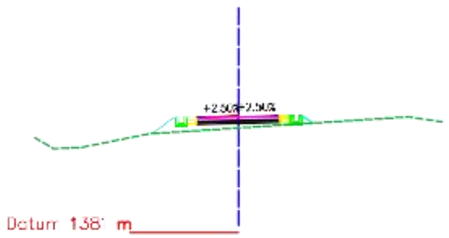
24		**RIVER TRAINING & PROTECTION WORKS**				
24.01	24.01(A-ii)	Providing and laying Gabion structure for retaining earth with diaphragm including rolling, cutting weaving , placing, laying sides and diaphragms with binding wire and filling boulders all complete as per Drawing and Technical Specification. Mesh wire- 10 Swg(0.0615 kg/m), Selvedge Wire 8 Swg (/m), binding wire 12 Swg (0.0409 kg/m) Hexagonal mesh Type 80 mm X 100 mm [Gabion Box size 2 X 1 X 1 m (11 sqm)]	cum	770.00	8,712.41	67,08,553.28
24.02	24.01(B-ii)	Providing and laying Gabion structure for retaining earth with diaphragm including rolling, cutting weaving , placing, laying sides and diaphragms with binding wire and filling boulders all complete as per Drawing and Technical Specification. Mesh wire- 10 Swg(0.0615 kg/m), Selvedge Wire 8 Swg (0.1057 kg/m), binding wire 12 Swg (0.0409 kg/m) Hexagonal mesh Type 100 mm X 120 mm [Gabion Box size 2 X 1 X 1 m (11 sqm)]	cum	45.00	9,145.28	4,11,537.77
24.03	24.05	Providing and laying of a geotextile filter between pitching and embankment slopes as per Drawing and Technical Specifications. [Laying and fixing of Geo-Textile all complete as per specification.]	sqm	514.00	12.51	6,428.78
24.04	24.16	Providing and laying Filter material underneath pitching in slopes complete as per drawing and Technical specification [Providing and laying Filter material underneath pitching in slopes]	cum	9.35	11,068.89	1,03,494.12
Sub Total						**72,30,013.96**
26		**MASONRY FOR STRUCTURES**				
26.01	26.06(A)	Providing and laying Stone Masonry work in cement mortar 1:6 in structure complete as per drawing and Technical Specifications [Random Rubble Masonry in Cement mortar (1:6)]	cum	1,818.16	16,943.63	3,08,06,202.08
Sub Total						**3,08,06,202.08**
Total						**12,53,38,234.89**

Design & Drawing by SW Road V2

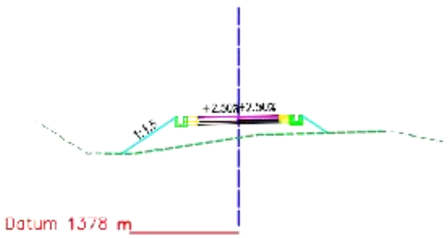
Ch: 0+020
GL:1391.604
DL 1392.535



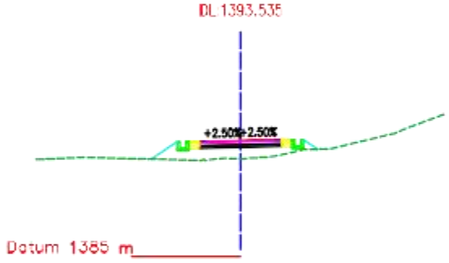
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GL 1388.707
DL 1389.547



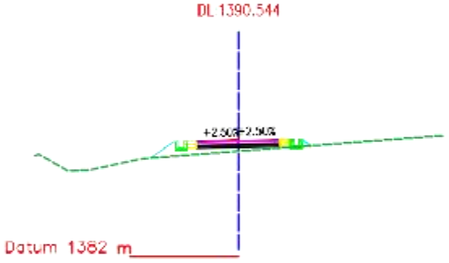
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DL 1386.555



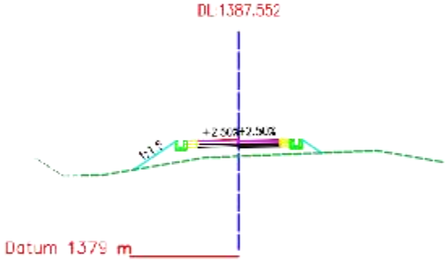
Ch: 0+010
GL 1392.204
DL 1393.535



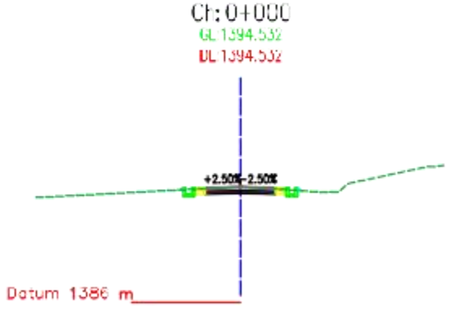
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GL 1389.760
DL 1390.544



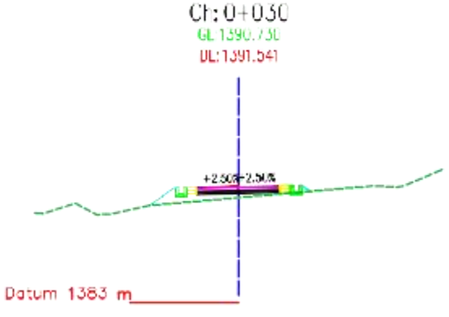
Ch: 0+070
GL:1386.40
DL 1387.552



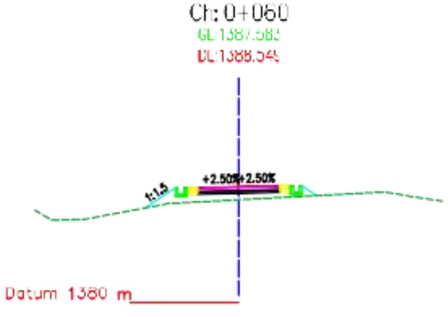
Ch: 0+000
GL 1394.532
DL 1394.532



Ch: 0+030
GL 1390.750
DL:1391.541



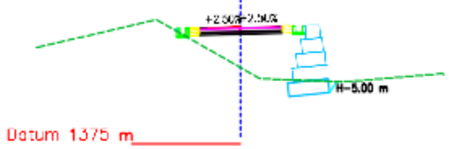
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GL 1387.583
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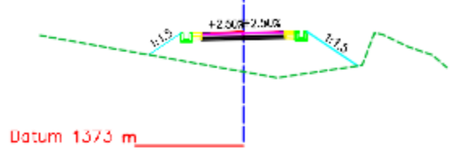
TRIBHUVAN UNIVERSITY INSTITUTE OF ENGINEERING THAPATHALI CAMPUS THAPATHALI, KATHMANDU & DEPARTMENT OF CIVIL ENGINEERING	FINAL YEAR PROJECT GROUP NO. 17	SURYABINAYAK-GUNDU	DATE	REVISION	SIGNATURE	CROSS-0SECTIONS DRAWN BY :GROUP 17 CHECKED BY : APPROVED BY :	SCALE 1:200 4 m 0 8 m	CROSS-SECTIONS 0+000-0+080	DATE: DRG NO : 1 SHEET NO : 1

Design & Drawing by SW Road V2

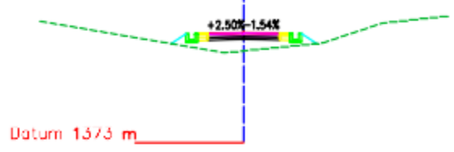
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DL:1383.564



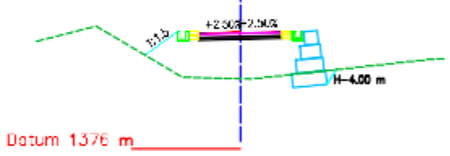
Ch:0+140
GL:1378.447
DL:1381.324



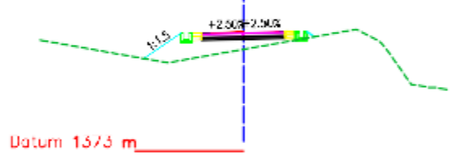
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GL:1379.757
DL:1381.103



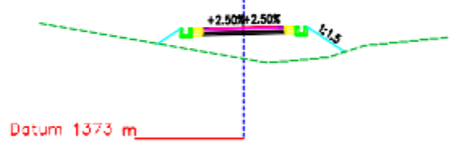
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GL:1381.165
DL:1384.56



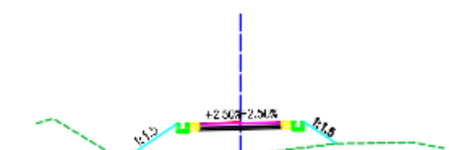
Ch:0+130
GL:1380.488
DL:1381.718



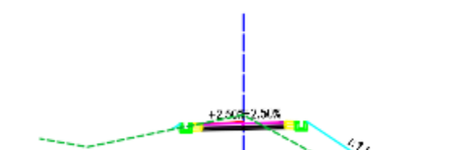
Ch:0+160
GL:1378.885
DL:1381.173



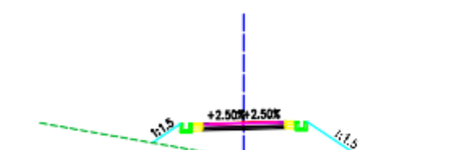
Ch:0+090
GL:1383.100
DL:1385.556



Ch:0+120
GL:1383.115
DL:1382.567



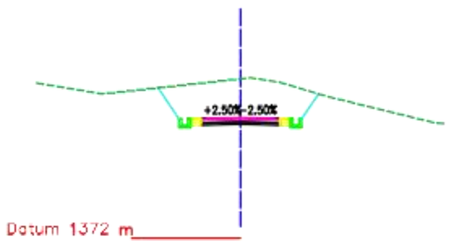
Ch:0+150
GL:1378.965
DL:1381.244



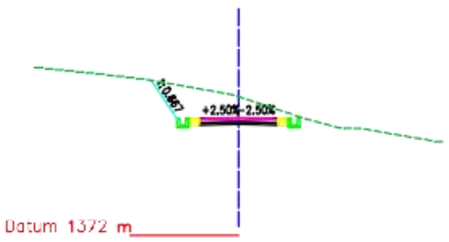
TRIBHUVAN UNIVERSITY INSTITUTE OF ENGINEERING THAPATHALI CAMPUS THAPATHALI, KATHMANDU & DEPARTMENT OF CIVIL ENGINEERING	FINAL YEAR PROJECT GROUP NO. 17	SURYABINAYAK-GUNDU	DATE	REVISION	SIGNATURE	CROSS-0SECTIONS DRAWN BY :GROUP 17 CHECKED BY : APPROVED BY :	SCALE 1:200 4 m 0 8 m	CROSS-SECTIONS 0+090-0+170	DATE: DRG NO : 1 SHEET NO : 2

Design & Drawing by SW Road V2

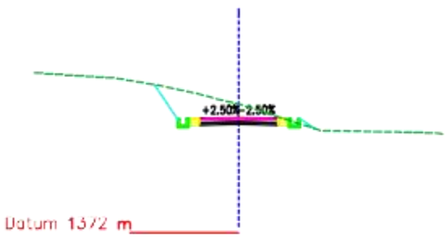
Ch: 0+200
GL:1385.738
DL:1380.89



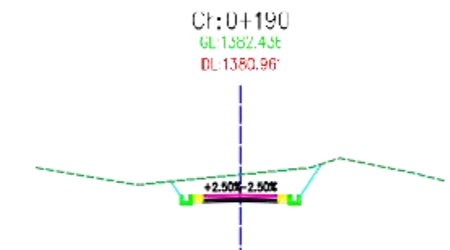
Ch: 0+230
GL:1382.226
DL:1380.679



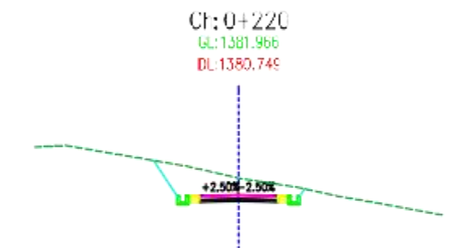
Ch: 0+260
GL:1381.474
DL:1380.467



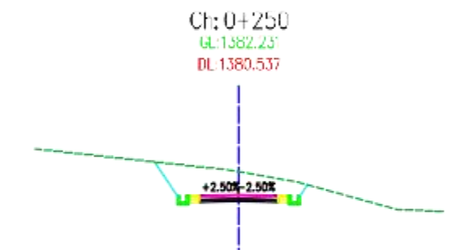
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GL:1382.436
DL:1380.96



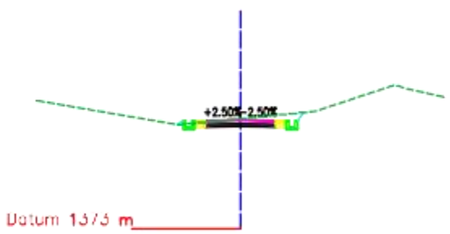
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GL:1381.966
DL:1380.749



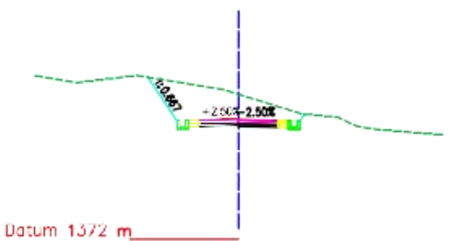
Ch: 0+250
GL:1382.23
DL:1380.537



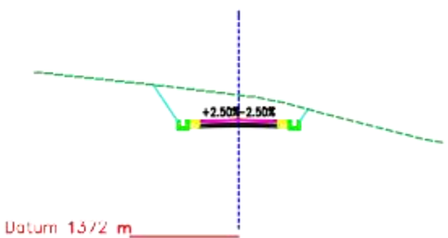
Ch: 0+180
GL:1381.097
DL:1381.032



Ch: 0+210
GL:1382.690
DL:1380.820



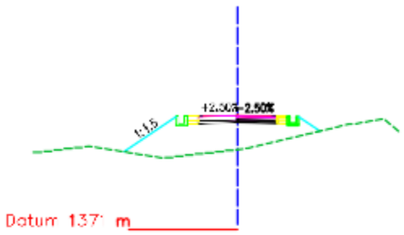
Ch: 0+240
GL:1382.474
DL:1380.608



TRIBHUVAN UNIVERSITY INSTITUTE OF ENGINEERING THAPATHALI CAMPUS THAPATHALI, KATHMANDU & DEPARTMENT OF CIVIL ENGINEERING	FINAL YEAR PROJECT GROUP NO. 17	SURYABINAYAK-GUNDU	DATE	REVISION	SIGNATURE	CROSS-0SECTIONS DRAWN BY :GROUP 17 CHECKED BY : APPROVED BY :	SCALE 1:200 4 m 0 8 m	CROSS-SECTIONS 0+180-0+260	DATE: DRG NO : 1 SHEET NO : 3

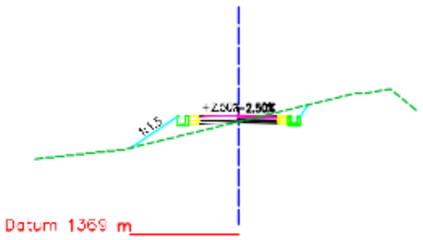
Design & Drawing by SW Road V2

Ch: 0+290
GL 1376.664
DL 1379.390



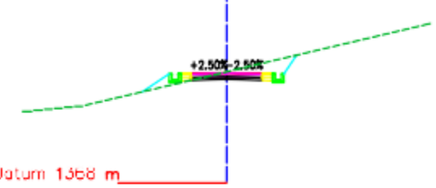
Datum 137[illegible] m

Ch: 0+320
GL 1377.305
DL 1377.756



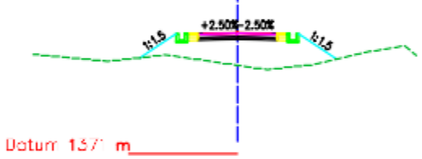
Datum 1369 m

Ch: 0+350
GL:1376.182
DL 1376.121



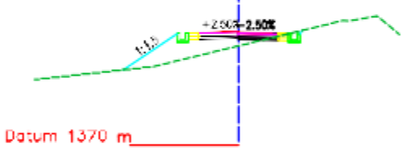
Datum 1368 m

Ch: 0+280
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DL 1379.93[illegible]



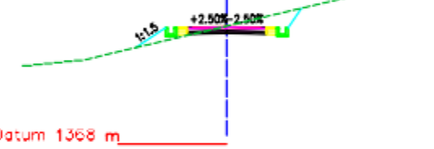
Datum 137[illegible] m

Ch: 0+310
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DL 1378.300



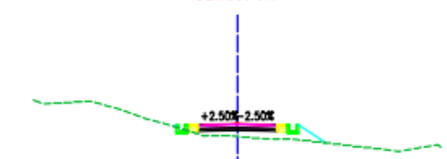
Datum 1370 m

Ch: 0+340
GL:1376.712
DL:1376.66[illegible]



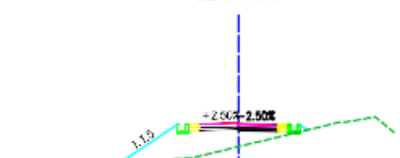
Datum 1368 m

Ch: 0+270
GL 1379.37[illegible]
DL:1380.316



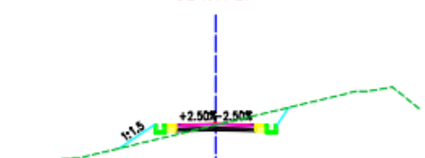
Datum 1372 m

Ch: 0+300
GL 1377.754
DL 1378.845



Datum 1370 m

Ch: 0+330
GL 1377.101
DL 1377.211

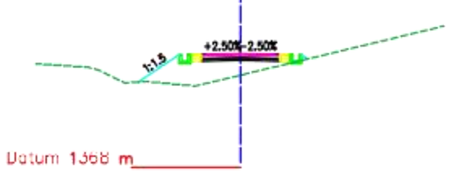


Datum 1369 m

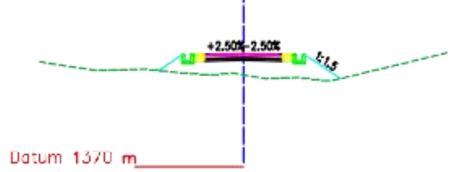
TRIBHUVAN UNIVERSITY INSTITUTE OF ENGINEERING THAPATHALI CAMPUS THAPATHALI, KATHMANDU & DEPARTMENT OF CIVIL ENGINEERING	FINAL YEAR PROJECT GROUP NO. 17	SURYABINAYAK-GUNDU	DATE	REVISION	SIGNATURE	CROSS-0SECTIONS DRAWN BY :GROUP 17 CHECKED BY : APPROVED BY :	SCALE 1:200 4 m 0 8 m	CROSS-SECTIONS 0+270-0+350	DATE: DRG NO : 1 SHEET NO : 4

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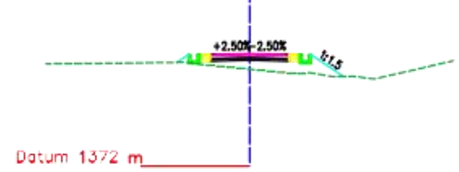
Ch:0+380
GL:1374.780
DL:1376.37



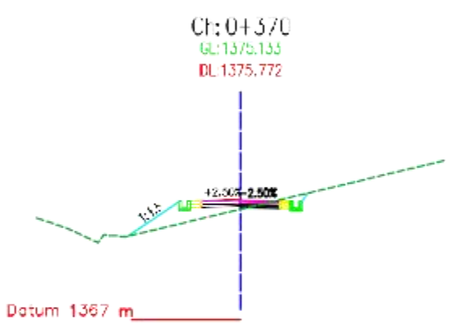
Ch:0+400
GL:1376.757
DL:1378.334



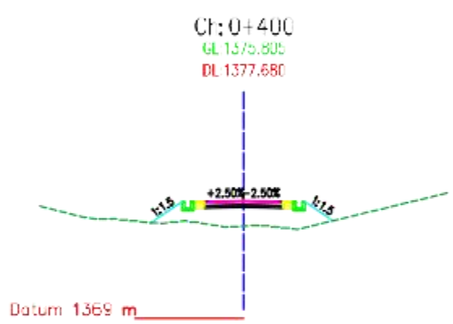
Ch:0+440
GL:1379.149
DL:1380.298



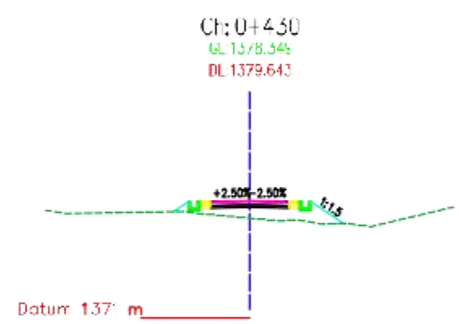
Ch:0+370
GL:1375.133
DL:1375.772



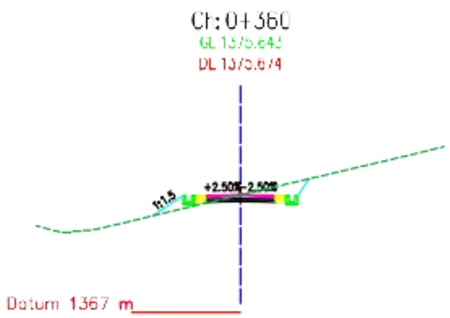
Ch:0+400
GL:1376.808
DL:1377.680



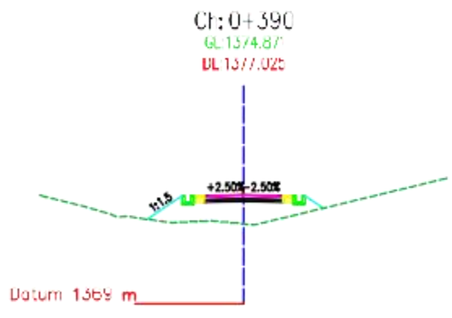
Ch:0+430
GL:1378.349
DL:1379.643



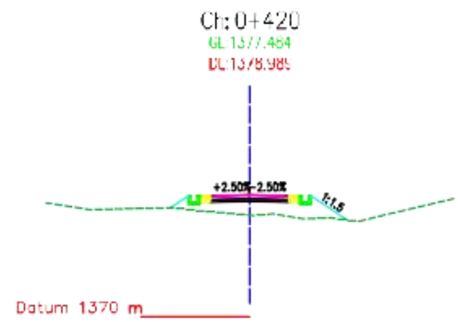
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DL:1376.674



Ch:0+390
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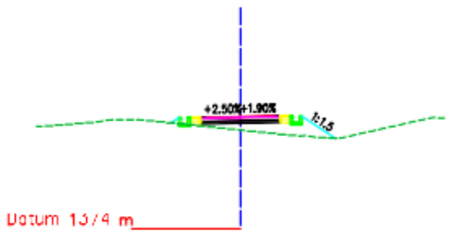
Ch:0+420
GL:1377.464
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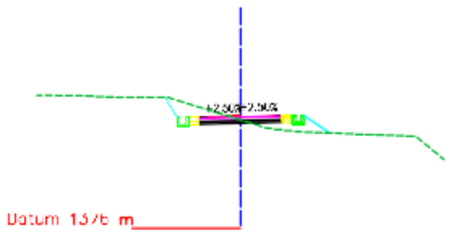
TRIBHUVAN UNIVERSITY INSTITUTE OF ENGINEERING THAPATHALI CAMPUS THAPATHALI, KATHMANDU & DEPARTMENT OF CIVIL ENGINEERING	FINAL YEAR PROJECT GROUP NO. 17	SURYABINAYAK-GUNDU	DATE	REVISION	SIGNATURE	CROSS-0SECTIONS DRAWN BY :GROUP 17 CHECKED BY : APPROVED BY :	SCALE 1:200 4 m 0 8 m	CROSS-SECTIONS 0+360-0+440	DATE: DRG NO : 1 SHEET NO : 5

Design & Drawing by SW Road V2

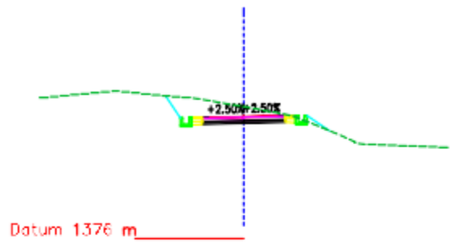
Ch: 0+470
GL:1381.269
DL:1382.26



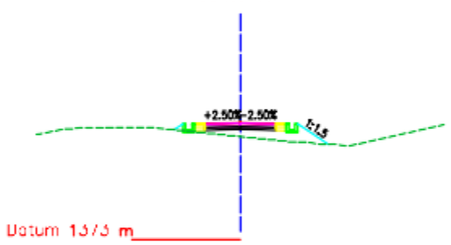
Ch: 0+500
GL 1383.982
DL 1384.225



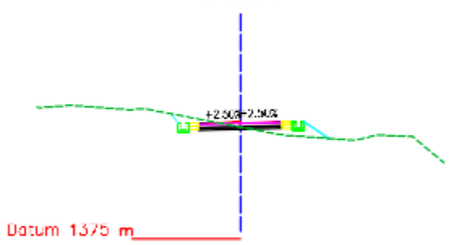
Ch: 0+530
GL:1385.72
DL 1384.987



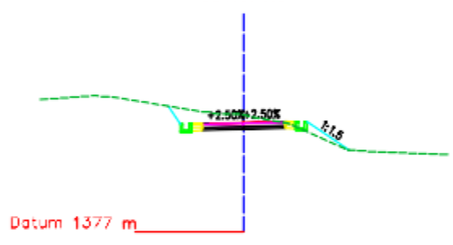
Ch: 0+460
GL 1380.574
DL:1381.607



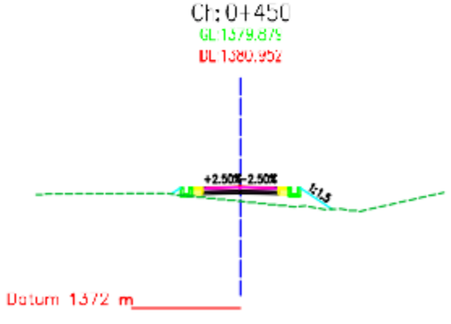
Ch: 0+490
GL 1383.242
DL 1383.570



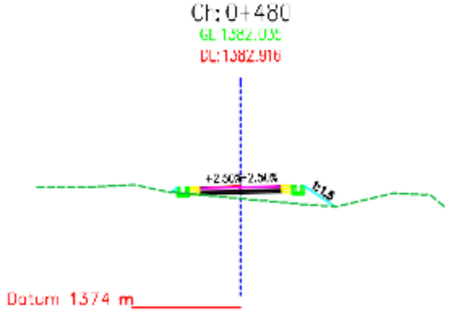
Ch: 0+520
GL:1385.65
DL 1385.052



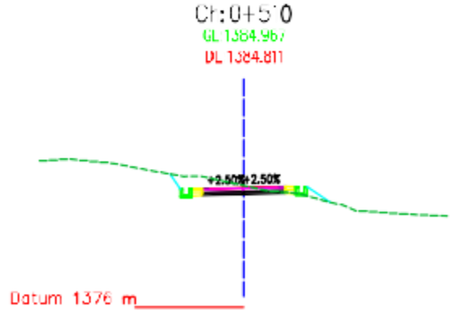
Ch: 0+450
GL:1379.879
DL 1380.952



Ch: 0+480
GL 1382.035
DL:1382.916



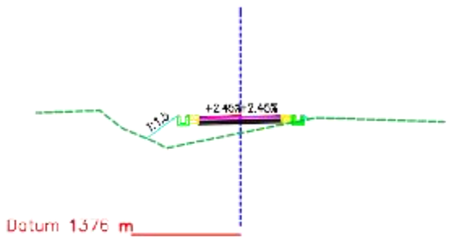
Ch: 0+510
GL 1384.967
DL 1384.811



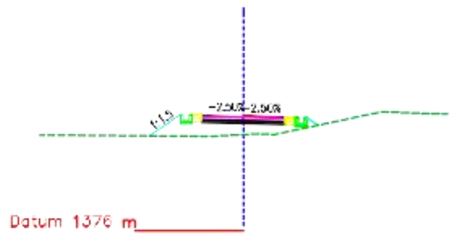
TRIBHUVAN UNIVERSITY INSTITUTE OF ENGINEERING THAPATHALI CAMPUS THAPATHALI, KATHMANDU & DEPARTMENT OF CIVIL ENGINEERING	FINAL YEAR PROJECT GROUP NO. 17	SURYABINAYAK-GUNDU	DATE	REVISION	SIGNATURE	CROSS-0SECTIONS DRAWN BY :GROUP 17 CHECKED BY : APPROVED BY :	SCALE 1:200 4 m 0 8 m	CROSS-SECTIONS 0+450-0+530	DATE: DRG NO : 1 SHEET NO : 6

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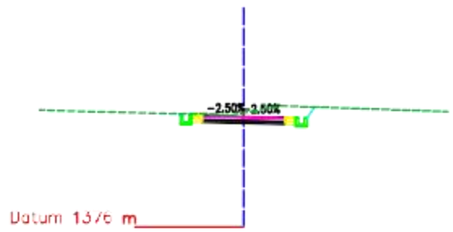
Ch: 0+560
GL:1385.488
DL:1384.700



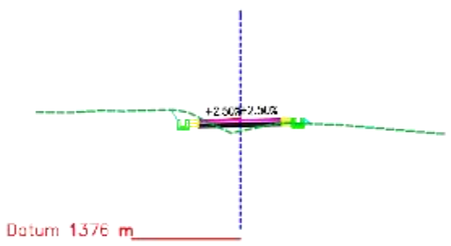
Ch: 0+590
GL:1385.088
DL:1384.414



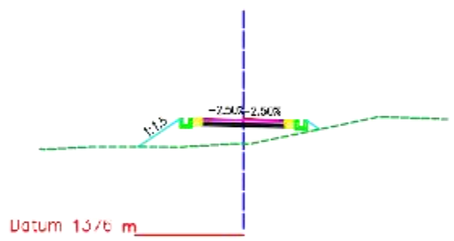
Ch: 0+620
GL:1384.595
DL:1384.128



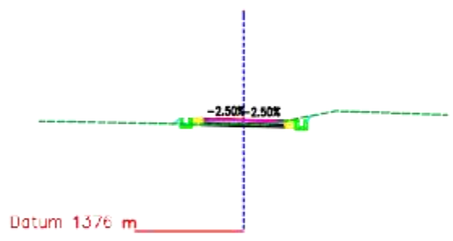
Ch: 0+550
GL:1385.950
DL:1384.798



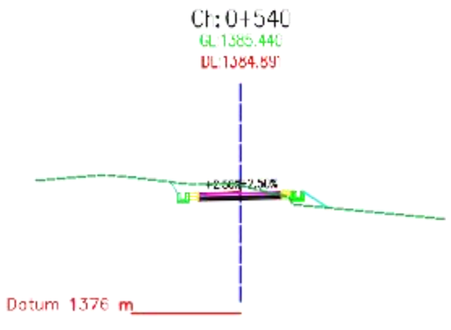
Ch: 0+580
GL:1382.739
DL:1384.509



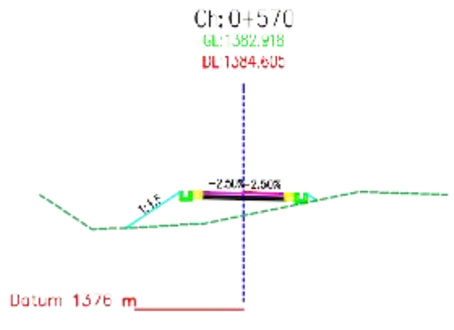
Ch: 0+610
GL:1383.952
DL:1384.223



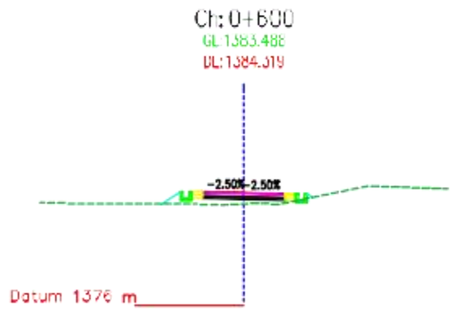
Ch: 0+540
GL:1385.440
DL:1384.697



Ch: 0+570
GL:1382.918
DL:1384.605



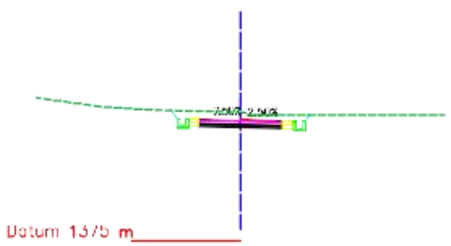
Ch: 0+600
GL:1383.488
DL:1384.319



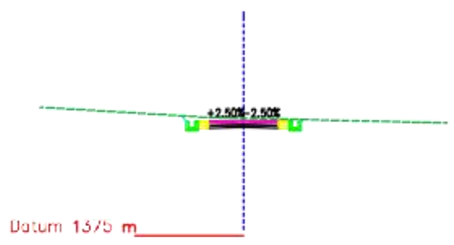
TRIBHUVAN UNIVERSITY INSTITUTE OF ENGINEERING THAPATHALI CAMPUS THAPATHALI, KATHMANDU & DEPARTMENT OF CIVIL ENGINEERING	FINAL YEAR PROJECT GROUP NO. 17	SURYABINAYAK-GUNDU	DATE	REVISION	SIGNATURE	CROSS-0SECTIONS DRAWN BY :GROUP 17 CHECKED BY : APPROVED BY :	SCALE 1:200 4 m 0 8 m	CROSS-SECTIONS 0+540-0+620	DATE: DRG NO : 1 SHEET NO : 7

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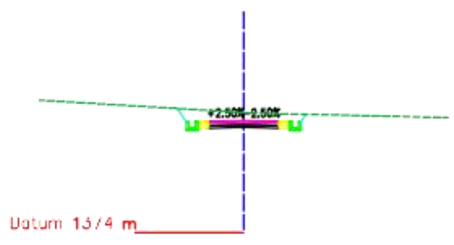
Ch:0+650
GL:1384.456
DL:1383.84



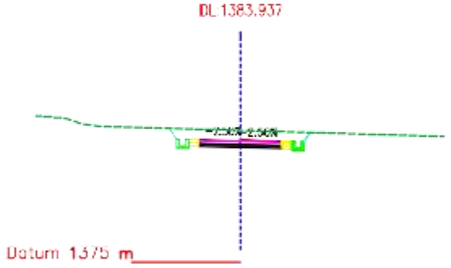
Ch:0+680
GL:1383.637
DL:1383.462



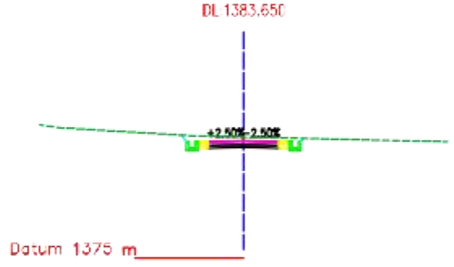
Ch:0+710
GL:1382.830
DL:1382.239



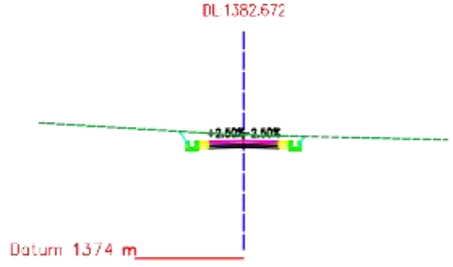
Ch:0+640
GL:1384.682
DL:1383.937



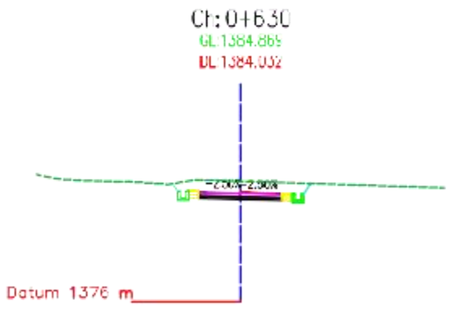
Ch:0+670
GL:1383.914
DL:1383.650



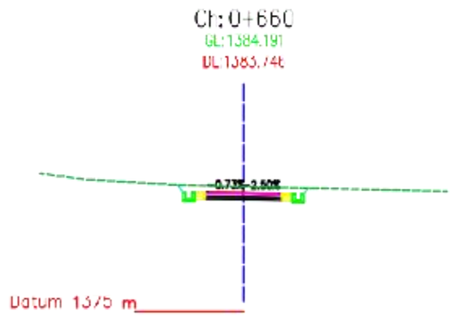
Ch:0+700
GL:1383.083
DL:1382.672



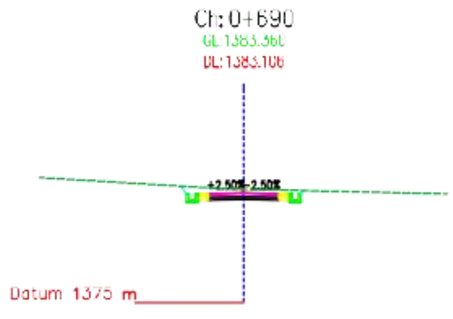
Ch:0+630
GL:1384.865
DL:1384.032



Ch:0+660
GL:1384.191
DL:1383.748



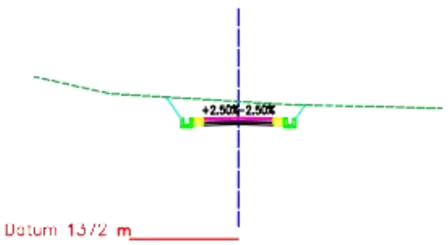
Ch:0+690
GL:1383.361
DL:1383.106



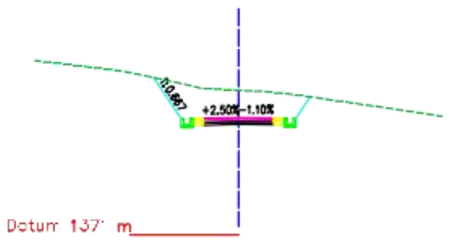
TRIBHUVAN UNIVERSITY INSTITUTE OF ENGINEERING THAPATHALI CAMPUS THAPATHALI, KATHMANDU & DEPARTMENT OF CIVIL ENGINEERING	FINAL YEAR PROJECT GROUP NO. 17	SURYABINAYAK-GUNDU	DATE	REVISION	SIGNATURE	CROSS-0SECTIONS DRAWN BY :GROUP 17 CHECKED BY : APPROVED BY :	SCALE 1:200 4 m 0 8 m	CROSS-SECTIONS 0+630-0+710	DATE: DRG NO : 1 SHEET NO : 8

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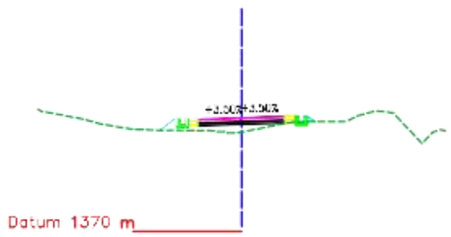
Ch: 0+740
GL:1382.121
DL 1380.935



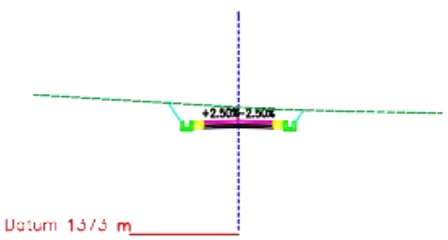
Ch: 0+770
GL:1381.646
DL 1379.638



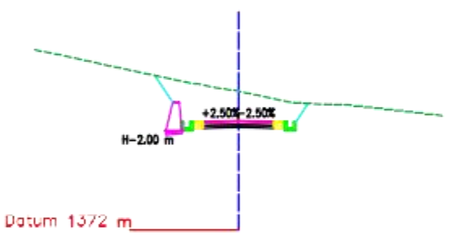
Ch: 0+800
GL:1377.313
DL 1378.377



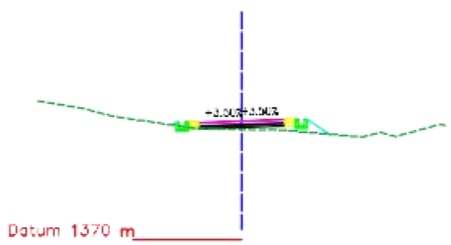
Ch: 0+730
GL:1382.357
DL:1381.372



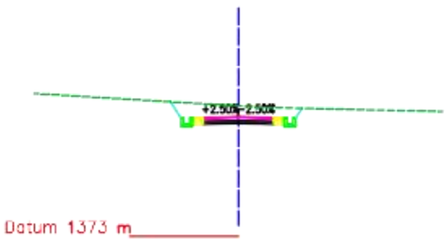
Ch: 0+760
GL:1382.235
DL 1380.072



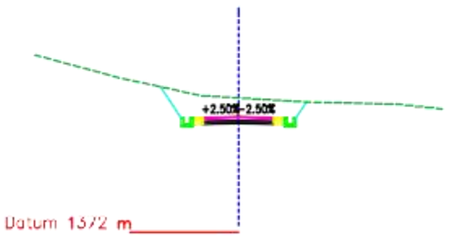
Ch: 0+790
GL:1378.113
DL:1378.77



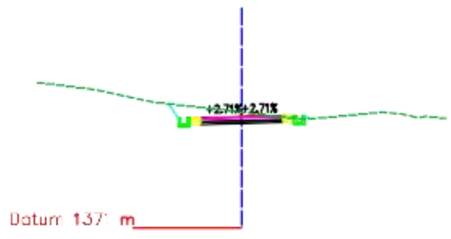
Ch: 0+720
GL:1382.594
DL:1381.805



Ch: 0+750
GL:1381.585
DL 1380.505



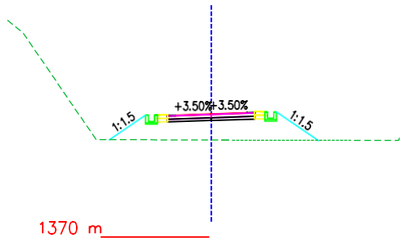
Ch: 0+780
GL:1379.53
DL 1379.20



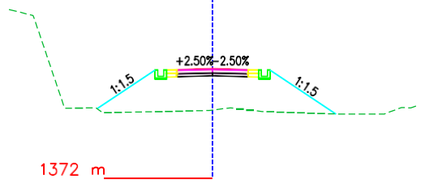
TRIBHUVAN UNIVERSITY INSTITUTE OF ENGINEERING THAPATHALI CAMPUS THAPATHALI, KATHMANDU & DEPARTMENT OF CIVIL ENGINEERING	FINAL YEAR PROJECT GROUP NO. 17	SURYABINAYAK-GUNDU	DATE	REVISION	SIGNATURE	CROSS-0SECTIONS DRAWN BY :GROUP 17 CHECKED BY : APPROVED BY :	SCALE 1:200 4 m 0 8 m	CROSS-SECTIONS 0+720-0+800	DATE: DRG NO : 1 SHEET NO : 9

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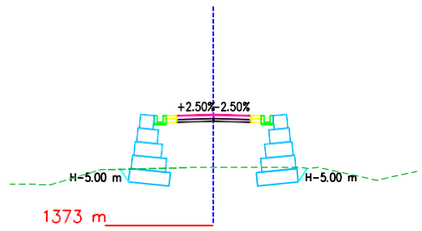
Ch: 0+830
GL:1377.045
DL:1378.977



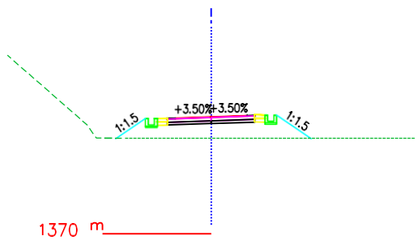
Ch: 0+860
GL:1377.012
DL:1380.058



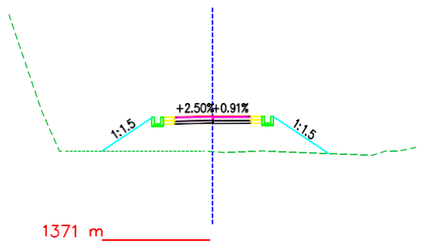
Ch: 0+890
GL:1377.275
DL:1381.139



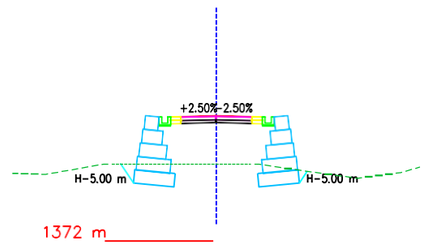
Ch: 0+820
GL:1377.042
DL:1378.616



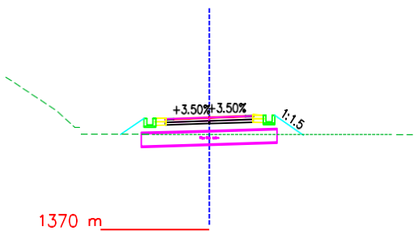
Ch: 0+850
GL:1377.091
DL:1379.697



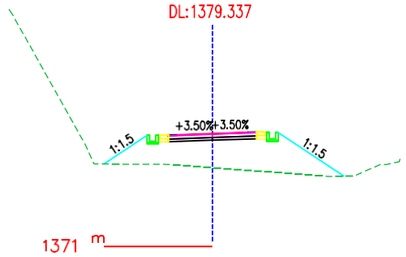
Ch: 0+880
GL:1377.247
DL:1380.779



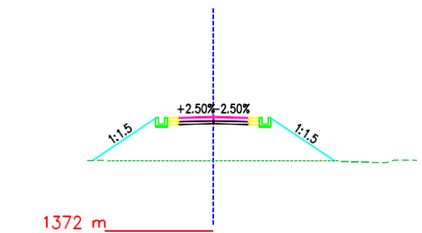
Ch: 0+810
GL:1377.030
DL:1378.318



Ch: 0+840
GL:1376.806
DL:1379.337



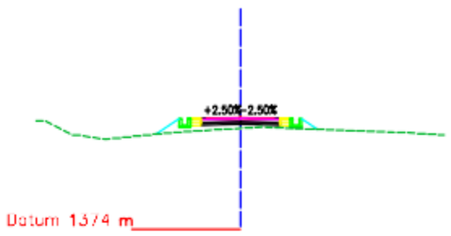
Ch: 0+870
GL:1377.207
DL:1380.418



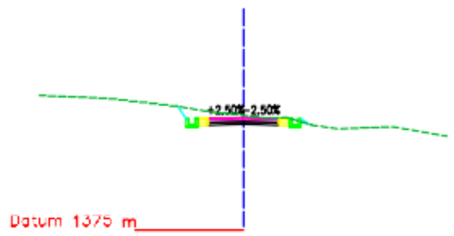
TRIBHUVAN UNIVERSITY INSTITUTE OF ENGINEERING THAPATHALI CAMPUS THAPATHALI, KATHMANDU & DEPARTMENT OF CIVIL ENGINEERING	FINAL YEAR PROJECT GROUP NO. 17	SURYABINAYAK-GUNDU	DATE	REVISION	SIGNATURE	CROSS-0SECTIONS DRAWN BY :GROUP 17 CHECKED BY : APPROVED BY :	SCALE 1:200 4 m 0 8 m	CROSS-SECTIONS 0+810-0+890	DATE: DRG NO : 1 SHEET NO : 10

Design & Drawing by SW Road V2

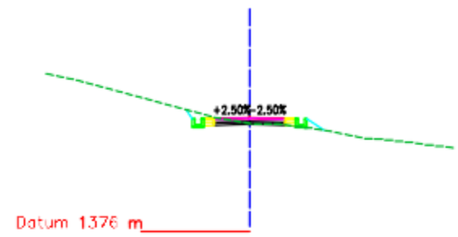
Ch:0+920
GL:1381.427
DL:1382.220



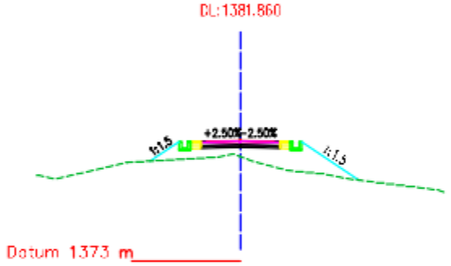
Ch:0+950
GL:1383.474
DL:1383.301



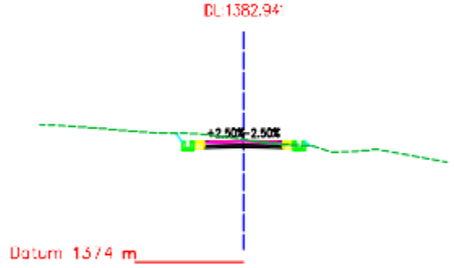
Ch:0+980
GL:1383.992
DL:1384.382



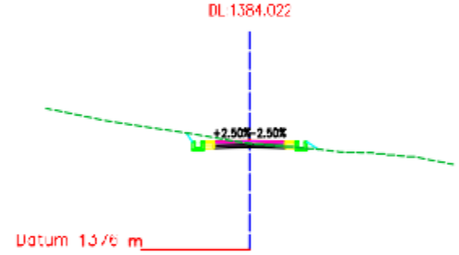
Ch:0+910
GL:1380.677
DL:1381.860



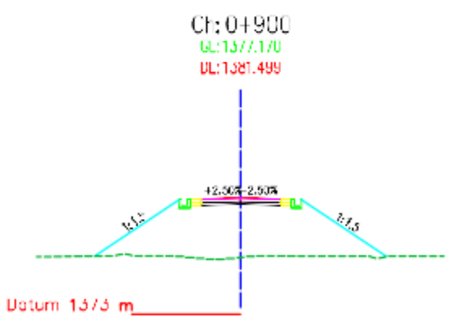
Ch:0+940
GL:1383.028
DL:1382.941



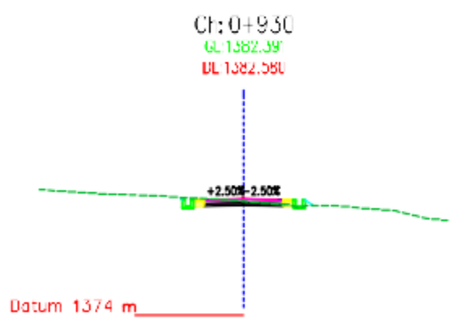
Ch:0+970
GL:1383.822
DL:1384.022



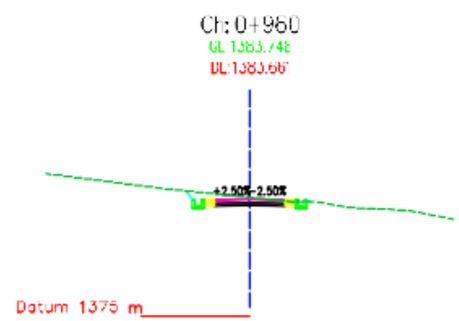
Ch:0+900
GL:1377.170
DL:1381.499



Ch:0+930
GL:1382.39
DL:1382.581



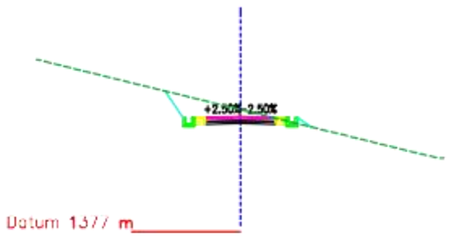
Ch:0+960
GL:1383.748
DL:1383.66



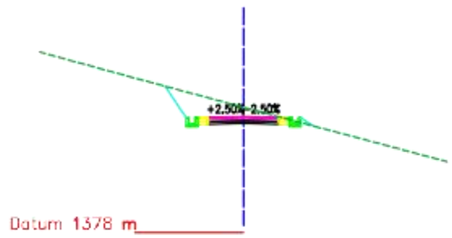
TRIBHUVAN UNIVERSITY INSTITUTE OF ENGINEERING THAPATHALI CAMPUS THAPATHALI, KATHMANDU & DEPARTMENT OF CIVIL ENGINEERING	FINAL YEAR PROJECT GROUP NO. 17	SURYABINAYAK-GUNDU	DATE	REVISION	SIGNATURE	CROSS-0SECTIONS DRAWN BY :GROUP 17 CHECKED BY : APPROVED BY :	SCALE 1:200 4 m 0 8 m	CROSS-SECTIONS 0+900-0+980	DATE: DRG NO : 1 SHEET NO : 11

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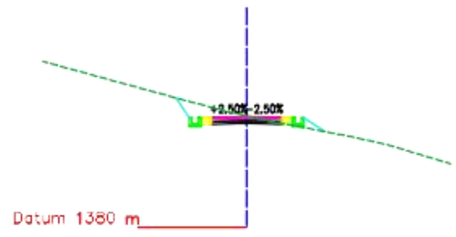
Ch:1+010
GL:1385.98
DL:1385.463



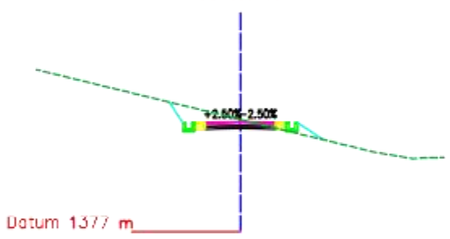
Ch:1+040
GL:1387.225
DL:1386.544



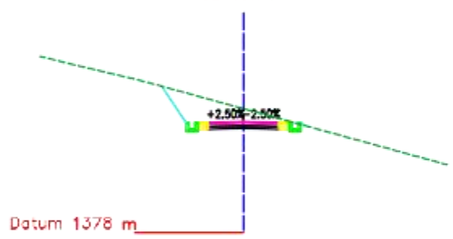
Ch:1+070
GL:1388.204
DL:1388.163



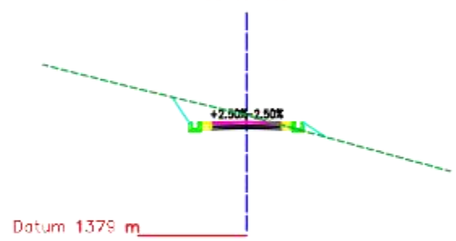
Ch:1+000
GL:1385.262
DL:1385.103



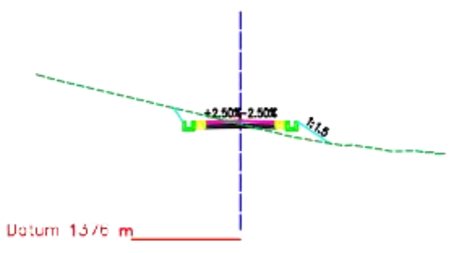
Ch:1+030
GL:1387.734
DL:1386.184



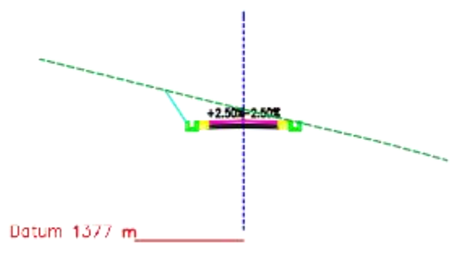
Ch:1+060
GL:1387.844
DL:1387.382



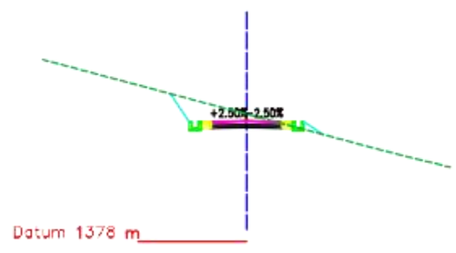
Ch:0+990
GL:1384.464
DL:1384.742



Ch:1+020
GL:1386.653
DL:1385.824



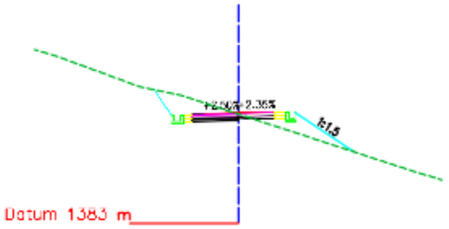
Ch:1+050
GL:1387.388
DL:1386.905



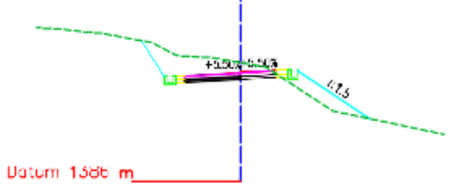
TRIBHUVAN UNIVERSITY INSTITUTE OF ENGINEERING THAPATHALI CAMPUS THAPATHALI, KATHMANDU & DEPARTMENT OF CIVIL ENGINEERING	FINAL YEAR PROJECT GROUP NO. 17	SURYABINAYAK-GUNDU	DATE	REVISION	SIGNATURE	CROSS-0SECTIONS DRAWN BY :GROUP 17 CHECKED BY : APPROVED BY :	SCALE 1:200 4 m 0 8 m	CROSS-SECTIONS 0+990-1+070	DATE: DRG NO : 1 SHEET NO : 12

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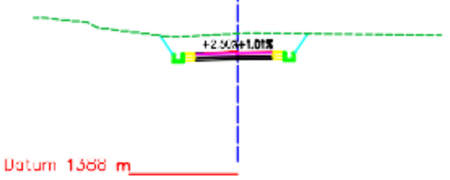
Ch:1+100
GL:1391.053
DL:1391.086



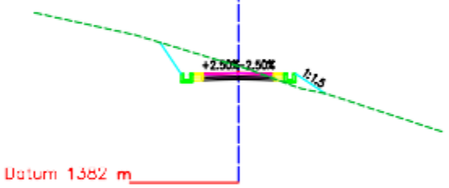
Ch:1+130
GL:1394.746
DL:1394.012



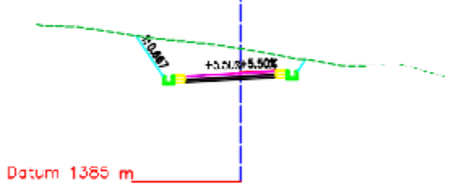
Ch:1+160
GL:1398.534
DL:1396.937



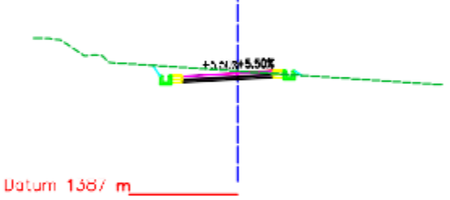
Ch:1+090
GL:1390.582
DL:1390.11



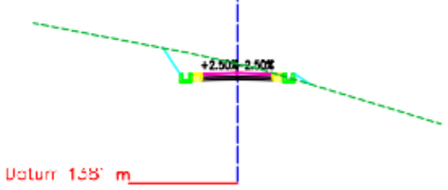
Ch:1+120
GL:1394.810
DL:1393.038



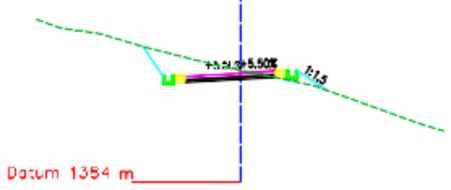
Ch:1+150
GL:1396.072
DL:1395.962



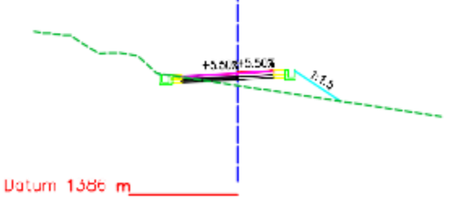
Ch:1+080
GL:1389.796
DL:1389.136



Ch:1+110
GL:1392.250
DL:1392.06'



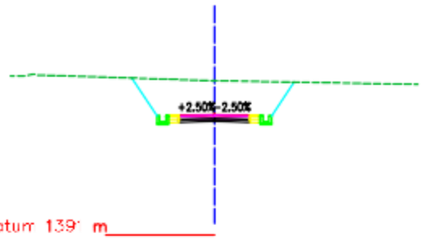
Ch:1+140
GL:1394.032
DL:1394.987



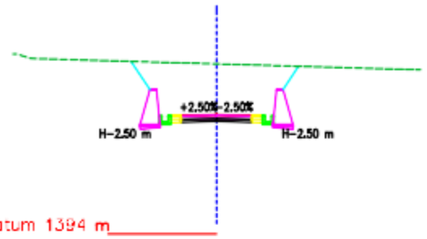
TRIBHUVAN UNIVERSITY INSTITUTE OF ENGINEERING THAPATHALI CAMPUS THAPATHALI, KATHMANDU & DEPARTMENT OF CIVIL ENGINEERING	FINAL YEAR PROJECT GROUP NO. 17	SURYABINAYAK-GUNDU	DATE	REVISION	SIGNATURE	CROSS-0SECTIONS DRAWN BY :GROUP 17 CHECKED BY : APPROVED BY :	SCALE 1:200 4 m 0 8 m	CROSS-SECTIONS 1+080-1+160	DATE: DRG NO : 1 SHEET NO : 13

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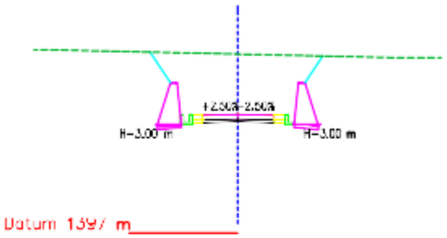
Ch:1+190
GL:1402.356
DL:1399.863



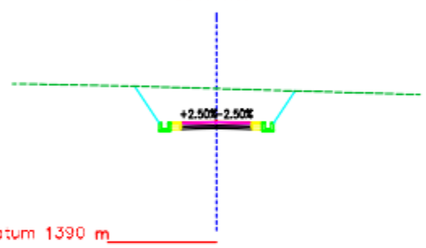
Ch:1+220
GL:1406.520
DL:1402.786



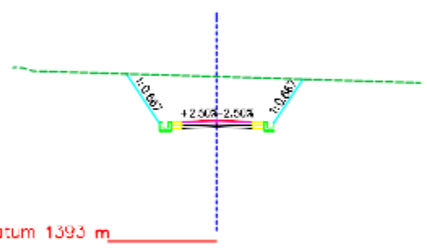
Ch:1+250
GL:1410.166
DL:1405.714



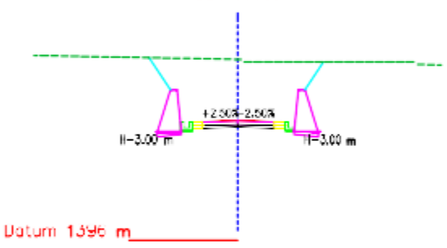
Ch:1+180
GL:1401.320
DL:1398.887



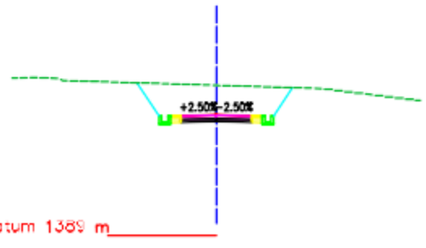
Ch:1+210
GL:1405.129
DL:1401.813



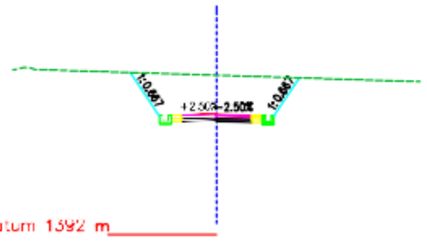
Ch:1+240
GL:1409.312
DL:1404.738



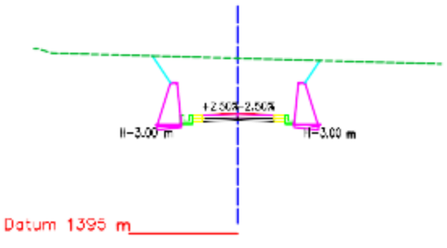
Ch:1+170
GL:1400.077
DL:1397.912



Ch:1+200
GL:1403.740
DL:1400.838



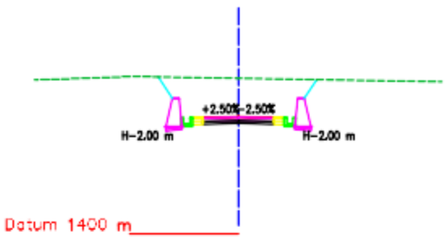
Ch:1+230
GL:1407.915
DL:1403.763



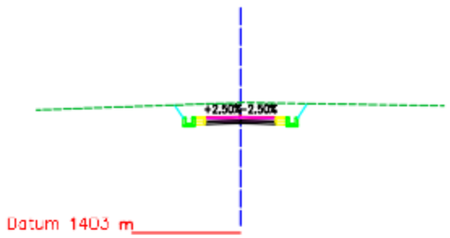
TRIBHUVAN UNIVERSITY INSTITUTE OF ENGINEERING THAPATHALI CAMPUS THAPATHALI, KATHMANDU & DEPARTMENT OF CIVIL ENGINEERING	FINAL YEAR PROJECT GROUP NO. 17	SURYABINAYAK-GUNDU	DATE	REVISION	SIGNATURE	CROSS-0SECTIONS DRAWN BY :GROUP 17 CHECKED BY : APPROVED BY :	SCALE 1:200 4 m 0 8 m	CROSS-SECTIONS 1+170-1+250	DATE: DRG NO : 1 SHEET NO : 14

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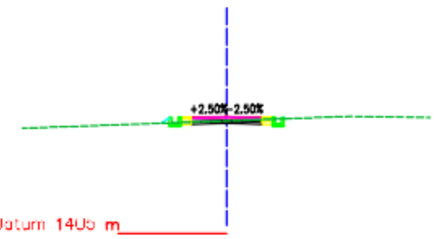
Ch:1+280
GL:1411.461
DL:1408.639



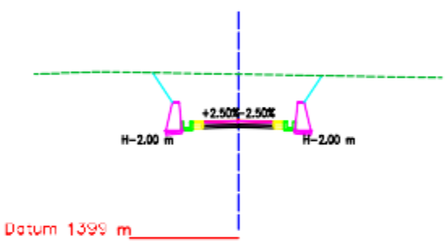
Ch:1+310
GL:1412.589
DL:1411.564



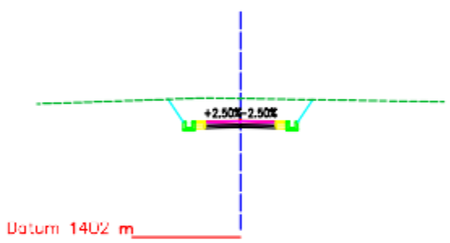
Ch:1+340
GL:1413.264
DL:1413.597



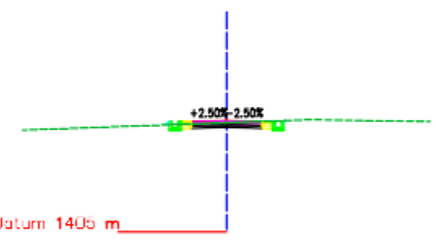
Ch:1+270
GL:1411.082
DL:1407.664



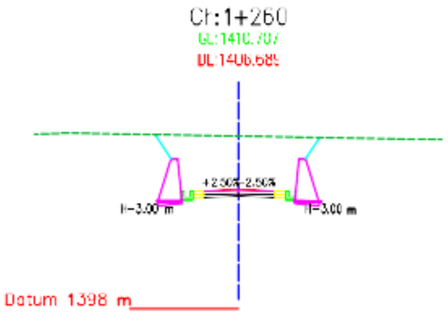
Ch:1+300
GL:1412.275
DL:1410.589



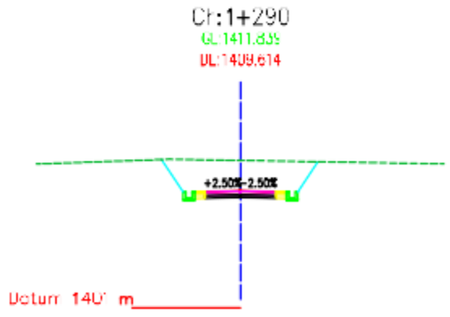
Ch:1+330
GL:1413.055
DL:1413.207



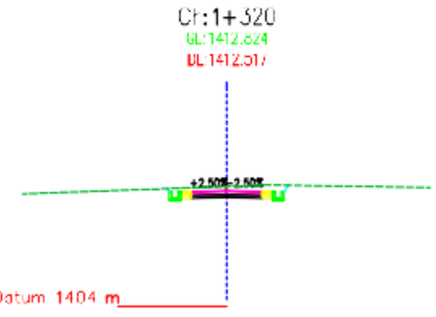
Ch:1+260
GL:1410.707
DL:1406.685



Ch:1+290
GL:1411.635
DL:1409.614



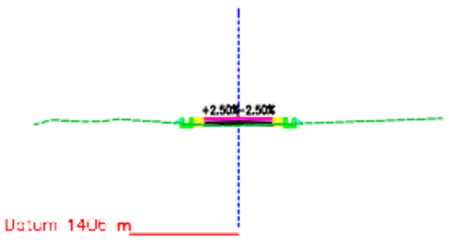
Ch:1+320
GL:1412.624
DL:1412.917



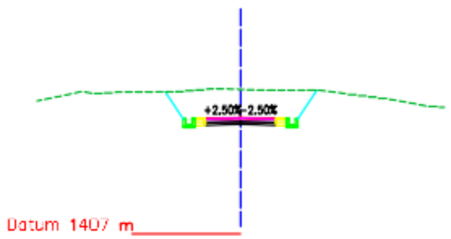
TRIBHUVAN UNIVERSITY INSTITUTE OF ENGINEERING THAPATHALI CAMPUS THAPATHALI, KATHMANDU & DEPARTMENT OF CIVIL ENGINEERING	FINAL YEAR PROJECT GROUP NO. 17	SURYABINAYAK-GUNDU	DATE	REVISION	SIGNATURE	CROSS-0SECTIONS DRAWN BY :GROUP 17 CHECKED BY : APPROVED BY :	SCALE 1:200 4 m 0 8 m	CROSS-SECTIONS 1+260-1+340	DATE: DRG NO : 1 SHEET NO : 15

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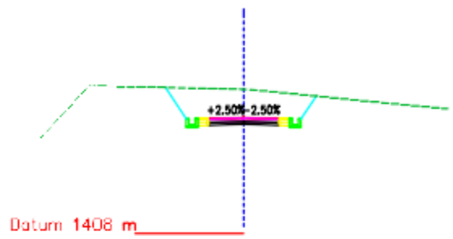
Ch:1+370
GL:1416.987
DL:1414.576



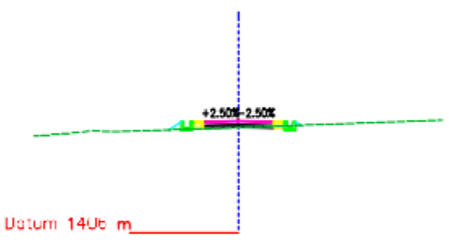
Ch:1+400
GL:1417.628
DL:1415.555



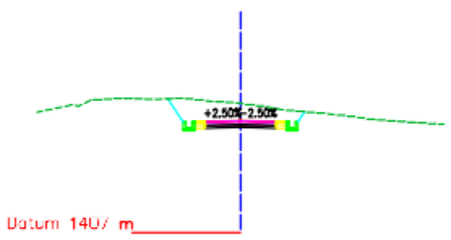
Ch:1+430
GL:1418.613
DL:1416.535



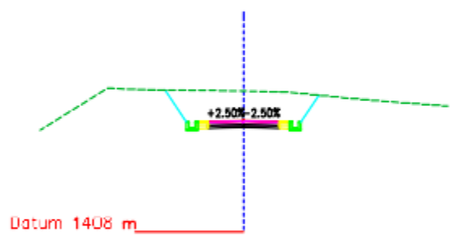
Ch:1+360
GL:1416.740
DL:1414.250



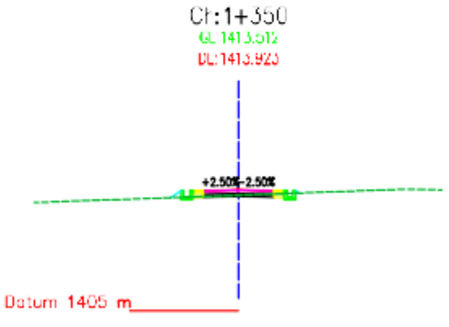
Ch:1+390
GL:1416.500
DL:1415.229



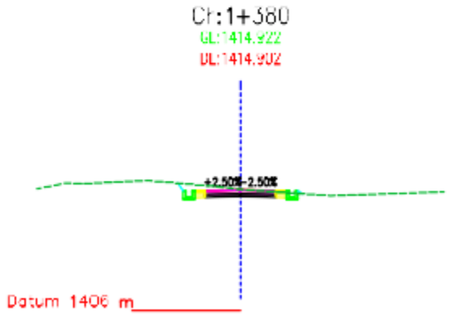
Ch:1+420
GL:1418.594
DL:1416.208



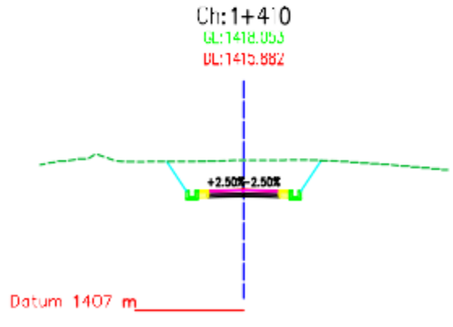
Ch:1+350
GL:1413.501
DL:1413.923



Ch:1+380
GL:1414.922
DL:1414.902



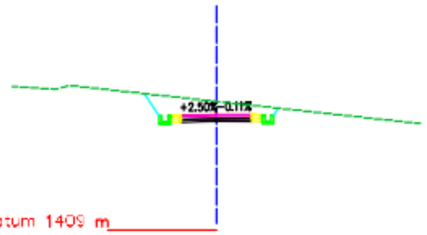
Ch:1+410
GL:1418.053
DL:1415.882



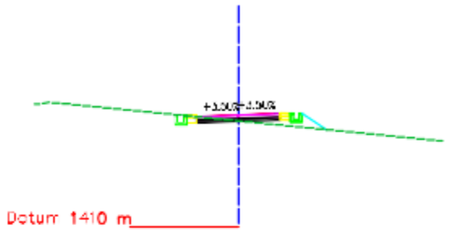
TRIBHUVAN UNIVERSITY INSTITUTE OF ENGINEERING THAPATHALI CAMPUS THAPATHALI, KATHMANDU & DEPARTMENT OF CIVIL ENGINEERING	FINAL YEAR PROJECT GROUP NO. 17	SURYABINAYAK-GUNDU	DATE	REVISION	SIGNATURE	CROSS-0SECTIONS DRAWN BY :GROUP 17 CHECKED BY : APPROVED BY :	SCALE 1:200 4 m 0 8 m	CROSS-SECTIONS 1+350-1+430	DATE: DRG NO : 1 SHEET NO : 16

Design & Drawing by SW Road V2

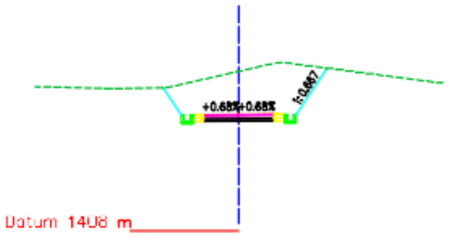
Ch:1+460
GL:1418.485
DL 1417.514



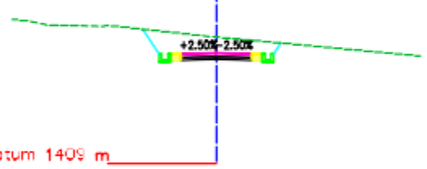
Ch:1+490
GL:1417.818
DL:1418.285



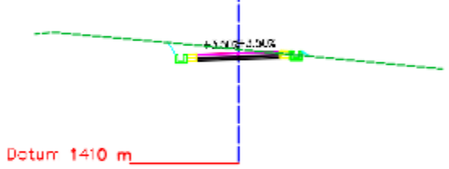
Ch:1+520
GL:1419.752
DL:1416.568



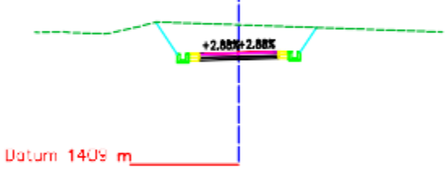
Ch:1+450
GL:1418.337
DL 1417.188



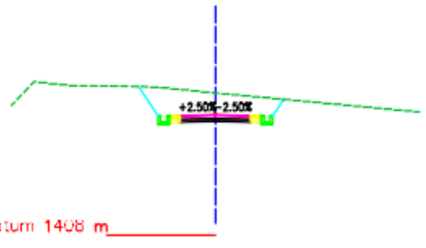
Ch:1+480
GL:1418.405
DL 1418.167



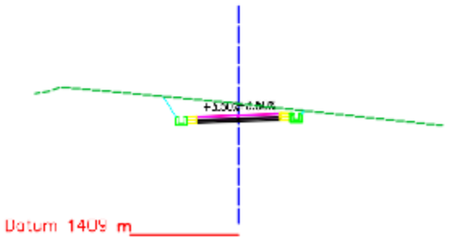
Ch:1+510
GL:1419.282
DL:1417.238



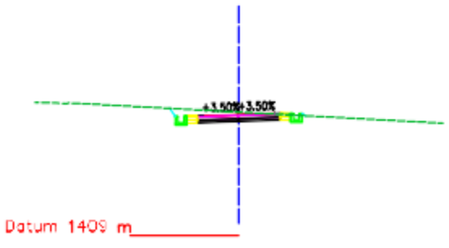
Ch:1+440
GL:1418.458
DL:1416.861



Ch:1+470
GL:1418.654
DL:1417.841



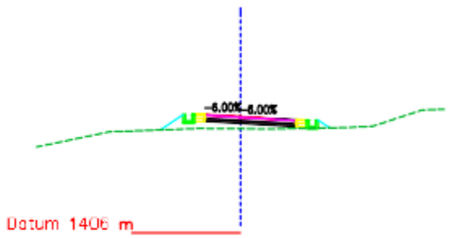
Ch:1+500
GL:1418.092
DL:1417.906



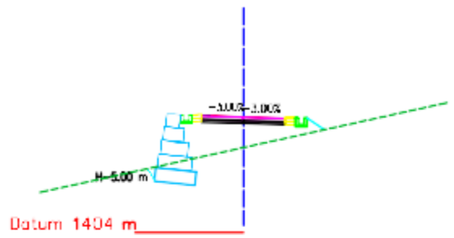
TRIBHUVAN UNIVERSITY INSTITUTE OF ENGINEERING THAPATHALI CAMPUS THAPATHALI, KATHMANDU & DEPARTMENT OF CIVIL ENGINEERING	FINAL YEAR PROJECT GROUP NO. 17	SURYABINAYAK-GUNDU	DATE	REVISION	SIGNATURE	CROSS-0SECTIONS DRAWN BY :GROUP 17 CHECKED BY : APPROVED BY :	SCALE 1:200 4 m 0 8 m	CROSS-SECTIONS 1+440-1+520	DATE: DRG NO : 1 SHEET NO : 17

Design & Drawing by SW Road V2

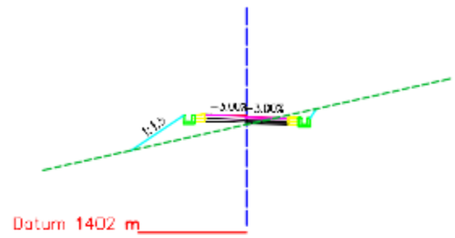
Ch:1+550
GL:1413.660
DL:1414.560



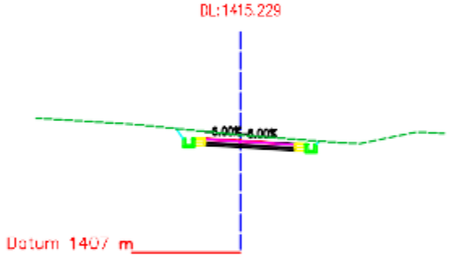
Ch:1+580
GL:1410.272
DL:1412.552



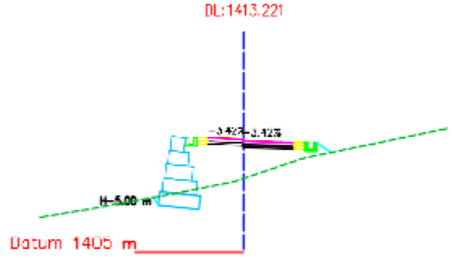
Ch:1+610
GL:1409.963
DL:1410.543



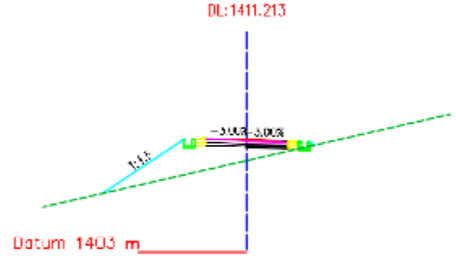
Ch:1+540
GL:1413.643
DL:1415.229



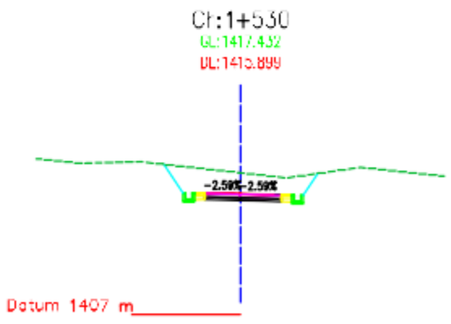
Ch:1+570
GL:1410.433
DL:1413.221



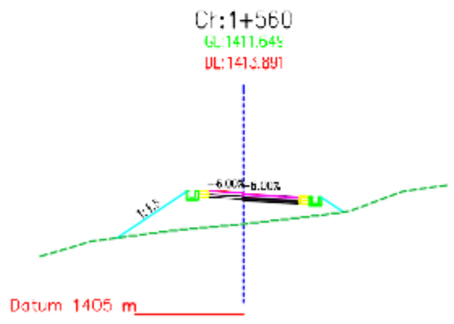
Ch:1+600
GL:1409.742
DL:1411.213



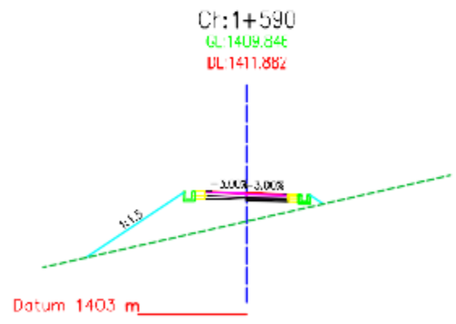
Ch:1+530
GL:1417.432
DL:1418.899



Ch:1+560
GL:1411.649
DL:1413.891



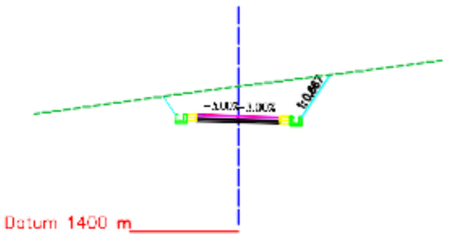
Ch:1+590
GL:1409.846
DL:1411.882



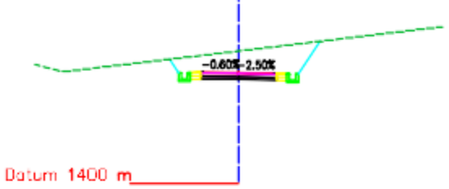
TRIBHUVAN UNIVERSITY INSTITUTE OF ENGINEERING THAPATHALI CAMPUS THAPATHALI, KATHMANDU & DEPARTMENT OF CIVIL ENGINEERING	FINAL YEAR PROJECT GROUP NO. 17	SURYABINAYAK-GUNDU	DATE	REVISION	SIGNATURE	CROSS-0SECTIONS DRAWN BY :GROUP 17 CHECKED BY : APPROVED BY :	SCALE 1:200 4 m 0 8 m	CROSS-SECTIONS 1+530-1+610	DATE: DRG NO : 1 SHEET NO : 18

Design & Drawing by SW Road V2

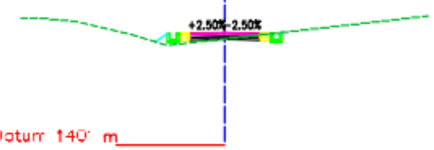
Ch:1+640
GL:1410.681
DL:1408.535



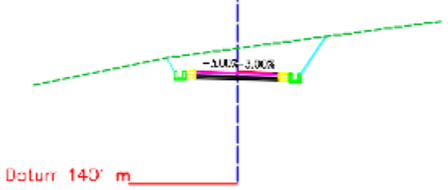
Ch:1+670
GL:1409.758
DL:1408.174



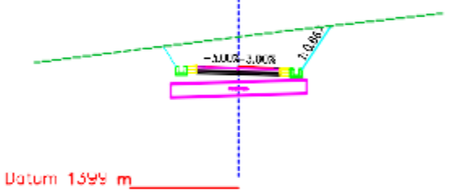
Ch:1+700
GL:1408.624
DL:1409.292



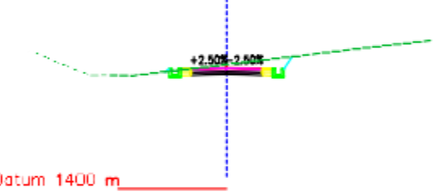
Ch:1+630
GL:1410.992
DL:1409.204



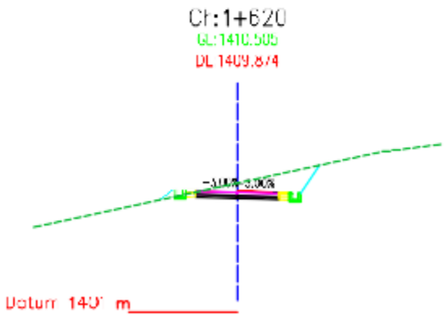
Ch:1+660
GL:1410.111
DL:1407.847



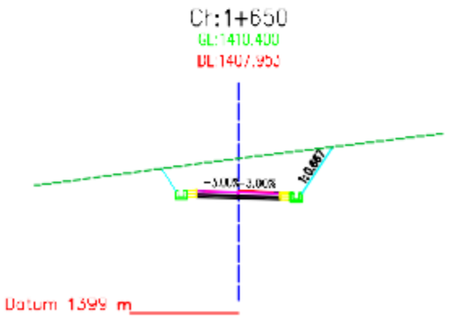
Ch:1+690
GL:1409.152
DL:1408.919



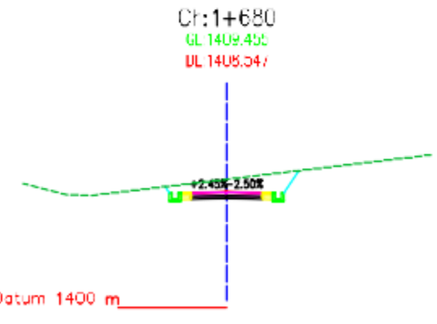
Ch:1+620
GL:1410.505
DL:1409.674



Ch:1+650
GL:1410.400
DL:1407.953



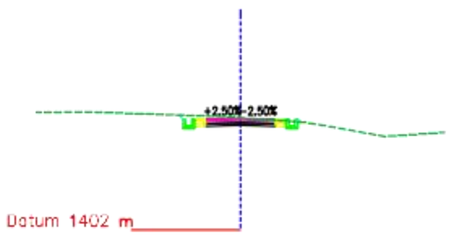
Ch:1+680
GL:1409.455
DL:1408.347



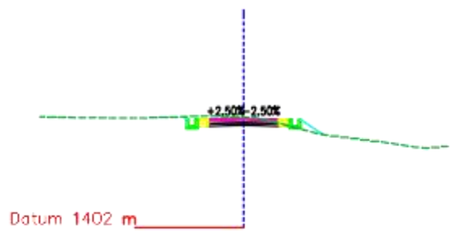
TRIBHUVAN UNIVERSITY INSTITUTE OF ENGINEERING THAPATHALI CAMPUS THAPATHALI, KATHMANDU & DEPARTMENT OF CIVIL ENGINEERING	FINAL YEAR PROJECT GROUP NO. 17	SURYABINAYAK-GUNDU	DATE	REVISION	SIGNATURE	CROSS-0SECTIONS DRAWN BY :GROUP 17 CHECKED BY : APPROVED BY :	SCALE 1:200 4 m 0 8 m	CROSS-SECTIONS 1+620-1+700	DATE: DRG NO : 1 SHEET NO : 19

Design & Drawing by SW Road V2

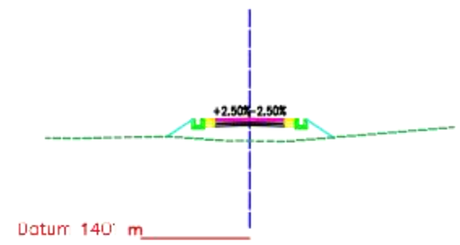
Ch:1+730
GL:1410.237
DL:1410.19



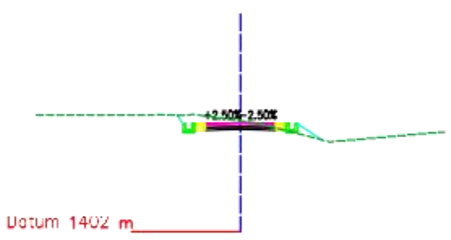
Ch:1+760
GL:1410.096
DL:1410.020



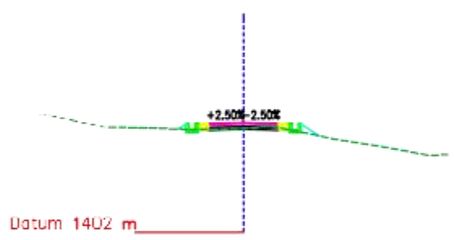
Ch:1+790
GL:1408.168
DL:1409.811



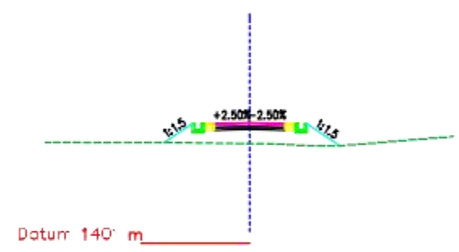
Ch:1+720
GL:1410.159
DL:1410.019



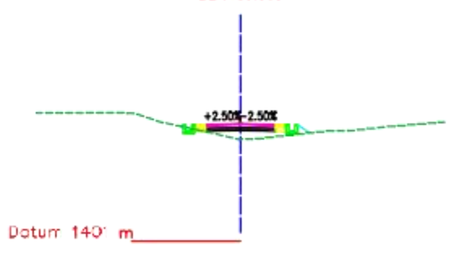
Ch:1+750
GL:1409.660
DL:1410.090



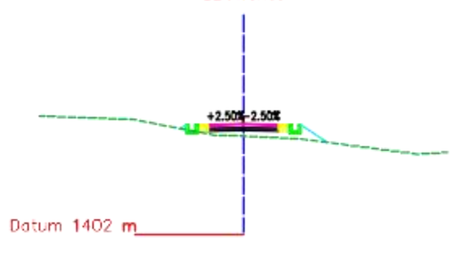
Ch:1+780
GL:1408.542
DL:1409.88



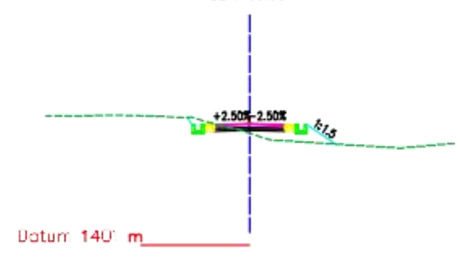
Ch:1+710
GL:1408.514
DL:1409.665



Ch:1+740
GL:1409.246
DL:1410.160



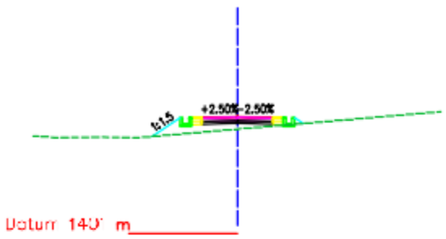
Ch:1+770
GL:1409.240
DL:1409.95



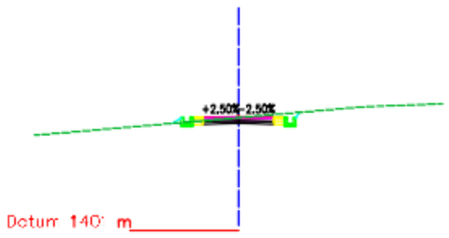
TRIBHUVAN UNIVERSITY INSTITUTE OF ENGINEERING THAPATHALI CAMPUS THAPATHALI, KATHMANDU & DEPARTMENT OF CIVIL ENGINEERING	FINAL YEAR PROJECT GROUP NO. 17	SURYABINAYAK-GUNDU	DATE	REVISION	SIGNATURE	CROSS-0SECTIONS DRAWN BY :GROUP 17 CHECKED BY : APPROVED BY :	SCALE 1:200 4 m 0 8 m	CROSS-SECTIONS 1+710-1+790	DATE: DRG NO : 1 SHEET NO : 20

Design & Drawing by SW Road V2

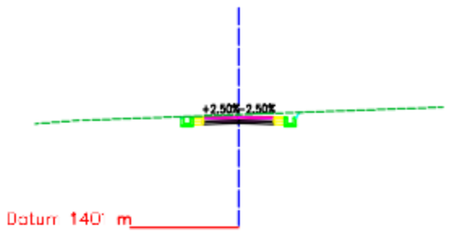
Ch:1+820
GL:1408.67
DL:1409.60



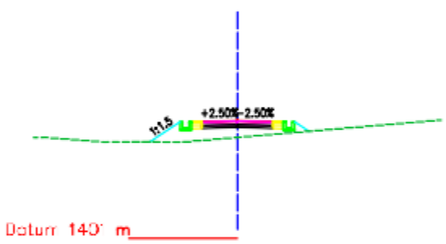
Ch:1+850
GL:1409.314
DL:1409.392



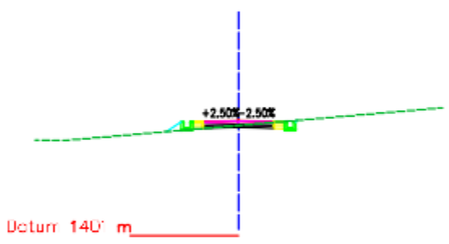
Ch:1+880
GL:1409.528
DL:1409.182



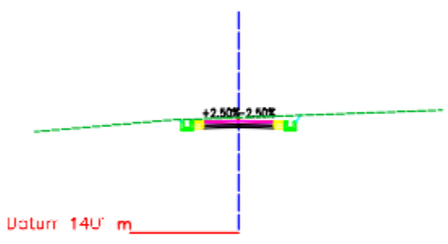
Ch:1+810
GL:1408.436
DL:1409.67



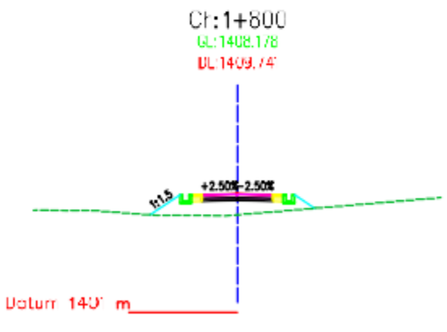
Ch:1+840
GL:1409.106
DL:1409.462



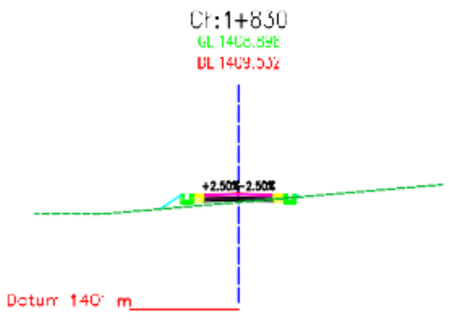
Ch:1+870
GL:1409.475
DL:1409.252



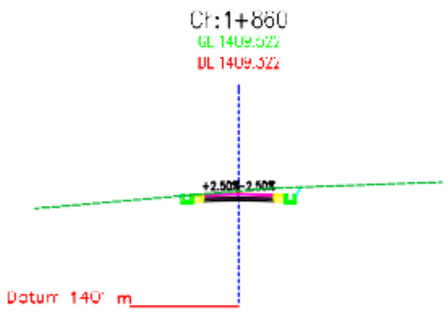
Ch:1+800
GL:1408.178
DL:1409.74



Ch:1+830
GL:1408.898
DL:1409.552



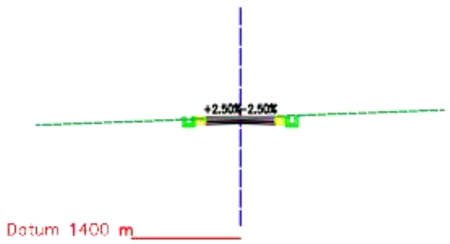
Ch:1+860
GL:1409.522
DL:1409.322



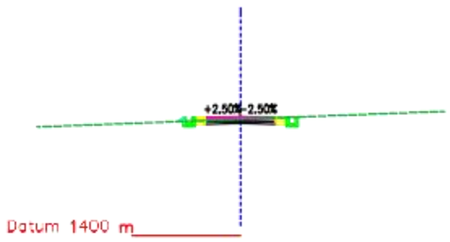
TRIBHUVAN UNIVERSITY INSTITUTE OF ENGINEERING THAPATHALI CAMPUS THAPATHALI, KATHMANDU & DEPARTMENT OF CIVIL ENGINEERING	FINAL YEAR PROJECT GROUP NO. 17	SURYABINAYAK-GUNDU	DATE	REVISION	SIGNATURE	CROSS-0SECTIONS DRAWN BY :GROUP 17 CHECKED BY : APPROVED BY :	SCALE 1:200 4 m 0 8 m	CROSS-SECTIONS 1+800-1+880	DATE: DRG NO : 1 SHEET NO : 21

Design & Drawing by SW Road V2

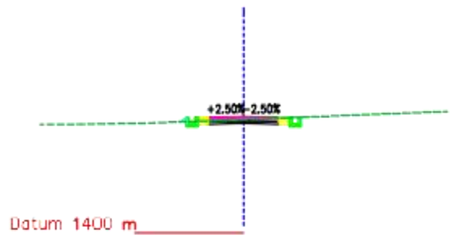
Ch:1+910
GL:1408.90
DL 1408.972



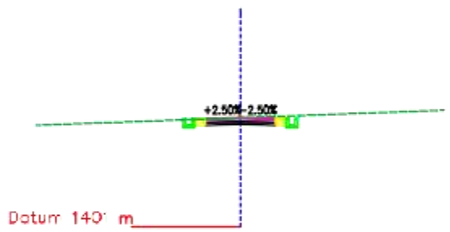
Ch:1+940
GL:1408.57
DL 1408.764



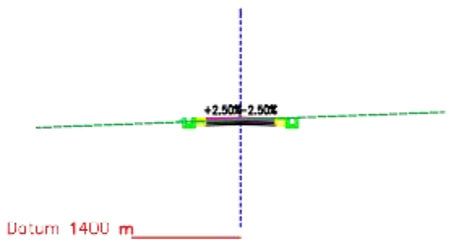
Ch:1+970
GL:1408.344
DL 1408.554



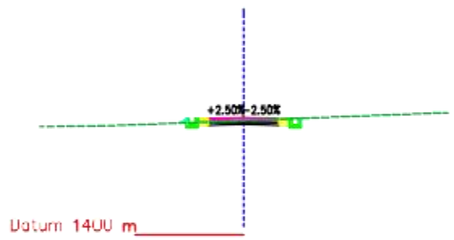
Ch:1+900
GL:1409.033
DL 1409.043



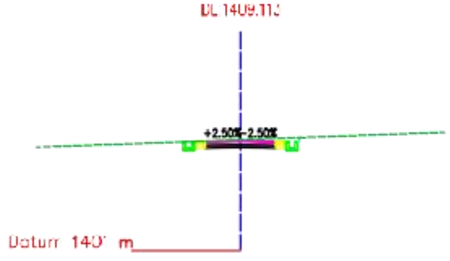
Ch:1+930
GL:1408.670
DL 1408.833



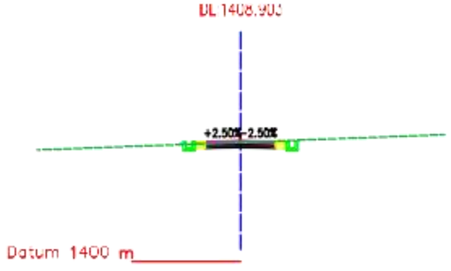
Ch:1+960
GL:1408.408
DL 1408.624



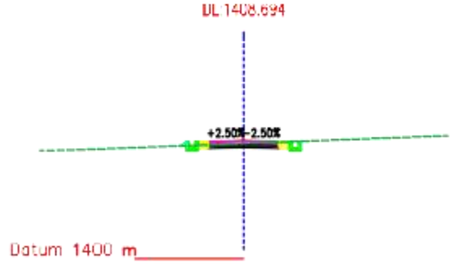
Ch:1+890
GL:1409.177
DL 1409.113



Ch:1+920
GL:1408.780
DL 1408.903



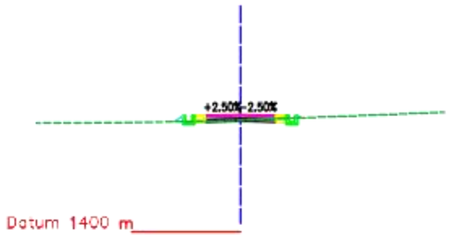
Ch:1+950
GL:1408.484
DL 1408.694



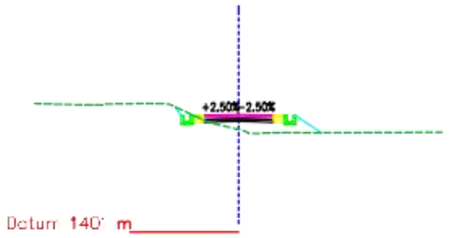
TRIBHUVAN UNIVERSITY INSTITUTE OF ENGINEERING THAPATHALI CAMPUS THAPATHALI, KATHMANDU & DEPARTMENT OF CIVIL ENGINEERING	FINAL YEAR PROJECT GROUP NO. 17	SURYABINAYAK-GUNDU	DATE	REVISION	SIGNATURE	CROSS-0SECTIONS DRAWN BY :GROUP 17 CHECKED BY : APPROVED BY :	SCALE 1:200 4 m 0 8 m	CROSS-SECTIONS 1+890-1+970	DATE: DRG NO : 1 SHEET NO : 22

Design & Drawing by SW Road V2

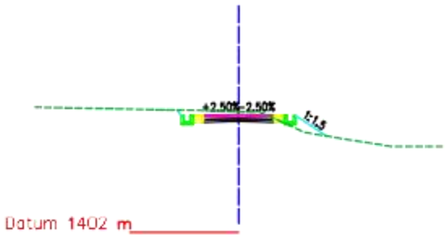
Ch: 2+000
GL 1408.222
DL 1408.640



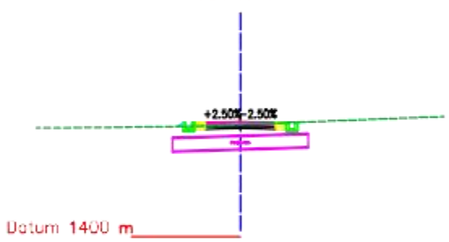
Ch: 2+030
GL 1408.547
DL 1409.650



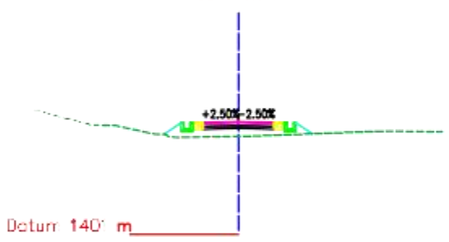
Ch: 2+060
GL:1410.890
DL:1410.669



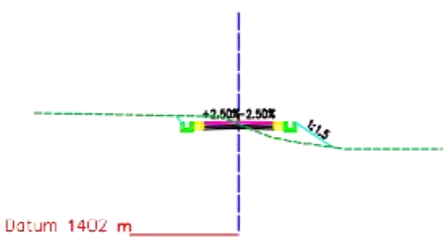
Ch: 1+990
GL 1408.25
DL 1408.464



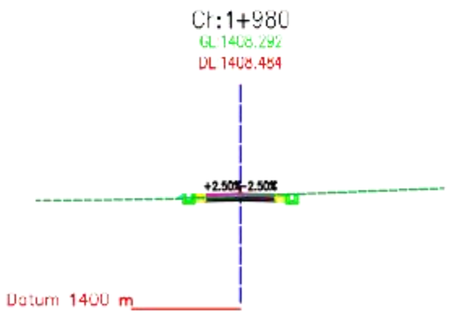
Ch: 2+020
GL 1408.222
DL:1409.310



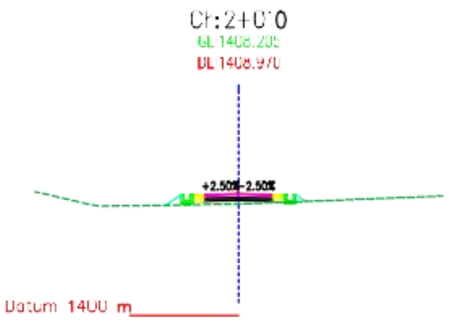
Ch: 2+050
GL:1410.746
DL:1410.329



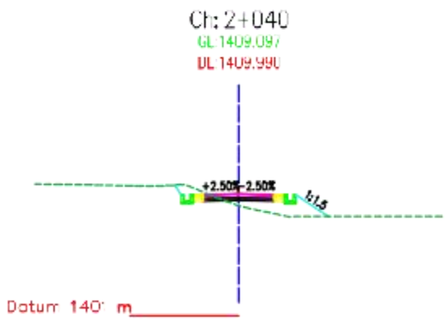
Ch: 1+980
GL 1408.292
DL 1408.484



Ch: 2+000
GL 1408.205
DL 1408.970



Ch: 2+040
GL 1409.097
DL 1409.990



TRIBHUVAN UNIVERSITY INSTITUTE OF ENGINEERING THAPATHALI CAMPUS THAPATHALI, KATHMANDU & DEPARTMENT OF CIVIL ENGINEERING	FINAL YEAR PROJECT GROUP NO. 17	SURYABINAYAK-GUNDU	DATE	REVISION	SIGNATURE	CROSS-0SECTIONS DRAWN BY :GROUP 17 CHECKED BY : APPROVED BY :	SCALE 1:200 4 m 0 8 m	CROSS-SECTIONS 1+980-2+060	DATE: DRG NO : 1 SHEET NO : 23

Design & Drawing by SW Road V2

Ch:2+090
GL:1411.618
DL:1411.689
+2.50%-2.50%
Datum 1403 m

Ch:2+120
GL:1412.216
DL:1413.058
+2.50%-2.50%
Datum 1405 m

Ch:2+150
GL:1413.432
DL:1414.736
+2.50%-2.50%
Datum 1406 m

Ch:2+080
GL:1411.599
DL:1411.349
+2.50%-2.50%
Datum 1403 m

Ch:2+110
GL:1412.028
DL:1412.507
+2.50%-2.50%
Datum 1404 m

Ch:2+140
GL:1412.979
DL:1414.178
+2.50%-2.50%
Datum 1406 m

Ch:2+070
GL:1411.101
DL:1411.009
+2.50%-2.50%
Datum 1403 m

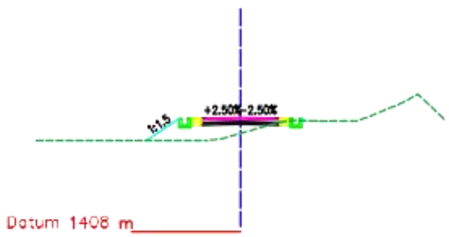
Ch:2+100
GL:1411.829
DL:1412.048
+2.50%-2.50%
Datum 1404 m

Ch:2+130
GL:1412.513
DL:1413.617
+2.50%-2.50%
Datum 1405 m

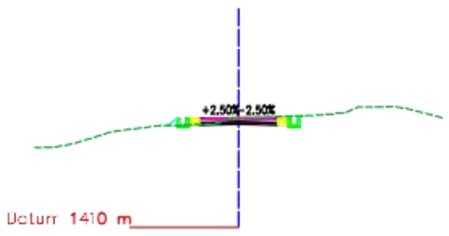
TRIBHUVAN UNIVERSITY INSTITUTE OF ENGINEERING THAPATHALI CAMPUS THAPATHALI, KATHMANDU & DEPARTMENT OF CIVIL ENGINEERING	FINAL YEAR PROJECT GROUP NO. 17	SURYABINAYAK-GUNDU	DATE	REVISION	SIGNATURE	CROSS-0SECTIONS DRAWN BY :GROUP 17 CHECKED BY : APPROVED BY :	SCALE 1:200 4 m 0 8 m	CROSS-SECTIONS 2+070-2+150	DATE: DRG NO : 1 SHEET NO : 24

Design & Drawing by SW Road V2

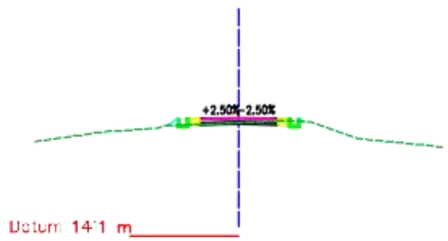
Ch:2+180
GL:1415.248
DL:1416.414



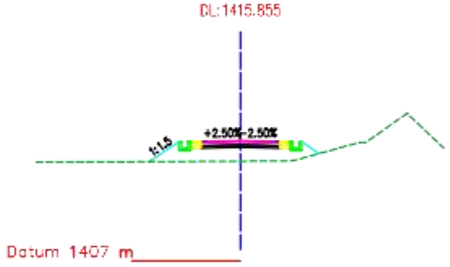
Ch:2+210
GL:1418.080
DL:1418.093



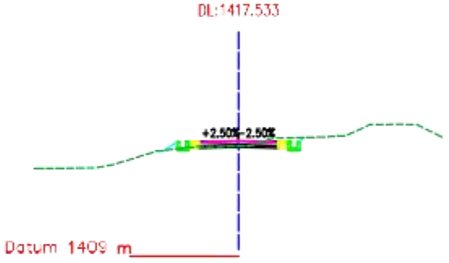
Ch:2+240
GL:1419.336
DL:1419.719



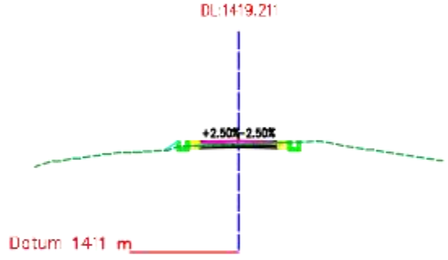
Ch:2+170
GL:1414.337
DL:1415.855



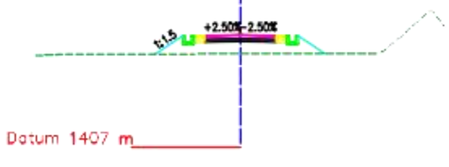
Ch:2+200
GL:1417.147
DL:1417.533



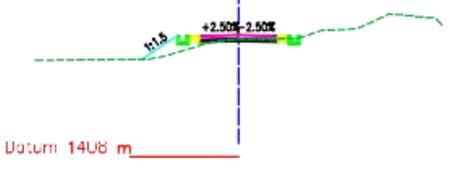
Ch:2+230
GL:1418.939
DL:1419.217



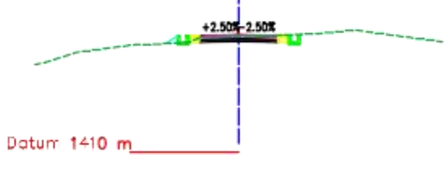
Ch:2+160
GL:1413.884
DL:1416.295



Ch:2+190
GL:1416.656
DL:1416.974



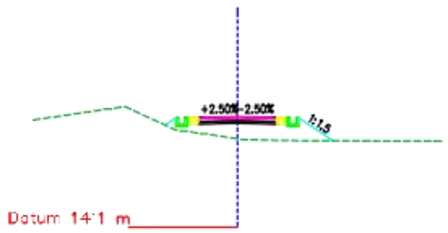
Ch:2+220
GL:1418.520
DL:1418.652



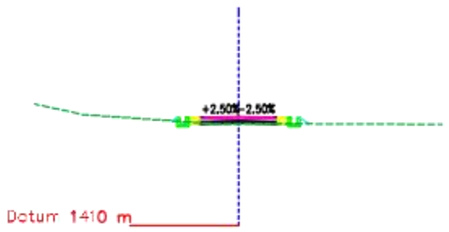
TRIBHUVAN UNIVERSITY INSTITUTE OF ENGINEERING THAPATHALI CAMPUS THAPATHALI, KATHMANDU & DEPARTMENT OF CIVIL ENGINEERING	FINAL YEAR PROJECT GROUP NO. 17	SURYABINAYAK-GUNDU	DATE	REVISION	SIGNATURE	CROSS-0SECTIONS DRAWN BY :GROUP 17 CHECKED BY : APPROVED BY :	SCALE 1:200 4 m 0 8 m	CROSS-SECTIONS 2+160-2+240	DATE: DRG NO : 1 SHEET NO : 25

Design & Drawing by SW Road V2

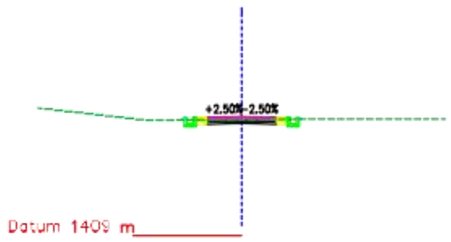
Ch: 2+270
GL:1417.508
DL:1419.148



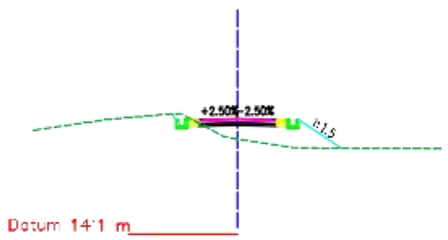
Ch: 2+300
GL:1417.494
DL:1418.035



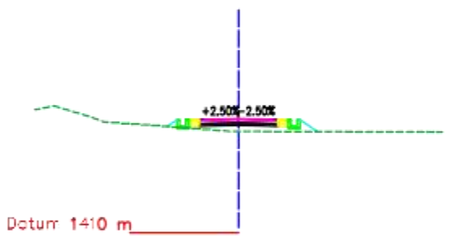
Ch: 2+330
GL:1417.544
DL:1417.768



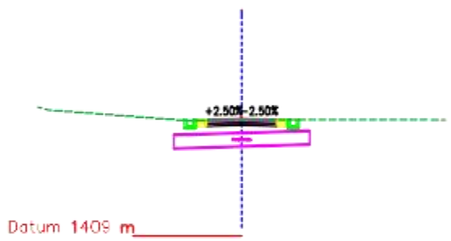
Ch: 2+260
GL:1418.024
DL:1419.519



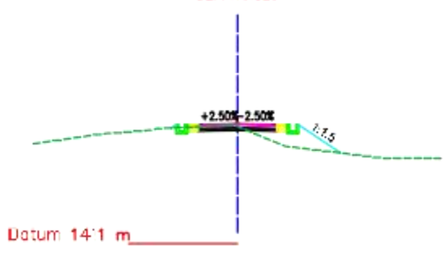
Ch: 2+290
GL:1417.472
DL:1418.406



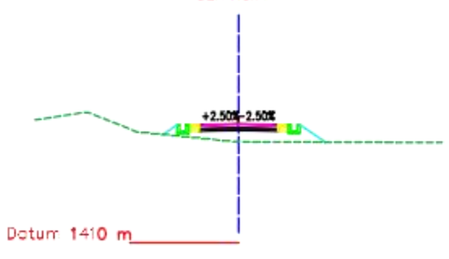
Ch: 2+320
GL:1417.526
DL:1417.634



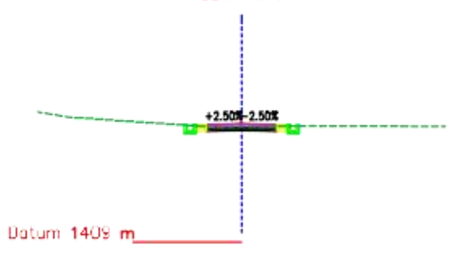
Ch: 2+250
GL:1419.645
DL:1419.826



Ch: 2+280
GL:1417.459
DL:1418.777



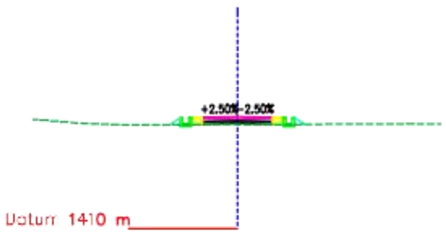
Ch: 2+310
GL:1417.507
DL:1417.774



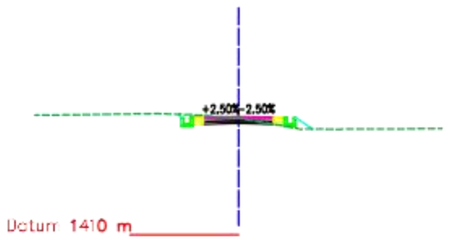
TRIBHUVAN UNIVERSITY INSTITUTE OF ENGINEERING THAPATHALI CAMPUS THAPATHALI, KATHMANDU & DEPARTMENT OF CIVIL ENGINEERING	FINAL YEAR PROJECT GROUP NO. 17	SURYABINAYAK-GUNDU	DATE	REVISION	SIGNATURE	CROSS-0SECTIONS DRAWN BY :GROUP 17 CHECKED BY : APPROVED BY :	SCALE 1:200 4 m 0 8 m	CROSS-SECTIONS 2+250-2+330	DATE: DRG NO : 1 SHEET NO : 26

Design & Drawing by SW Road V2

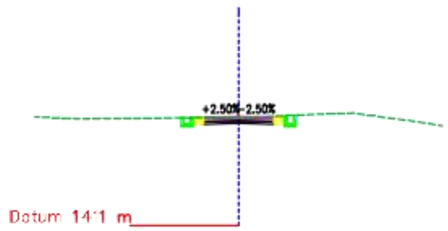
Ch: 2+360
GL:1417.661
DL:1418.229



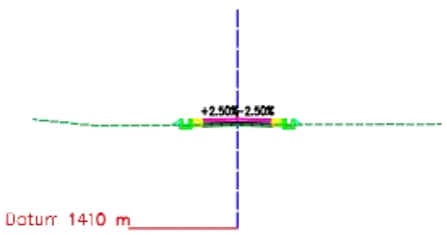
Ch: 2+390
GL:1418.577
DL:1418.690



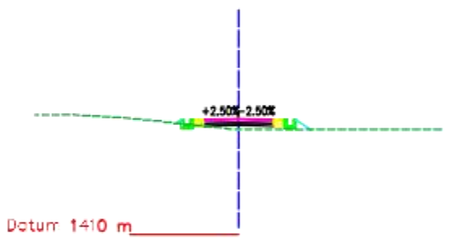
Ch: 2+420
GL:1418.982
DL:1419.022



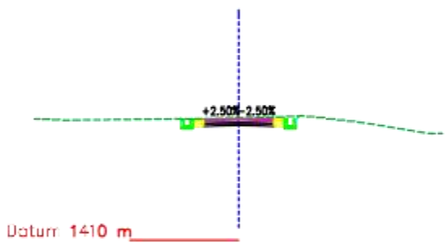
Ch: 2+350
GL:1417.606
DL:1418.075



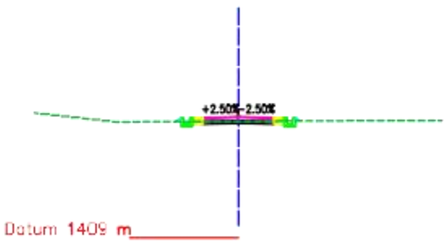
Ch: 2+380
GL:1417.759
DL:1418.536



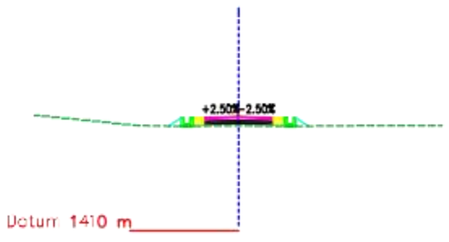
Ch: 2+410
GL:1418.976
DL:1418.943



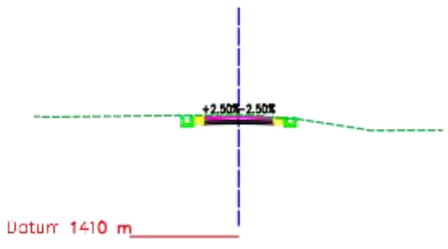
Ch: 2+340
GL:1417.564
DL:1417.922



Ch: 2+370
GL:1417.696
DL:1418.353



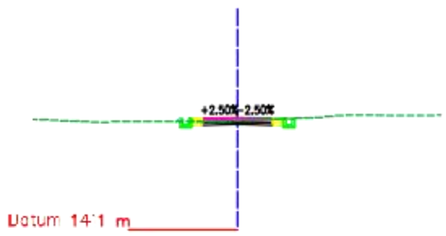
Ch: 2+400
GL:1418.972
DL:1418.834



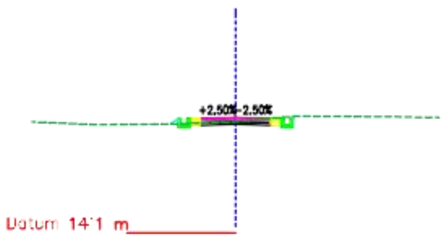
TRIBHUVAN UNIVERSITY INSTITUTE OF ENGINEERING THAPATHALI CAMPUS THAPATHALI, KATHMANDU & DEPARTMENT OF CIVIL ENGINEERING	FINAL YEAR PROJECT GROUP NO. 17	SURYABINAYAK-GUNDU	DATE	REVISION	SIGNATURE	CROSS-0SECTIONS DRAWN BY :GROUP 17 CHECKED BY : APPROVED BY :	SCALE 1:200 4 m 0 8 m	CROSS-SECTIONS 2+340-2+420	DATE: DRG NO : 1 SHEET NO : 27

Design & Drawing by SW Road V2

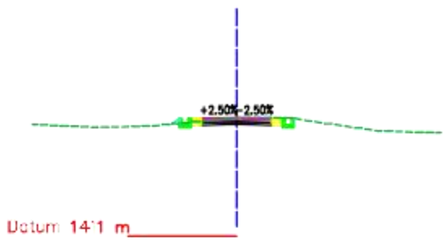
Ch: 2+450
GL:1419.036
DL:1419.256



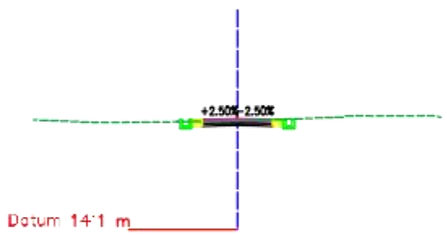
Ch: 2+480
GL:1419.783
DL:1419.490



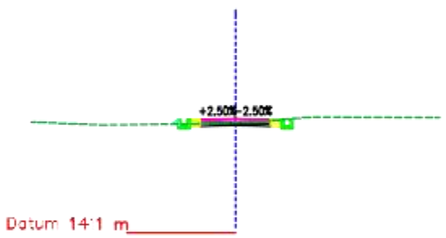
Ch: 2+510
GL:1419.691
DL:1419.717



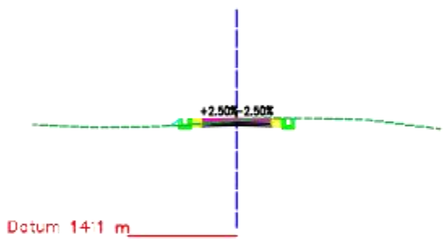
Ch: 2+440
GL:1419.004
DL:1419.178



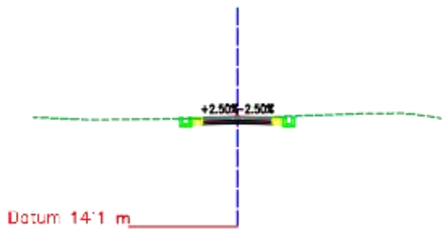
Ch: 2+470
GL:1419.736
DL:1419.412



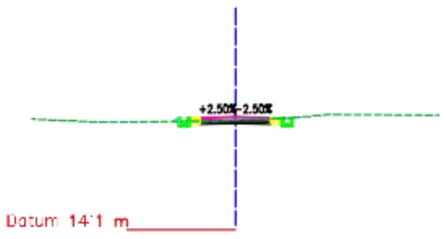
Ch: 2+500
GL:1419.522
DL:1419.646



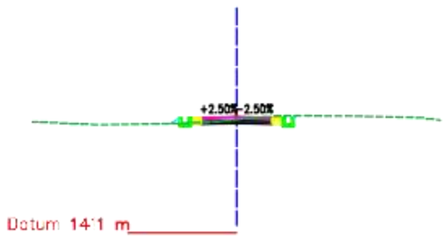
Ch: 2+430
GL:1418.997
DL:1419.100



Ch: 2+460
GL:1419.054
DL:1419.334



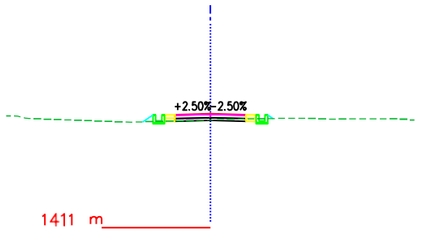
Ch: 2+490
GL:1419.353
DL:1419.568



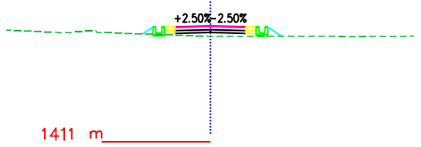
TRIBHUVAN UNIVERSITY INSTITUTE OF ENGINEERING THAPATHALI CAMPUS THAPATHALI, KATHMANDU & DEPARTMENT OF CIVIL ENGINEERING	FINAL YEAR PROJECT GROUP NO. 17	SURYABINAYAK-GUNDU	DATE	REVISION	SIGNATURE	CROSS-0SECTIONS DRAWN BY :GROUP 17 CHECKED BY : APPROVED BY :	SCALE 1:200 4 m 0 8 m	CROSS-SECTIONS 2+430-2+510	DATE: DRG NO : 1 SHEET NO : 28

Design & Drawing by SW Road V2

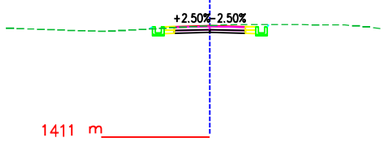
Ch: 2+540
GL:1418.934
DL:1419.390



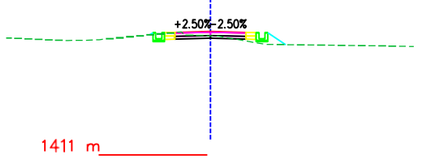
Ch: 2+530
GL:1418.759
DL:1419.553



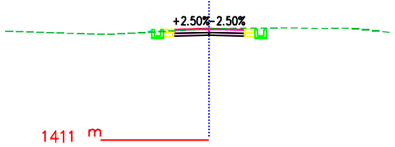
Ch: 2+551.19
GL:1419.207
DL:1419.207



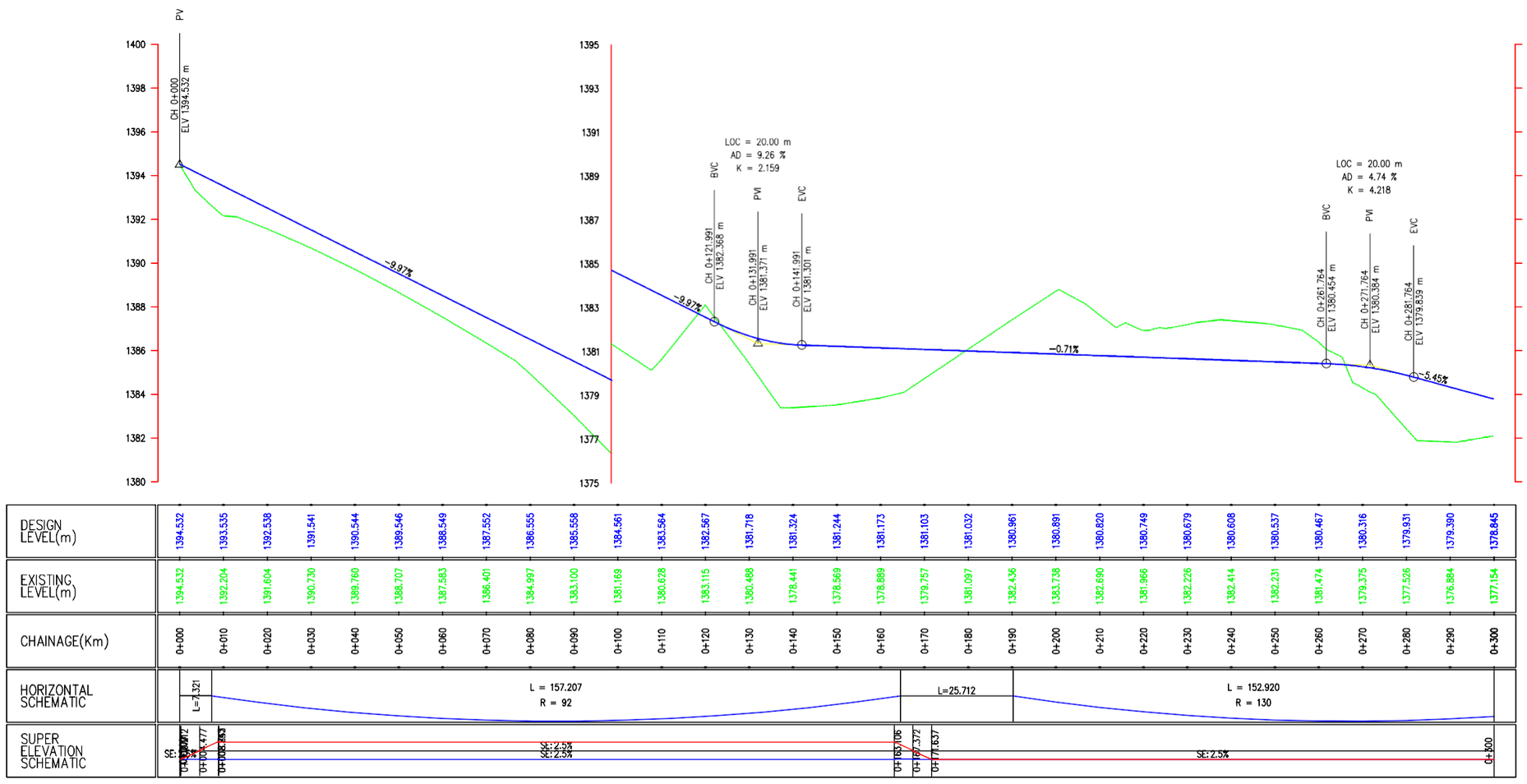
Ch: 2+520
GL:1419.369
DL:1419.691



Ch: 2+550
GL:1419.189
DL:1419.226

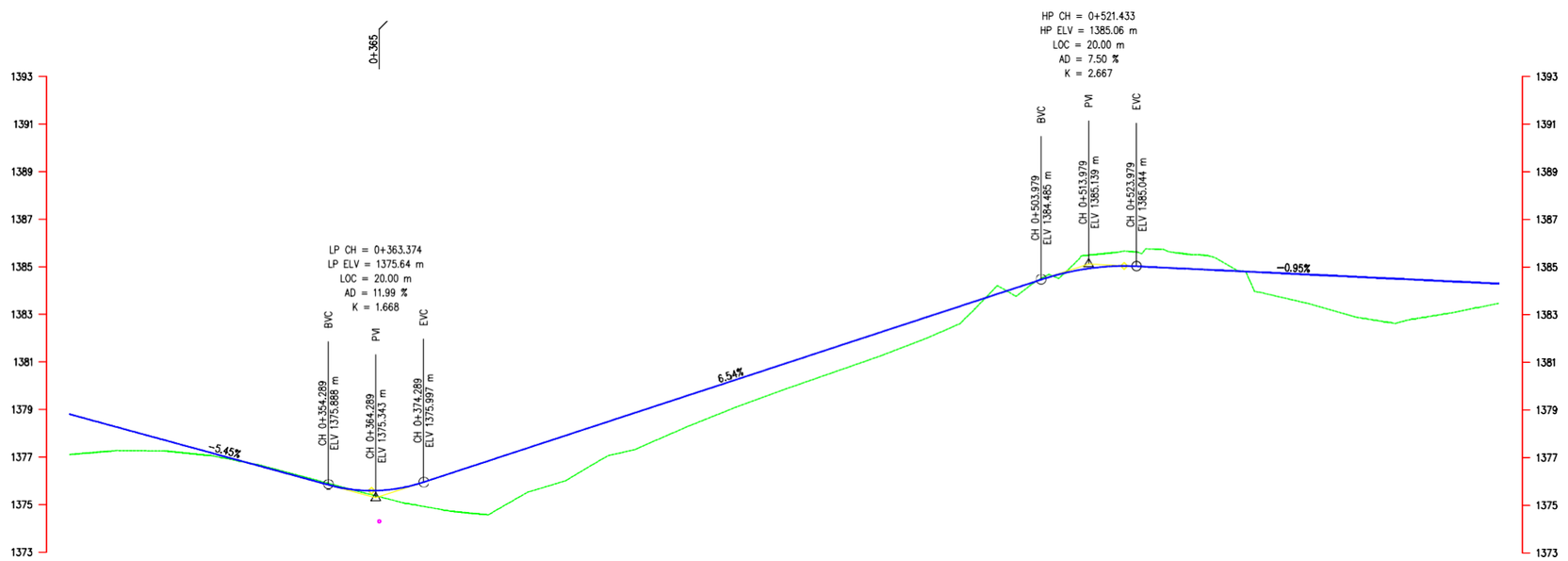


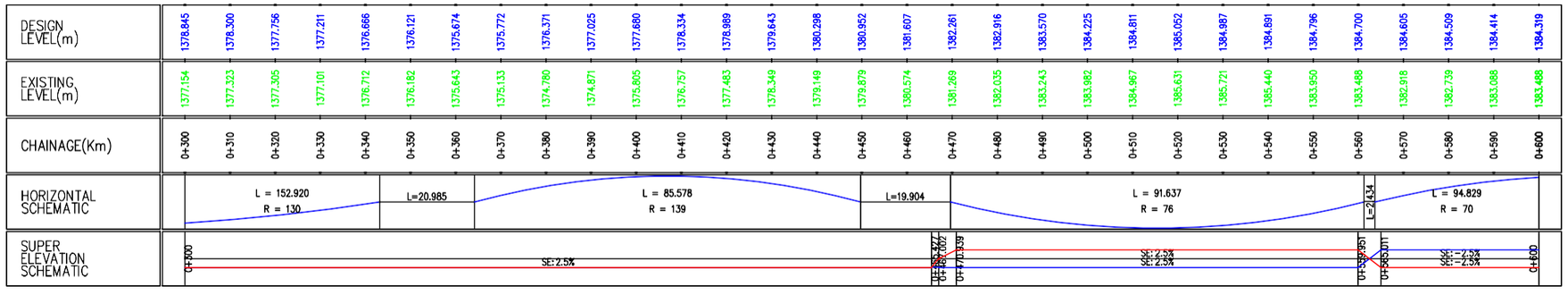
TRIBHUVAN UNIVERSITY INSTITUTE OF ENGINEERING THAPATHALI CAMPUS THAPATHALI, KATHMANDU & DEPARTMENT OF CIVIL ENGINEERING	FINAL YEAR PROJECT GROUP NO. 17	SURYABINAYAK-GUNDU	DATE	REVISION	SIGNATURE	CROSS-0SECTIONS DRAWN BY :GROUP 17 CHECKED BY : APPROVED BY :	SCALE 1:200 4 m 0 8 m	CROSS-SECTIONS 2+520-2+551.19	DATE: DRG NO : 1 SHEET NO : 29



CHAINAGE : 0+000 - 0+300

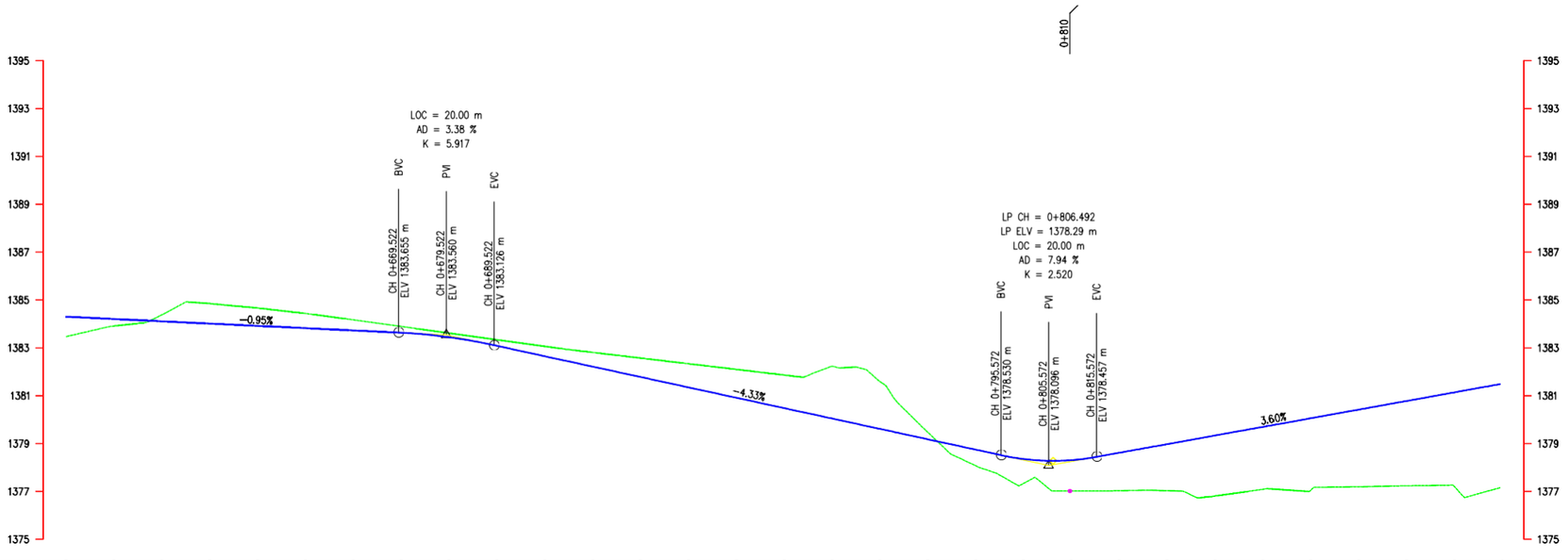
IP No.	CO-ORDINATES(m)		WCB (D-M-S)	DIST. (m)	DEF. ANG. (D-M-S)	TAN. LEN. (m)	SPIRAL CURVE DATA		SIMPLE CURVE DATA			CHAINAGE					
	EASTING	NORTHING					LENGTH (m)	SHIFT (m)	R (m)	E (m)	LC (m)	BC-Spiral	BC-Simple	MC	EC-Simple	EC-Spiral	IP
1	3059799.445	344356.497	225-20-20.98	0.000	00-00-0.00	0.000	0.000	0.000	0.000	0.000	0.000	0+000.000	0+000.000	0+000.000	0+000.000	0+000.000	0+000.000
3	3059719.086	344277.083	127-26-1.48	112.979	97-54-19.50	105.658	0.000	0.000	92.000	48.098	157.207	0+007.321	0+007.321	0+085.924	0+164.528	0+164.528	0+112.979
4	3059892.241	344144.534	60-02-10.68	218.064	67-23-50.80	86.695	0.000	0.000	130.000	26.256	152.920	0+190.240	0+190.240	0+266.700	0+343.160	0+343.160	0+276.935



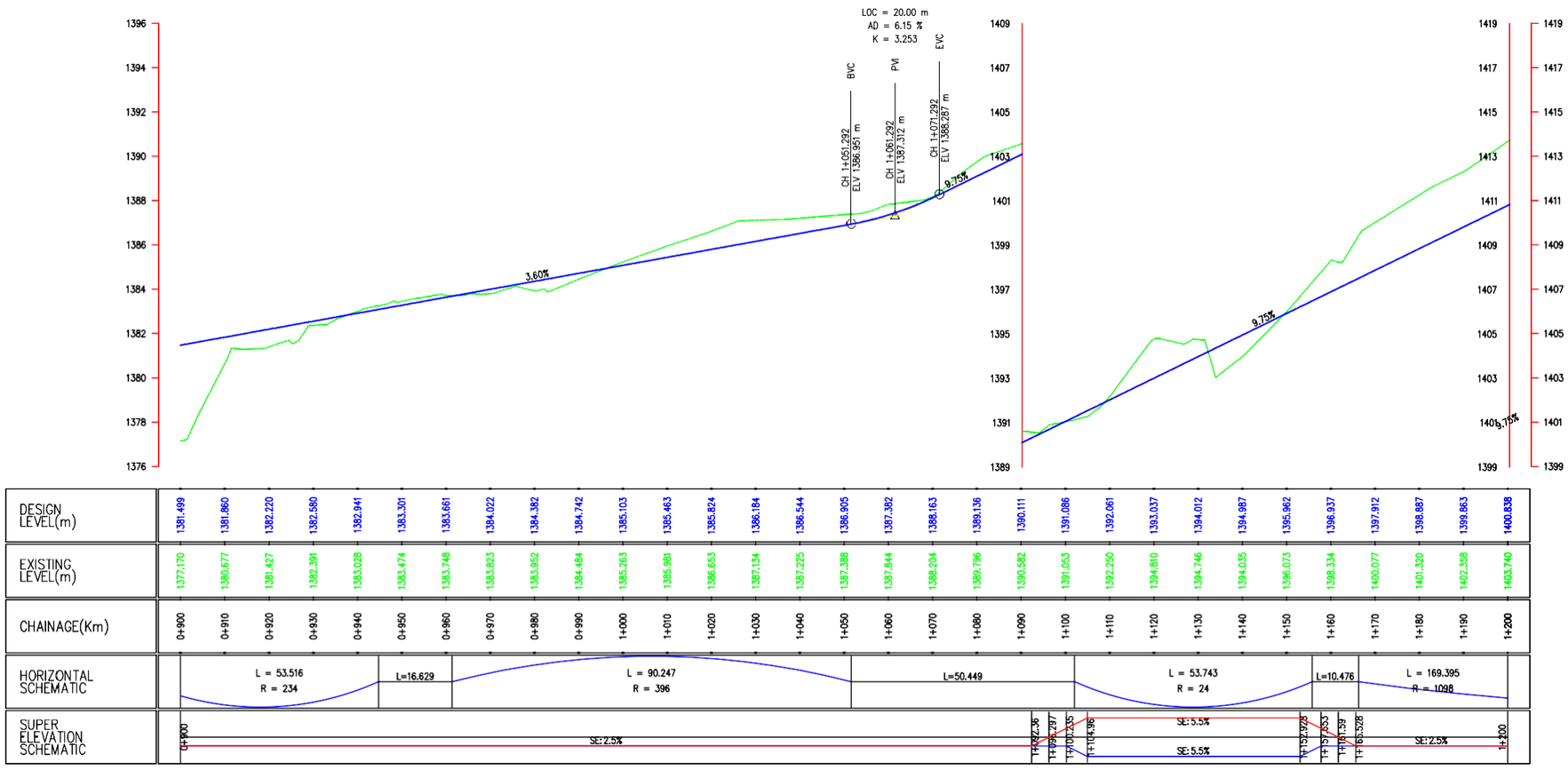


CHAINAGE : 0+300 - 0+600

IP No.	CO-ORDINATES(m)		WCB (D-M-S)	DIST. (m)	DEF. ANG. (D-M-S)	TAN. LEN. (m)	SPIRAL CURVE DATA		SIMPLE CURVE DATA			CHAINAGE					
	EASTING	NORTHING					LENGTH (m)	SHIFT (m)	R (m)	E (m)	LC (m)	BC-Spiral	BC-Simple	MC	EC-Simple	EC-Spiral	IP
4	3059892.241	344144.534	60-02-10.68	218.064	67-23-50.80	86.695	0.000	0.000	130.000	26.256	152.920	0+190.240	0+190.240	0+266.700	0+343.160	0+343.160	0+276.935
5	3060023.815	344220.387	95-18-41.58	151.874	35-16-30.90	44.194	0.000	0.000	139.000	6.856	85.578	0+364.144	0+364.144	0+406.933	0+449.722	0+449.722	0+408.338
6	3060139.730	344209.611	26-13-36.84	116.414	69-05-4.74	52.316	0.000	0.000	76.000	16.266	91.637	0+469.627	0+469.627	0+515.445	0+561.264	0+561.264	0+521.943
7	3060188.806	344309.229	103-50-44.18	111.051	77-37-7.34	56.300	0.000	0.000	70.000	19.832	94.829	0+563.698	0+563.698	0+611.113	0+658.527	0+658.527	0+619.998

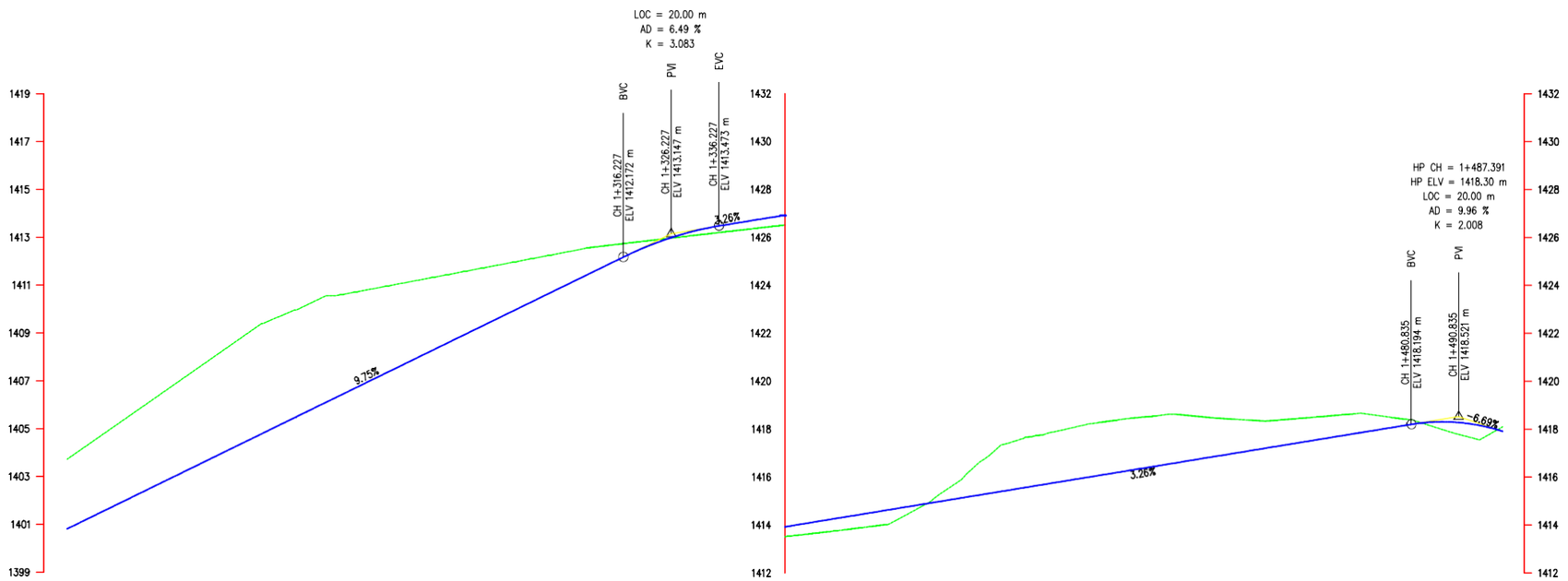

0+810
LOC = 20.00 m
AD = 3.38 %
K = 5.917
BVC
PVI
EVC
CH 0+669.522
ELV 1383.655 m
CH 0+679.522
ELV 1383.560 m
CH 0+689.522
ELV 1383.126 m
LP CH = 0+806.492
LP ELV = 1378.29 m
LOC = 20.00 m
AD = 7.94 %
K = 2.520
CH 0+795.572
ELV 1378.530 m
CH 0+805.572
ELV 1378.096 m
CH 0+815.572
ELV 1378.457 m
-0.95%
-4.33%
3.60%
1395
1393
1391
1389
1387
1385
1383
1381
1379
1377
1375

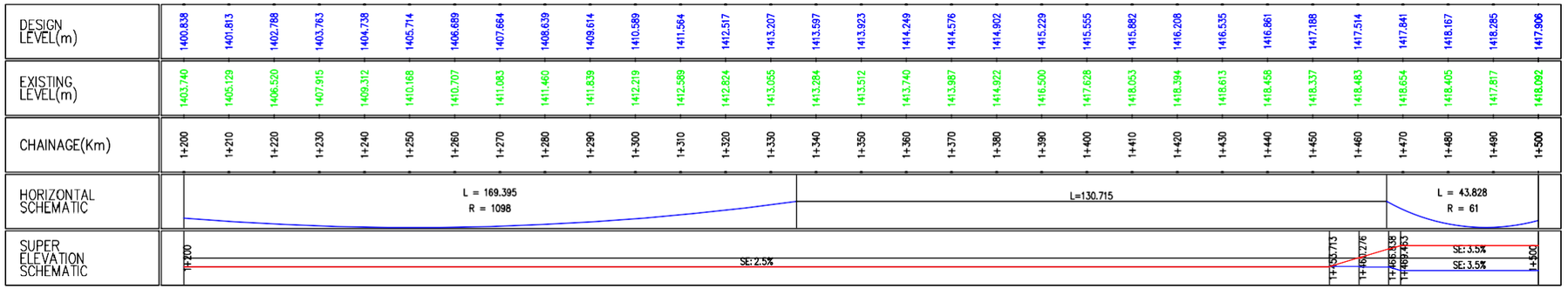
| | | |



CHAINAGE : 0+900 - 1+200

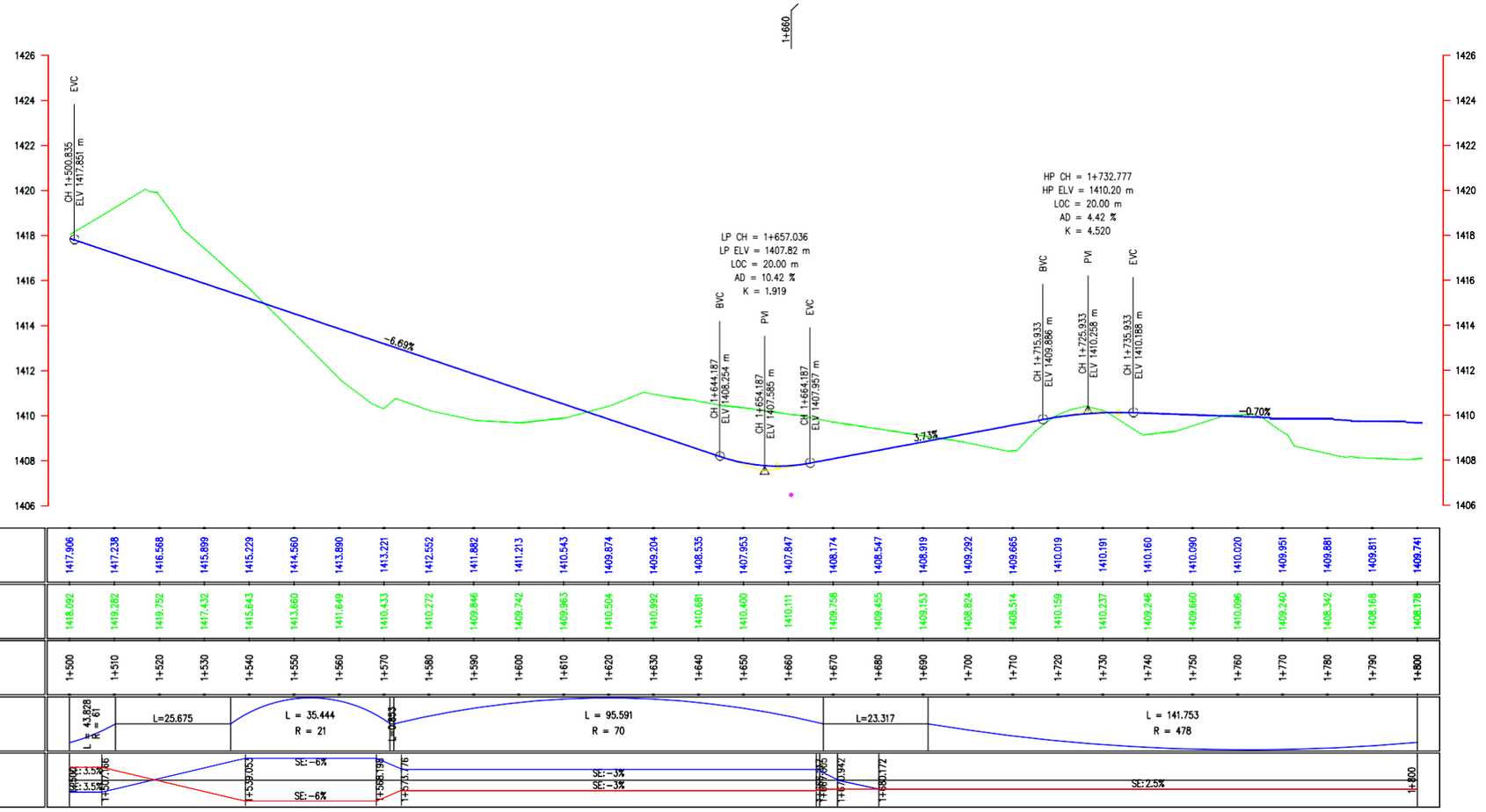
IP No.	CO-ORDINATES(m)		WCB (D-M-S)	DIST. (m)	DEF. ANG. (D-M-S)	TAN. LEN. (m)	SPIRAL CURVE DATA		SIMPLE CURVE DATA			CHAINAGE					
	EASTING	NORTHING					LENGTH (m)	SHIFT (m)	R (m)	E (m)	LC (m)	BC-Spiral	BC-Simple	MC	EC-Simple	EC-Spiral	IP
9	3060464.061	344345.055	24-18-58.08	109.821	13-06-12.80	26.875	0.000	0.000	234.000	1.538	53.516	0+891.232	0+891.232	0+917.990	0+944.748	0+944.748	0+918.107
10	3060500.636	344425.999	37-22-25.02	88.824	13-03-26.94	45.320	0.000	0.000	396.000	2.585	90.247	0+961.377	0+961.377	1+006.500	1+051.624	1+051.624	1+006.696
11	3060588.839	344541.473	269-04-17.59	145.307	128-18-7.43	49.538	0.000	0.000	24.000	31.046	53.743	1+102.072	1+102.072	1+128.944	1+155.815	1+155.815	1+151.611
12	3060443.978	344539.126	260-13-55.92	144.880	08-50-21.67	84.866	0.000	0.000	1098.000	3.275	169.395	1+166.291	1+166.291	1+250.989	1+335.686	1+335.686	1+251.157





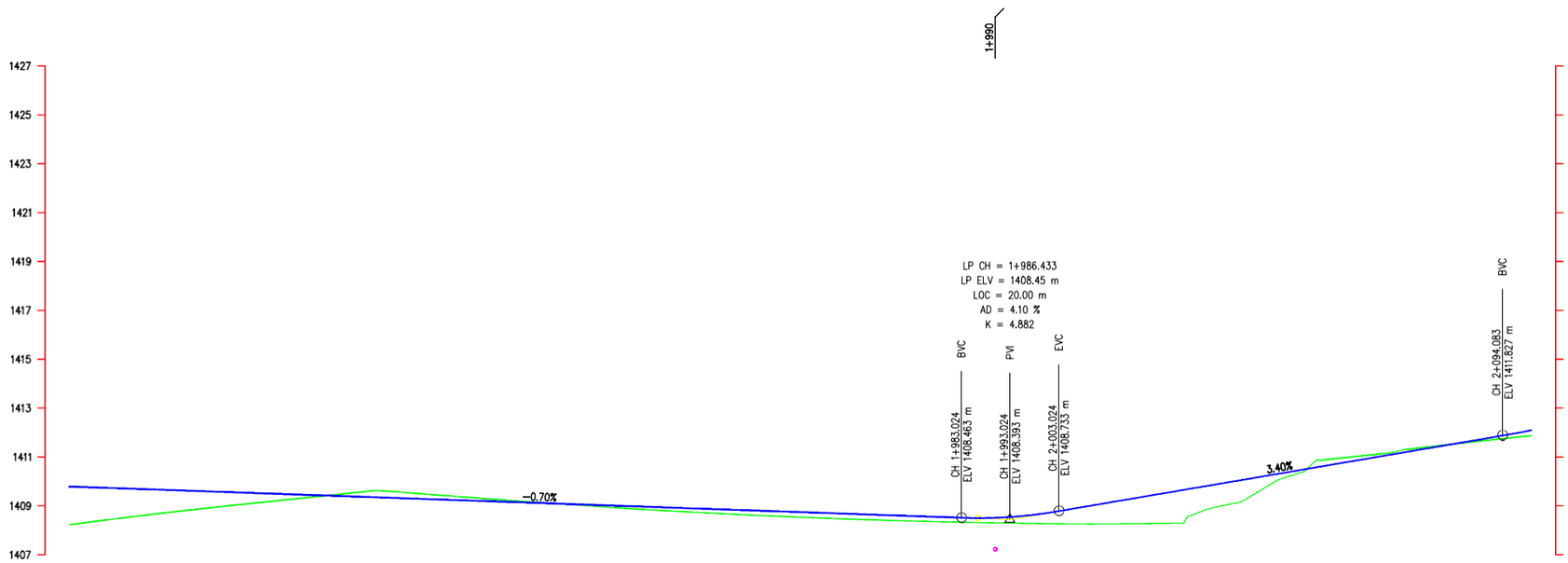
CHAINAGE : 1+200 - 1+500

IP No.	CO-ORDINATES(m)		WCB (D-M-S)	DIST. (m)	DEF. ANG. (D-M-S)	TAN. LEN. (m)	SPIRAL CURVE DATA		SIMPLE CURVE DATA			CHAINAGE					
	EASTING	NORTHING					LENGTH (m)	SHIFT (m)	R (m)	E (m)	LC (m)	BC-Spiral	BC-Simple	MC	EC-Simple	EC-Spiral	IP
12	3060443.978	344539.126	260-13-55.92	144.880	08-50-21.67	84.866	0.000	0.000	1098.000	3.275	169.395	1+166.291	1+166.291	1+250.989	1+335.686	1+335.686	1+251.157
13	3060208.947	344498.665	219-03-57.00	238.489	41-09-58.92	22.908	0.000	0.000	61.000	4.160	43.828	1+466.401	1+466.401	1+488.315	1+510.229	1+510.229	1+489.309

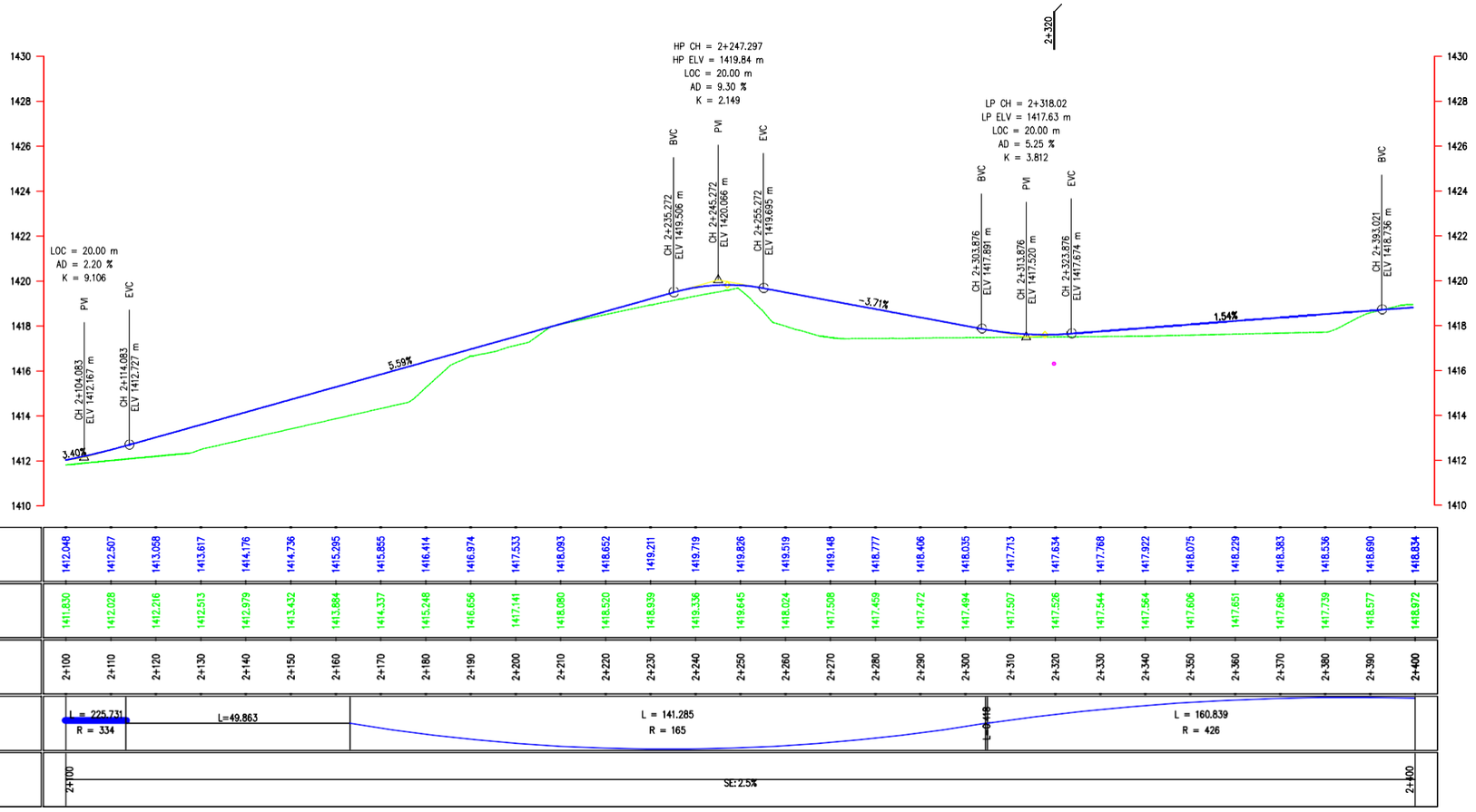


CHAINAGE : 1+500 – 1+800

IP No.	CO-ORDINATES(m)		WCB (D-M-S)	DIST. (m)	DEF. ANG. (D-M-S)	TAN. LEN. (m)	SPIRAL CURVE DATA		SIMPLE CURVE DATA			CHAINAGE					
	EASTING	NORTHING					LENGTH (m)	SHIFT (m)	R (m)	E (m)	LC (m)	BC-Spiral	BC-Simple	MC	EC-Simple	EC-Spiral	IP
13	3060208.947	344498.665	219-03-57.00	238.489	41-09-58.92	22.908	0.000	0.000	61.000	4.160	43.828	1+466.401	1+466.401	1+488.315	1+510.229	1+510.229	1+489.309
15	3060163.448	344442.610	315-46-16.69	72.196	96-42-19.68	23.614	0.000	0.000	21.000	10.601	35.444	1+535.903	1+535.903	1+553.626	1+571.348	1+571.348	1+559.517
16	3060106.671	344500.936	34-00-50.02	81.398	78-14-33.34	56.931	0.000	0.000	70.000	20.228	95.591	1+572.201	1+572.201	1+619.997	1+667.792	1+667.792	1+629.132
17	3060191.502	344626.638	17-01-21.22	151.648	16-59-28.81	71.401	0.000	0.000	478.000	5.303	141.753	1+691.109	1+691.109	1+761.986	1+832.862	1+832.862	1+762.510

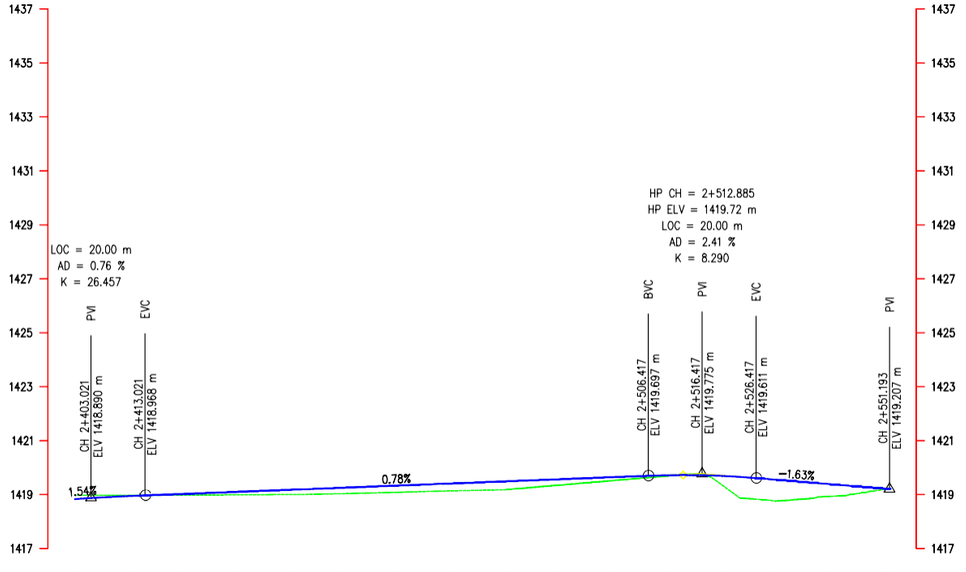

1+990
LP CH = 1+986.433
LP ELV = 1408.45 m
LOC = 20.00 m
AD = 4.10 %
K = 4.882
BVC
PVI
EVC
CH 1+983.024
ELV 1408.463 m
CH 1+993.024
ELV 1408.395 m
CH 2+003.024
ELV 1408.733 m
BVC
CH 2+094.083
ELV 1411.827 m
-0.70%
3.40%
1427
1425
1423
1421
1419
1417
1415
1413
1411
1409
1407

| | | |



CHAINAGE : 2+100 - 2+400

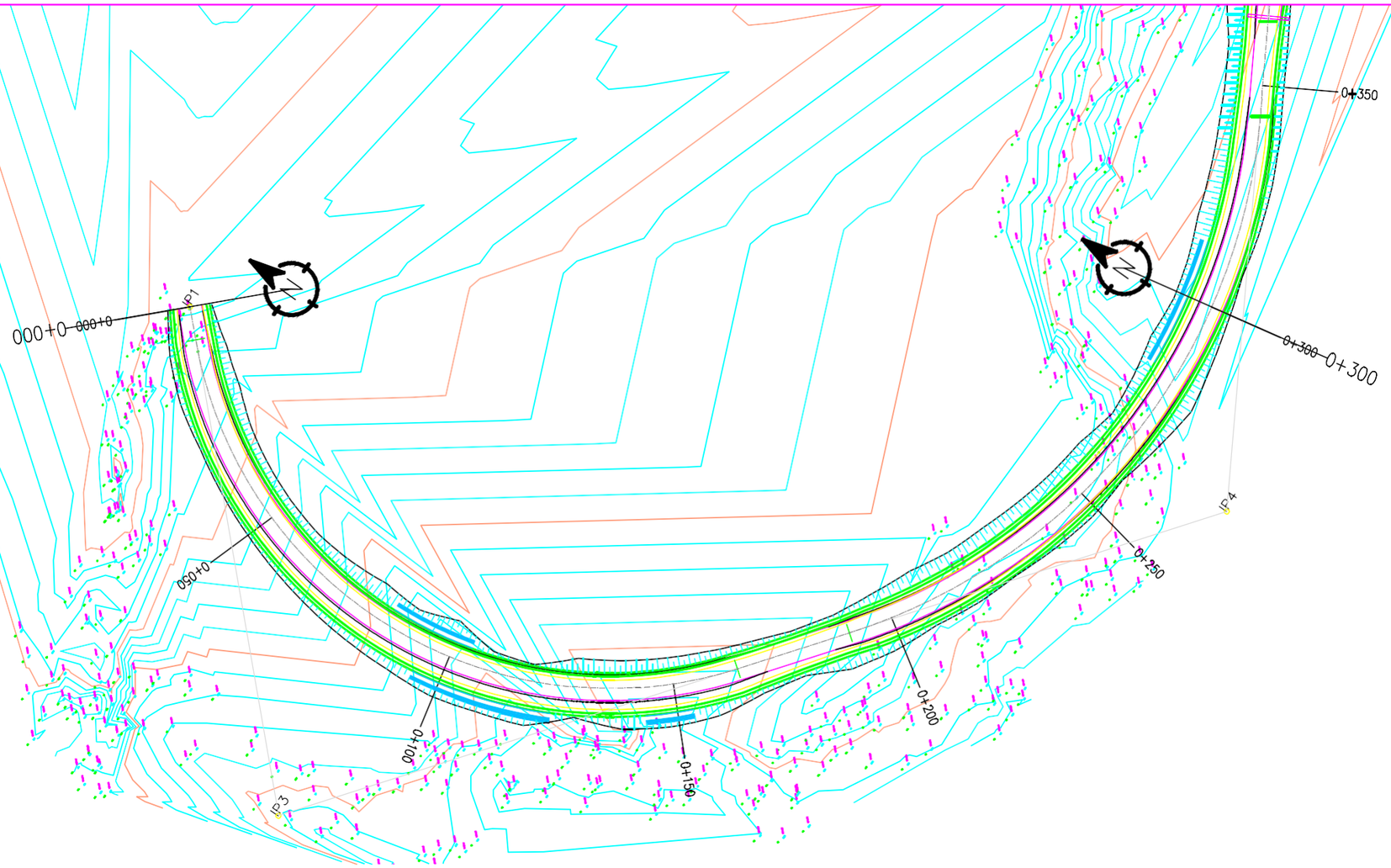
IP No.	CO-ORDINATES(m)		WCB (D-M-S)	DIST. (m)	DEF. ANG. (D-M-S)	TAN. LEN. (m)	SPIRAL CURVE DATA		SIMPLE CURVE DATA			CHAINAGE					
	EASTING	NORTHING					LENGTH (m)	SHIFT (m)	R (m)	E (m)	LC (m)	BC-Spiral	BC-Simple	MC	EC-Simple	EC-Spiral	IP
18	3060262.782	344859.455	55-44-43.62	243.484	38-43-22.40	117.367	0.000	0.000	334.000	20.021	225.731	1+887.579	1+887.579	2+000.444	2+113.310	2+113.310	2+004.946
19	3060463.244	344995.968	06-41-4.47	242.531	49-03-39.15	75.301	0.000	0.000	165.000	16.370	141.285	2+163.172	2+163.172	2+233.815	2+304.458	2+304.458	2+238.473
20	3060481.532	345152.007	28-19-1.02	157.107	21-37-56.55	81.389	0.000	0.000	426.000	7.705	160.839	2+304.875	2+304.875	2+385.295	2+465.714	2+465.714	2+386.264



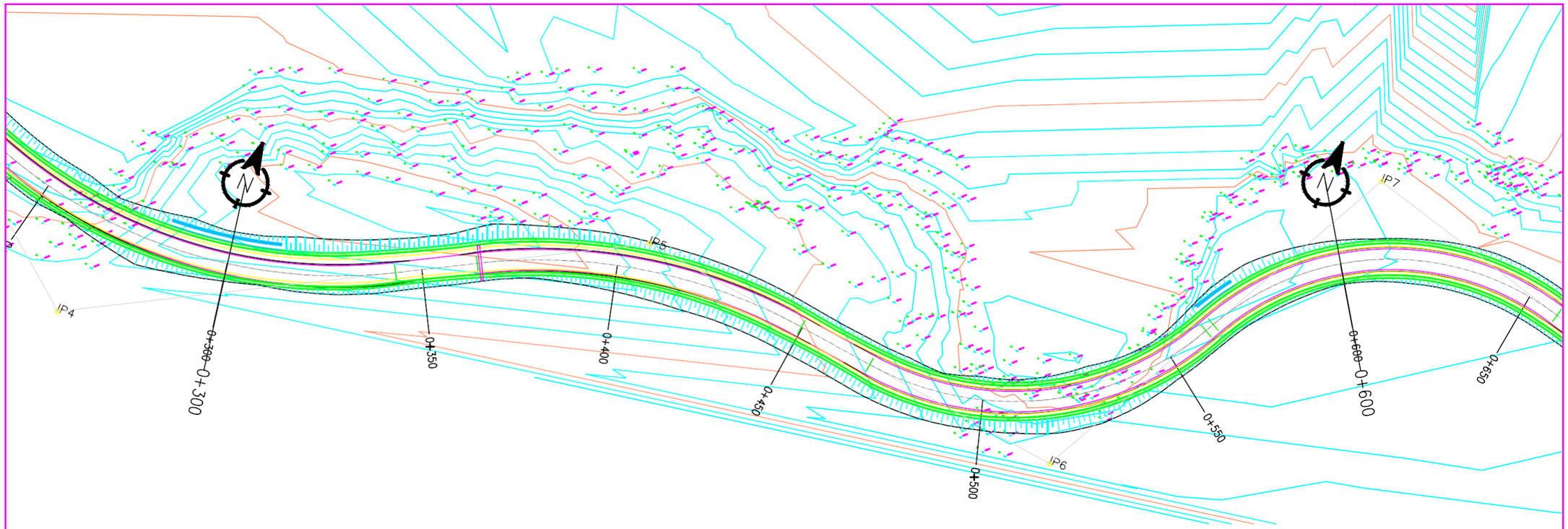
DESIGN LEVEL(m)	1418.834	1418.943	1419.022	1419.100	1419.178	1419.256	1419.334	1419.412	1419.490	1419.568	1419.646	1419.717	1419.691	1419.553	1419.390	1419.226	1419.207
EXISTING LEVEL(m)	1418.972	1418.976	1418.982	1418.991	1419.004	1419.036	1419.084	1419.138	1419.183	1419.353	1419.522	1419.691	1419.369	1418.759	1418.934	1419.189	1419.207
CHAINAGE(Km)	2+400	2+410	2+420	2+430	2+440	2+450	2+460	2+470	2+480	2+490	2+500	2+510	2+520	2+530	2+540	2+550	2+551.193
HORIZONTAL SCHEMATIC				L = 160.839 R = 426									L=85.479				
SUPER ELEVATION SCHEMATIC	2+400												SE: 2.5%			2+551.193	

CHAINAGE : 2+400 - 2+551.193

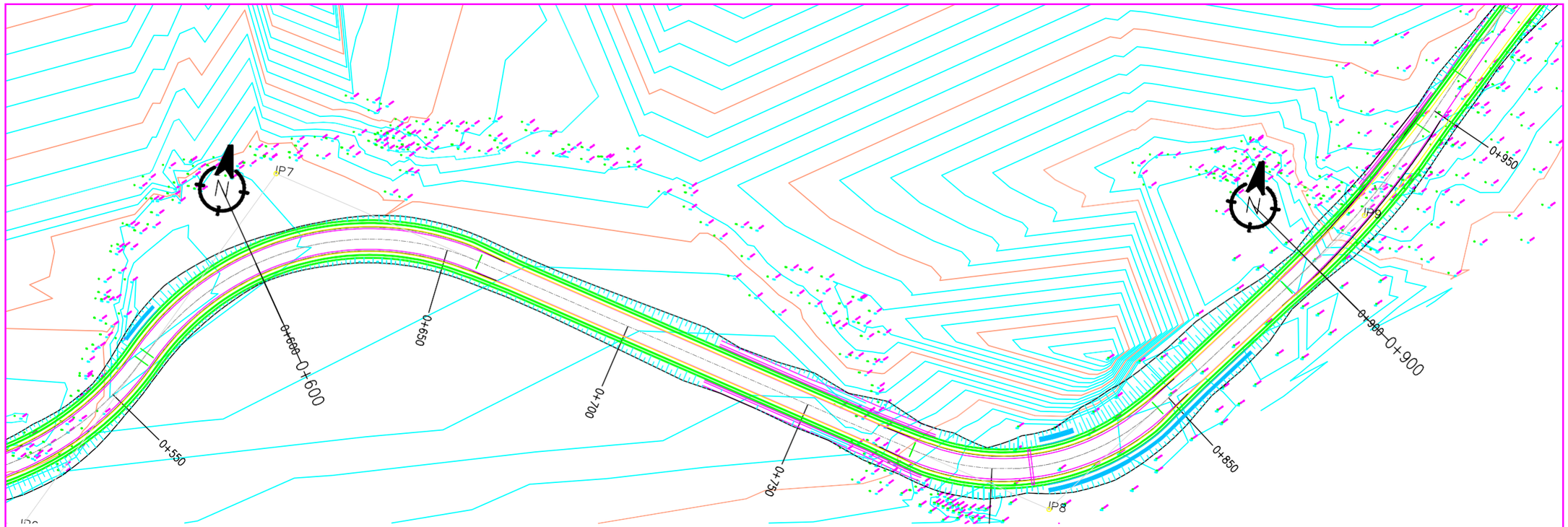
IP No.	CO-ORDINATES(m)		WCB (D-M-S)	DIST. (m)	DEF. ANG. (D-M-S)	TAN. LEN. (m)	SPIRAL CURVE DATA		SIMPLE CURVE DATA			CHAINAGE					
	EASTING	NORTHING					LENGTH (m)	SHIFT (m)	R (m)	E (m)	LC (m)	BC-Spiral	BC-Simple	MC	EC-Simple	EC-Spiral	IP
20	3060481.532	345152.007	28-19-1.02	157.107	21-37-56.55	81.389	0.000	0.000	426.000	7.705	160.839	2+304.875	2+304.875	2+385.295	2+465.714	2+465.714	2+386.264



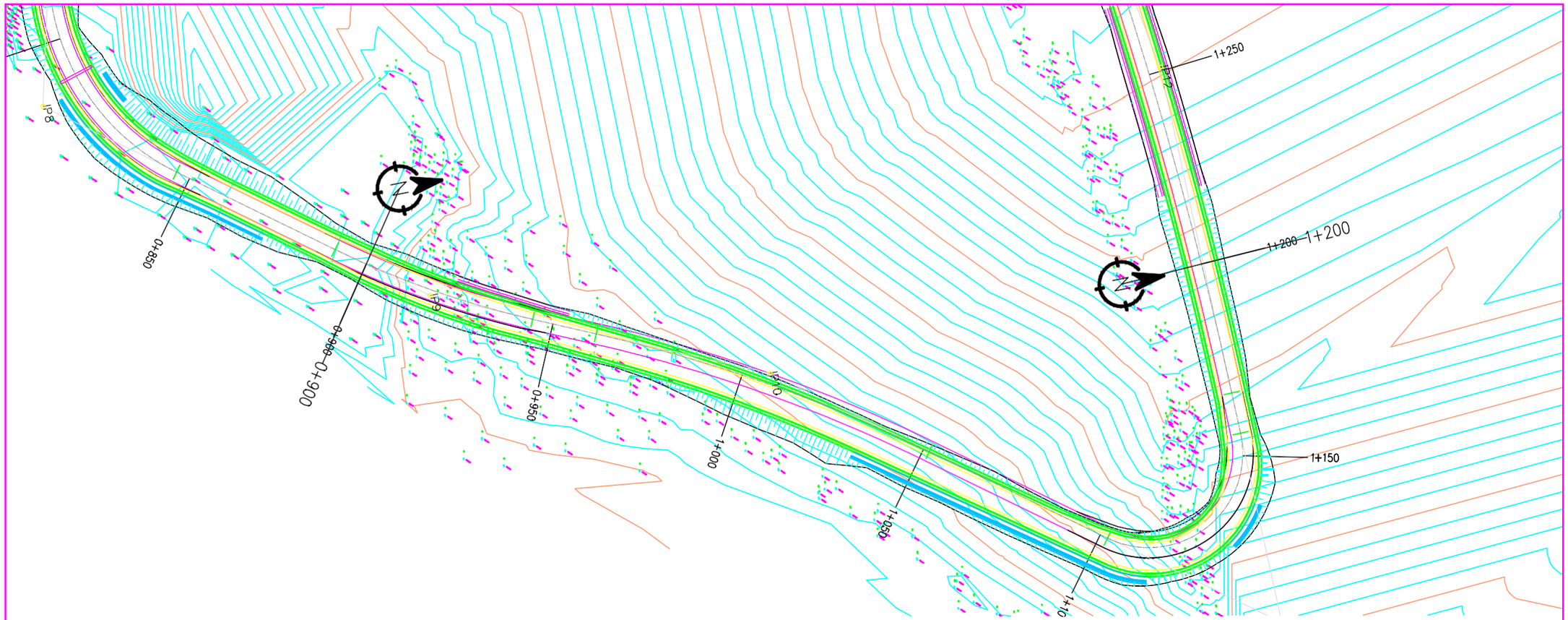
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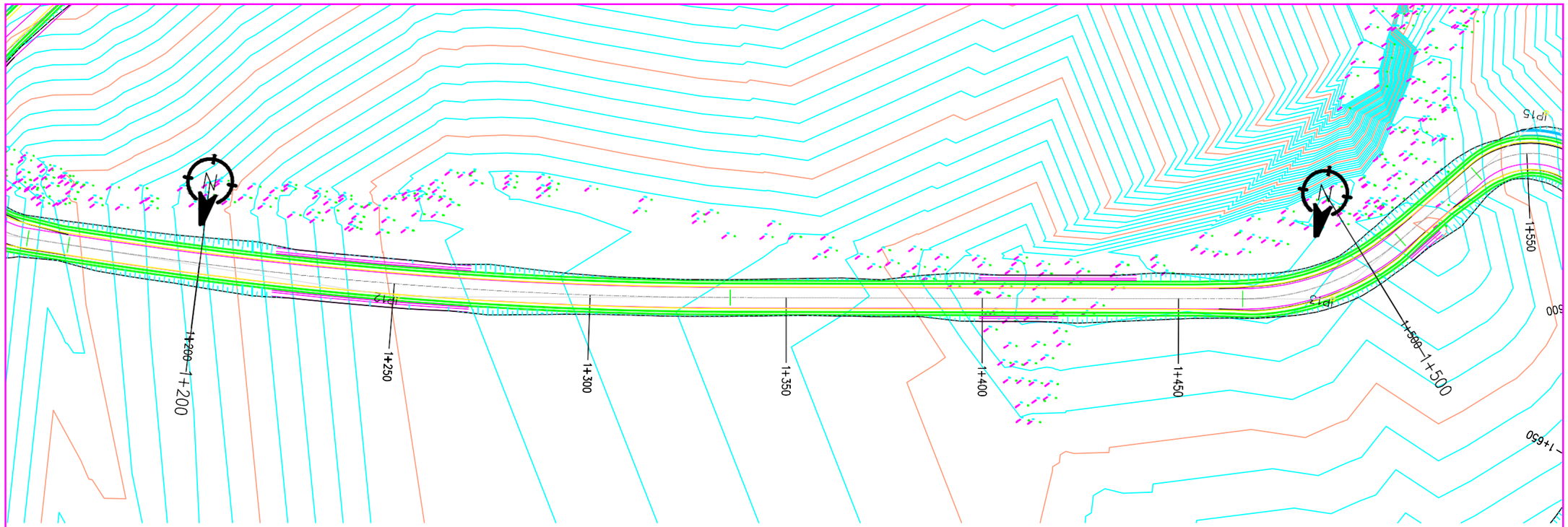
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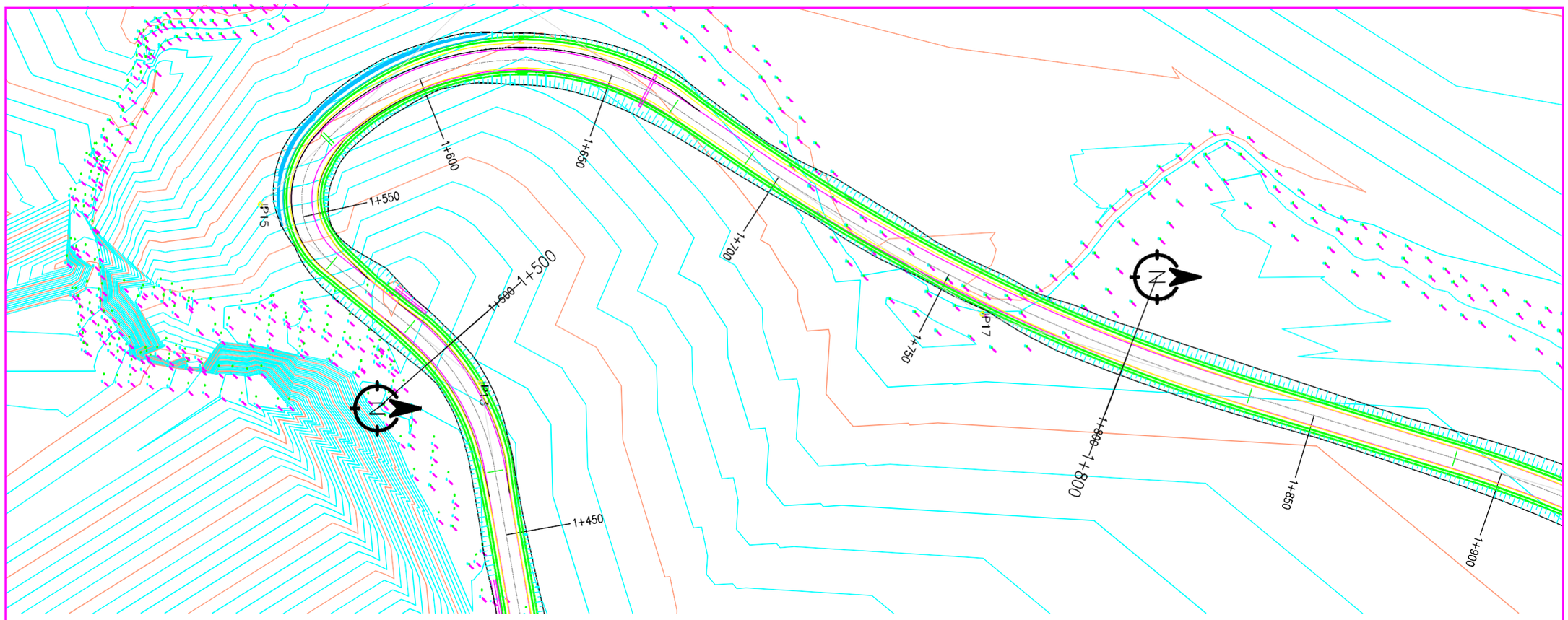
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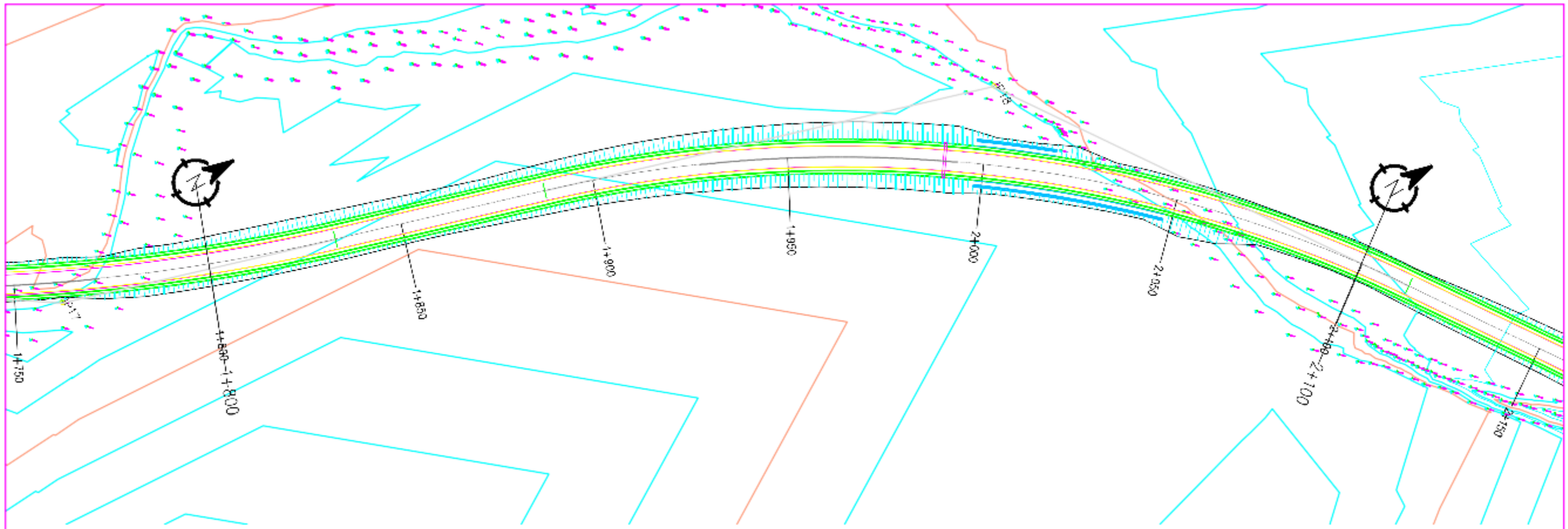
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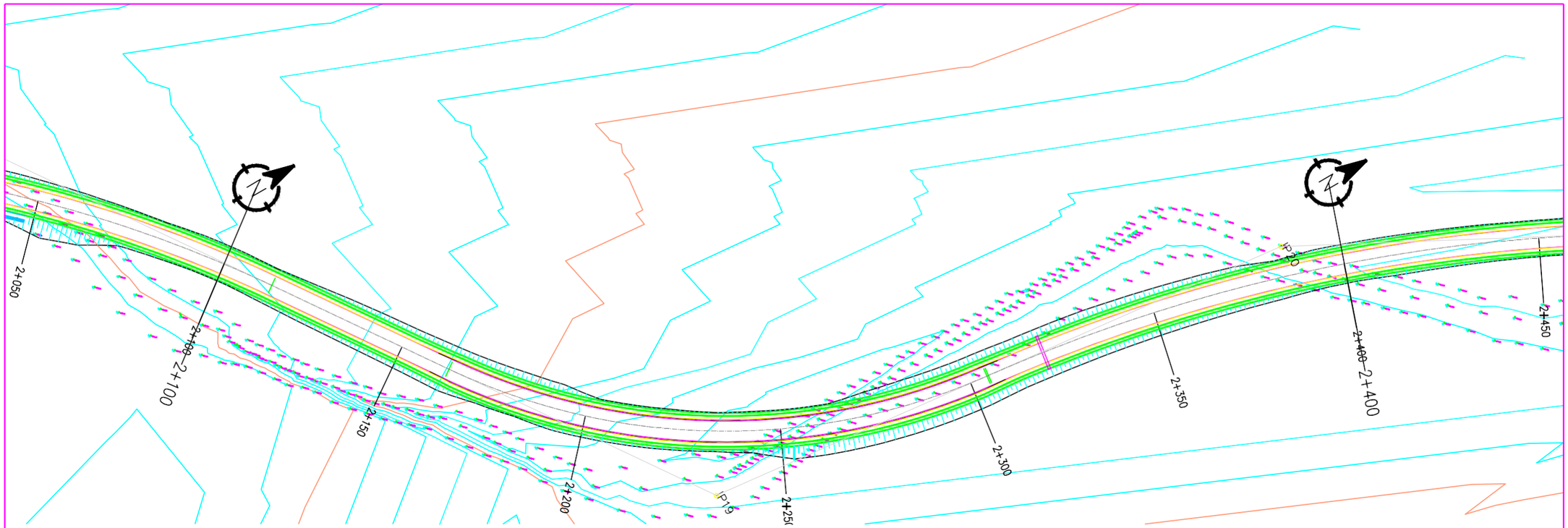
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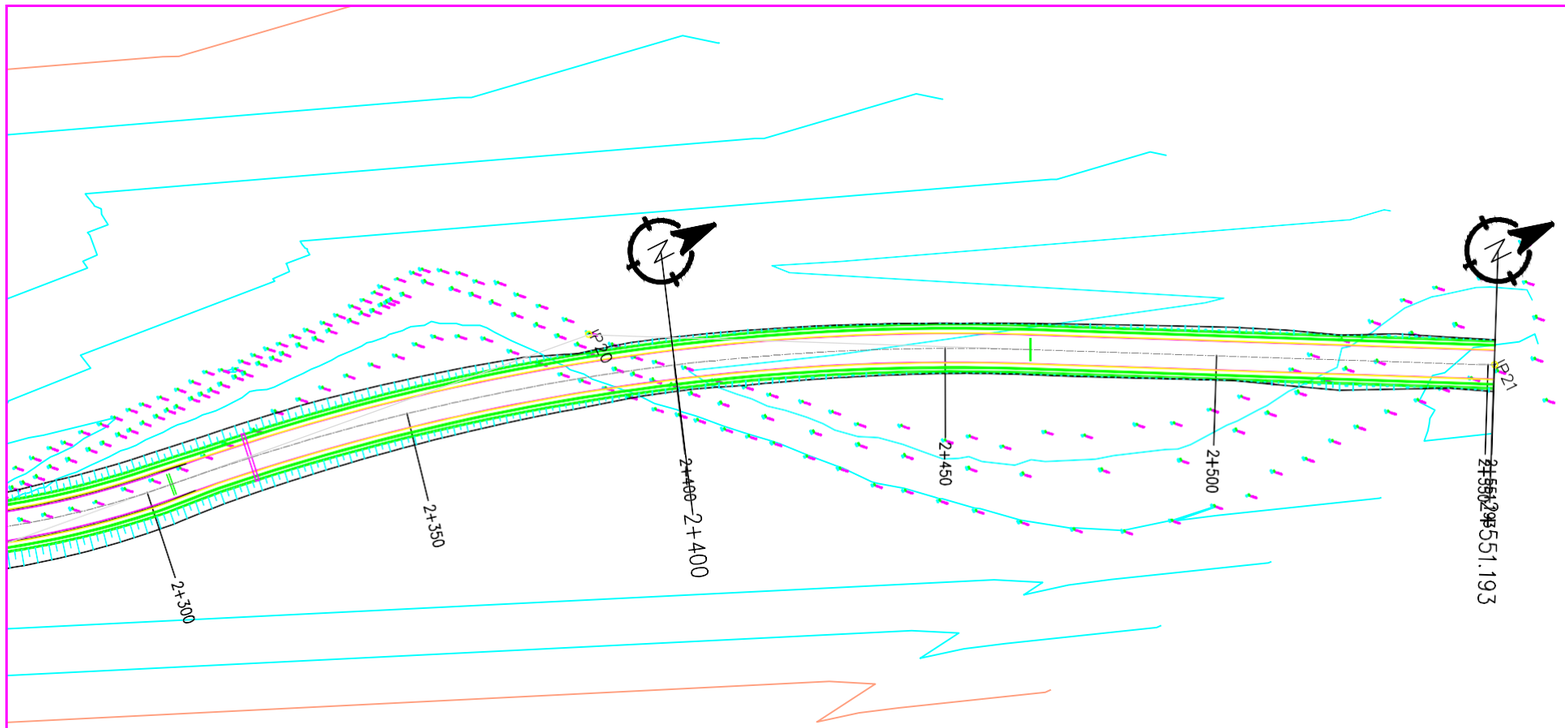
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